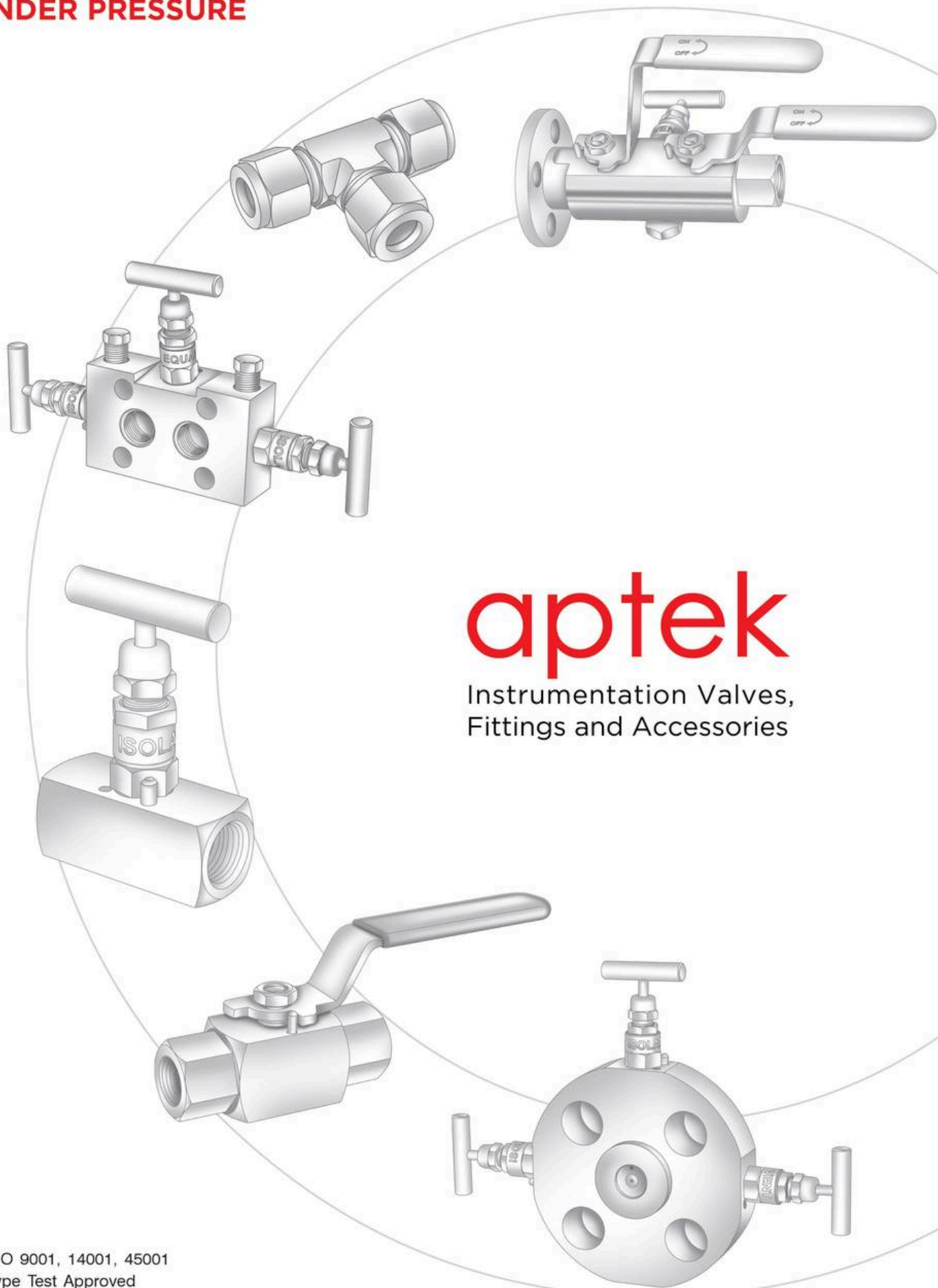


PERFORMANCE
UNDER PRESSURE



aptek

Instrumentation Valves,
Fittings and Accessories

ISO 9001, 14001, 45001
Type Test Approved
PED
CRN
EIL

About us

Aptek Instrumentation Pvt. Ltd. is an ISO 9001: 2015 company manufacturing a wide range of Instrumentation valves, tube fittings, various Instrumentation accessories..

The company started on a very small scale in 1995, is now having latest manufacturing & testing facility. The company is located in Ahmedabad a top industrial center & rapidly growing metro of India.

As we say "PERFORMANCE UNDER PRESSURE" our products perform really well, not only under pressure but also under different conditions & applications.

As a result "APTEK" is now a preferred brand for many reputed OEM's, consultants, Agents, Trading houses & end users globally.

Our Main products are 2,3,5 manifold Valves, Low, High Pressure Ball Valves, Monoflange instrumentation valves, Needle Valves, Gauge Valves, Double Block & Bleed Valve, Instrumentation Tube Fittings, Instrumentation Pipe Fittings, Pressure Gauge Accessories, Thermowells.



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General Technical Information

Applications

Aptek products are mainly instrumentation accessories used for critical applications in the field of oil and gas, Petrochem, Chemical, Pharma, Food, Pulp & paper & many more where pressure parameters are measured or monitored.

Different applications are

- Isolation (Throttling & ON- OFF)
- Isolation & drain
- Isolation & calibration
- Differential pressure measurement
- Distribution

Standards

Aptek follows ISO 9001:2015, ISO 14001-2015 & ISO 45001-2018 quality system for manufacturing & supply of all the products. The entire product range is certified under PED 2014/68/EU. Fitting are Type Test approved in accordance with ASTM F1387. Valve & Manifolds approved in accordance with MSS-SP-99

Manufacturing & Testing Standards

- MSS - SP 99 for instrument valves
- MSS - SP - 110 for Instruments ball valves
- ASTM F1387- Performance of Tube Fittings
- ASME B 16.5 - For Pipe Flanges
- ASME B 16.34 - For Valves
- ANSI/ASME B 1.20.1- For Pipe Threads

Testing Procedure

All the Aptek products are tested as per the relevant standards for hydro & pneumatic pressure.

Certification

Each & every product is supplied with the certificates for the wetted parts & the pressure test. The general format is EN 10204 3.1

Marking

All Aptek products are supplied with laser marking. The marking is done in accordance with MSS- SP -25

Our Certifications

ISO 9001-2015



ISO 45001-2018 TUV



ISO 14001-2015 TUV



ASTM F1387 Type Test Fittings



MSS-SP-99 Type Test Valves



PED Fittings



PED Valves & Manifolds



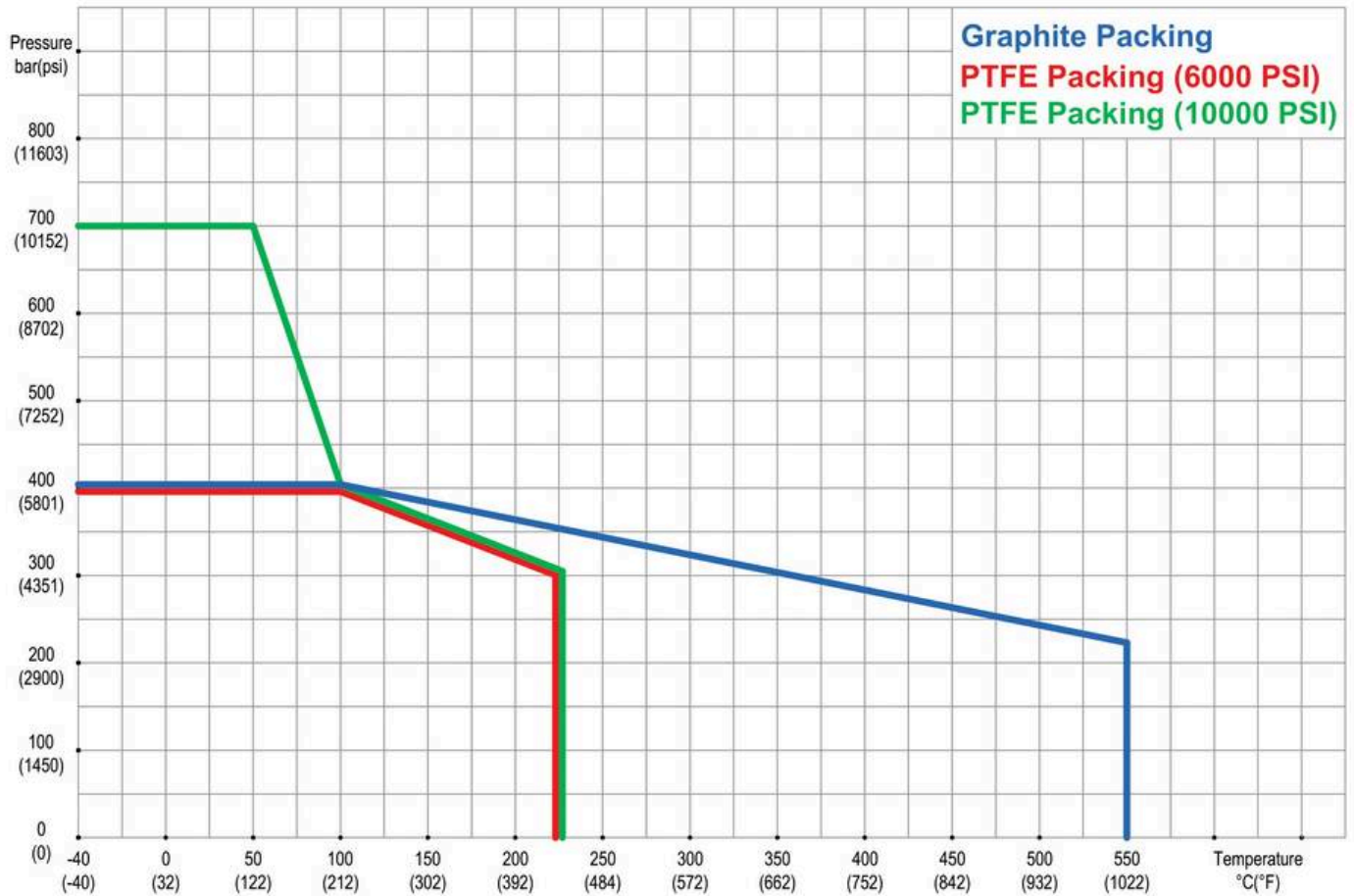
CRN - ABSA



EIL



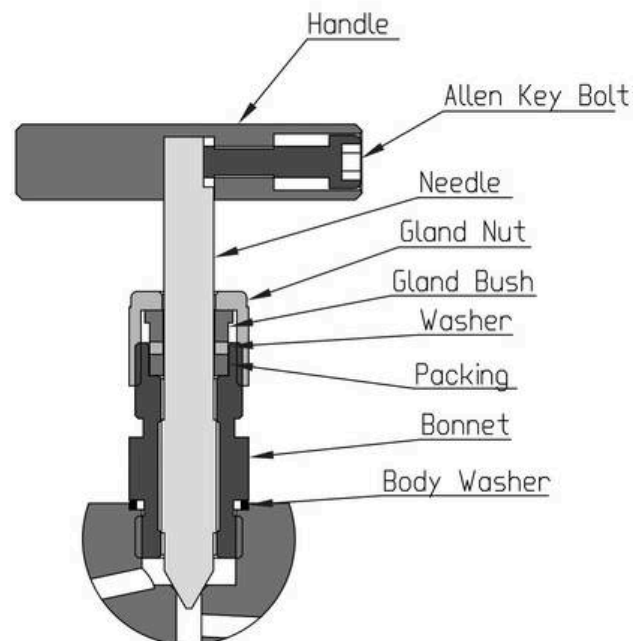
Technical information on valve heads of Needle valves,
Gauge valves & manifold valves



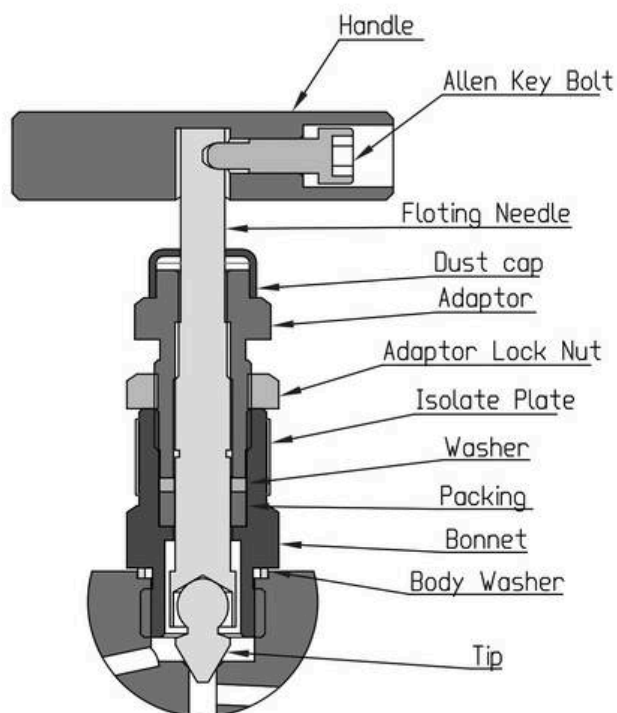
P-T CHART FOR NEEDLE VALVE & MANIFOLD VALVE

Valve Heads or Bonnet Assemblies

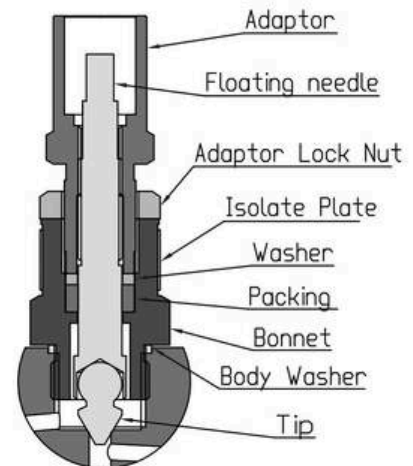
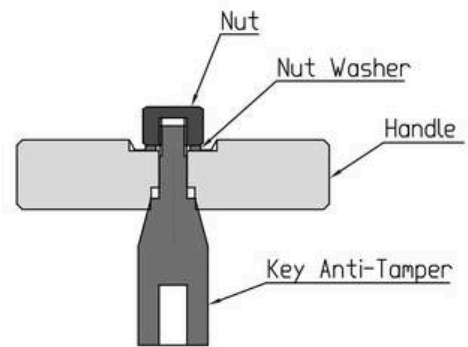
Type A (Single Bonnet Assembly)



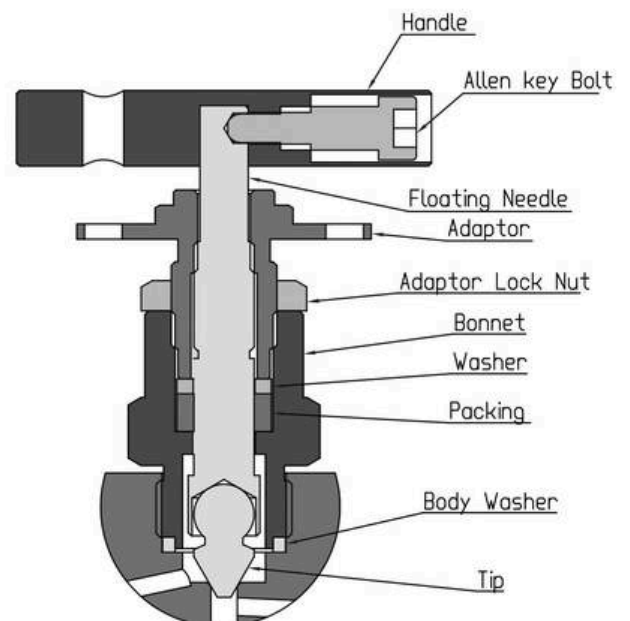
Type B (Double Bonnet Assembly)

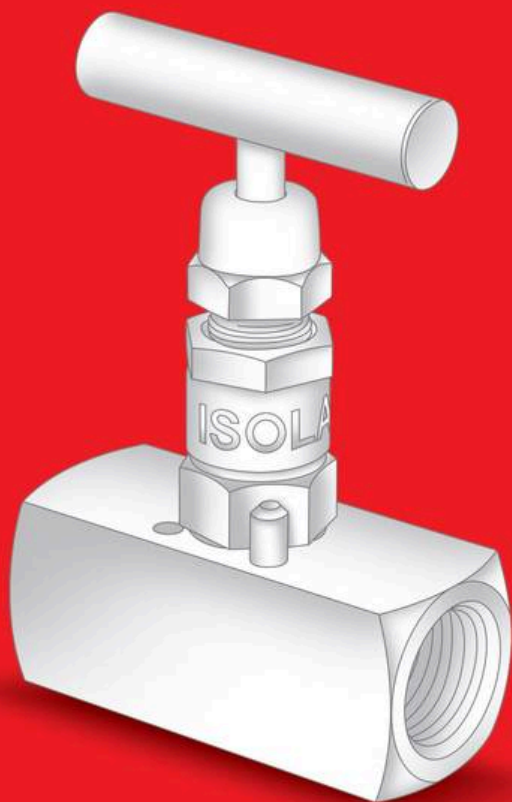


Type C (Anti-Tamper Bonnet Assembly)



Type D (Lockable Bonnet Assembly)





Needle Valves

Needle Valve Selection Chart

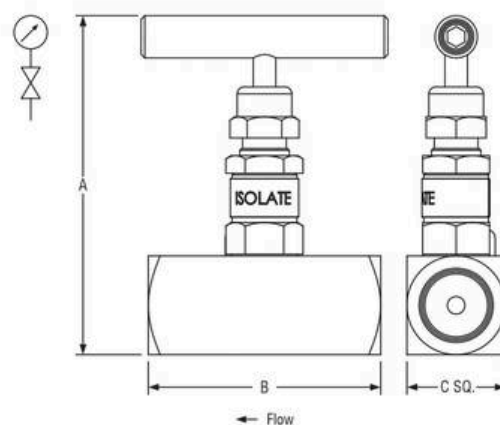
A	NV	1	FF	N	12	SS	T	OPTION
A = Aptek	NV = Needle Valve	1 = 6000 psi, Double Bonnet Assembly 2 = 6000 psi, Single Bonnet Assembly 3 = Angle Valve 4 = 10,000 psi	FF = Female X Female MF = Male X Female MM = Male X Male CF = Compression fitting (tube ends) SW = Socket Weld BW = Butt Weld	N = NPT G = BSPP GT = BSPT M = Metric Blank = for CF / SW / BW end	18 = 1/8" 14 = 1/4" 38 = 3/8" 12 = 1/2" 34 = 3/4" 01 = 1" 06M = 6 mm OD tube end 08M = 8 mm OD tube end 10M = 10 mm OD tube end 12M = 12 mm OD tube end	SS = SS 316 SSL = SS 316L SSD = SS 316/316L Dual SS4 = SS 304 MO = Monel 400 HC = Hastelloy C 276 DU = Duplex SS SD = Super Duplex IN6 = Inconel-625 IN8 = Inconel-825 CS = Carbon Steel BR = Brass	T = PTFE Packing G = Graphite Packing	NA = NACE MR 01-75 (Latest Edition) OXY = Cleaned for Oxygen Service RK = Round Knurled Metal Handle PM = Panel Mounting FS = Fire Safe AT = Anti Temper LH = Lockable Handle

These needle valves are an ideal option for isolating or throttling applications. The self centering, non rotating stem ensures perfect seating. The rigid design gives repetitive bubble tight shut off, easy maintenance & long trouble free life.

Standard Specifications:

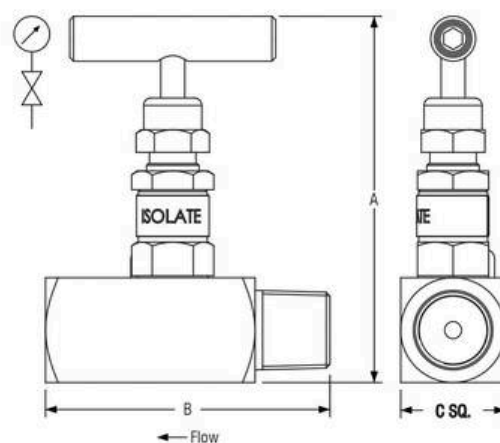
Maximum working Pressure: 413 bar at 38°C (6000 psi at 100°F) (see Pressure & Temperature Chart on page No. 07 for higher temperature applications). MOC in accordance with NACE MR0175 is available on request.

ANV 1 FF



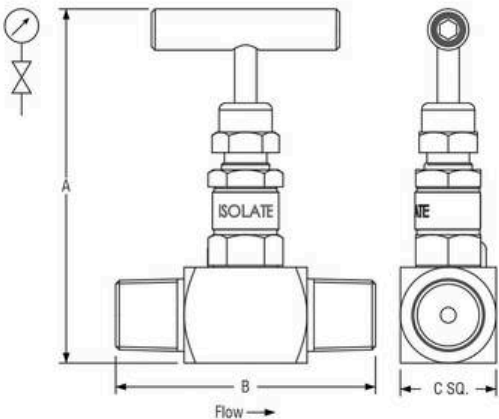
SIZE	A	B	CSQ	WEIGHT (kg)
1/4"	93	55	25	0.36
3/8"	93	55	25	0.37
1/2"	95	66	28	0.46
3/4"	109	70	38	0.8
1"	117	80	45	1.3

ANV 1 MF



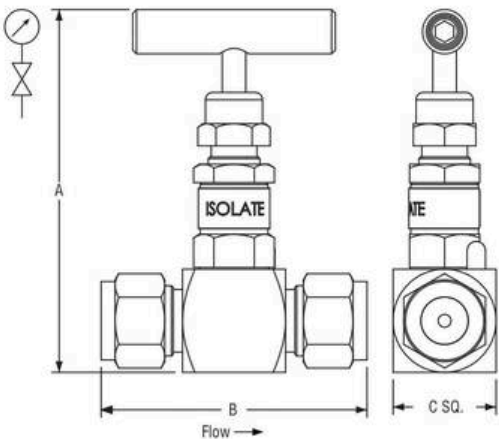
SIZE	A	B	CSQ	WEIGHT (kg)
1/4"	93	65	25	0.45
3/8"	93	65	25	0.45
1/2"	95	76	28	0.56
3/4"	109	80	38	0.9
1"	117	95	45	1.4

ANV 1 MM



SIZE	A	B	CSQ	WEIGHT (kg)
1/4"	93	65	25	0.4
3/8"	93	65	25	0.4
1/2"	95	76	28	0.5
3/4"	109	80	38	0.8
1"	117	95	45	1.3

ANV 1 CF



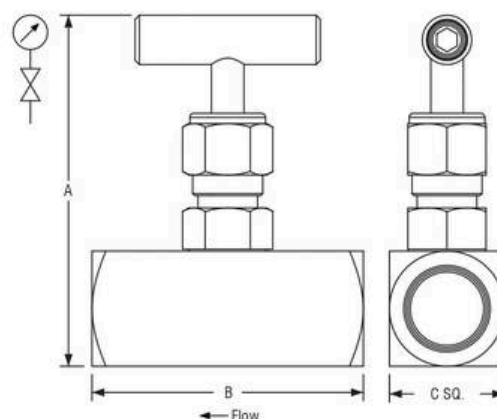
SIZE	A	B	CSQ	WEIGHT (kg)
1/4"	93	70	25	0.3
3/8"	93	72	25	0.32
1/2"	95	76	28	0.4
6 mm	93	70	25	0.3
10 mm	93	72	25	0.4
12 mm	95	76	28	0.4

These valves provide economical & reliable option for isolation & throttling applications. The bonnet assembly has a compact design with one piece rotating stem which is V tipped for accurate flow control. Easy maintenance & long trouble free life.

Standard Specifications:

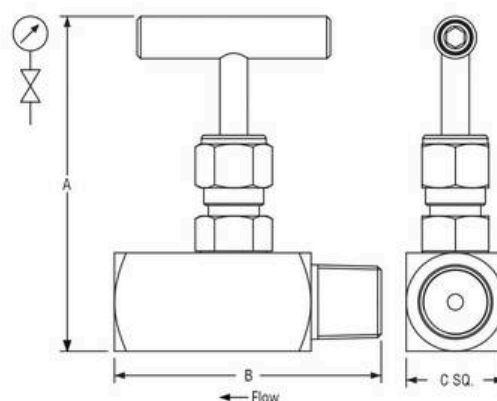
Maximum working Pressure: 413 bar at 38°C (6000 psi at 100°F) (see Pressure & Temperature Chart on page No. 07 for higher temperature applications). MOC in accordance with NACE MR0175 is available on request.

ANV 2 FF



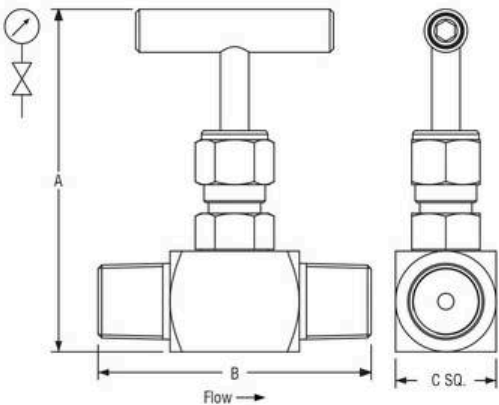
SIZE	A	B	CSQ	WEIGHT (kg)
1/4"	90	55	25	0.4
3/8"	90	55	25	0.4
1/2"	93	66	28	0.5
3/4"	100	70	38	0.8
1"	110	80	45	1.4

ANV 2 MF



SIZE	A	B	CSQ	WEIGHT (kg)
1/4"	90	65	25	0.45
3/8"	90	65	25	0.45
1/2"	93	76	28	0.56
3/4"	100	80	38	0.9
1"	110	95	45	1.5

ANV 2 MM

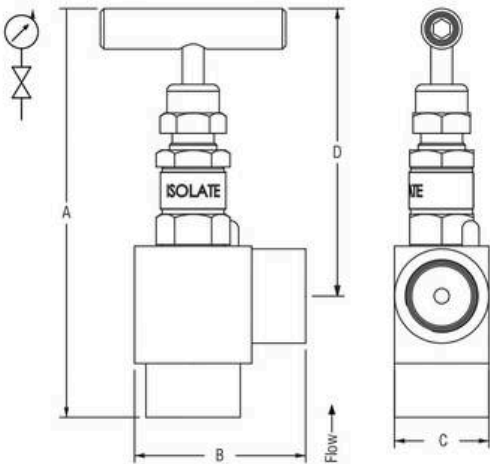


SIZE	A	B	CSQ	WEIGHT (kg)
1/4"	90	65	25	0.45
3/8"	90	65	25	0.45
1/2"	93	76	28	0.56
3/4"	100	80	38	0.9
1"	110	95	45	1.4

These valves provide angular end connections of 90 degree for specific application. The other design features for this applications are similar to type ANV 1

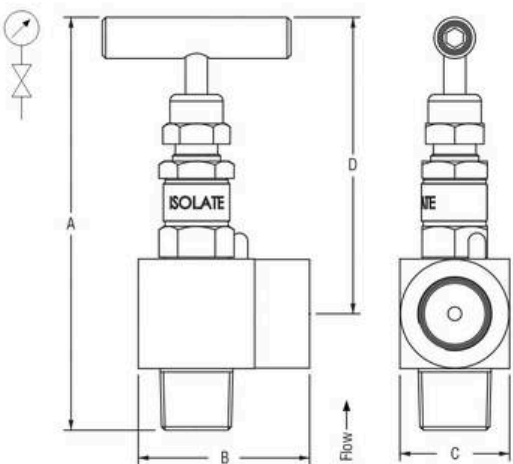
Standard Specifications:
Maximum working Pressure: 413 bar at 38°C (6000 psi at 100°F) (see Pressure & Temperature Chart on page No. 07 for higher temperature applications).
MOC in accordance with NACE MR0175 is available on request
Valves for 10000 PSI working pressure are also available

ANV 3 FF



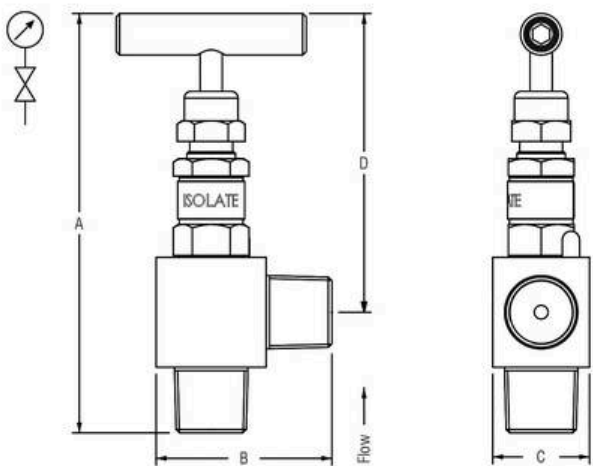
SIZE	A	B	C	D	WEIGHT (kg)
1/4"	101	38	25	76	0.45
3/8"	101	38	25	76	0.45
1/2"	115	50	28	76	0.57

ANV 3 MF



SIZE	A	B	C	D	WEIGHT (kg)
1/4"	101	38	25	76	0.45
3/8"	101	38	25	76	0.45
1/2"	115	50	28	76	0.57

ANV 3 MM



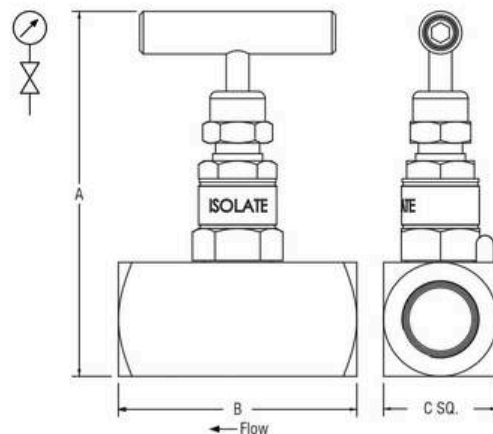
SIZE	A	B	C	D	WEIGHT (kg)
1/4"	101	38	25	76	0.45
3/8"	101	38	25	76	0.45
1/2"	115	50	28	76	0.57

These valves are designed for high pressure applications. These valves are also used in high pressure testing equipments. Other design features are similar to ANV 1

Standard Specifications:

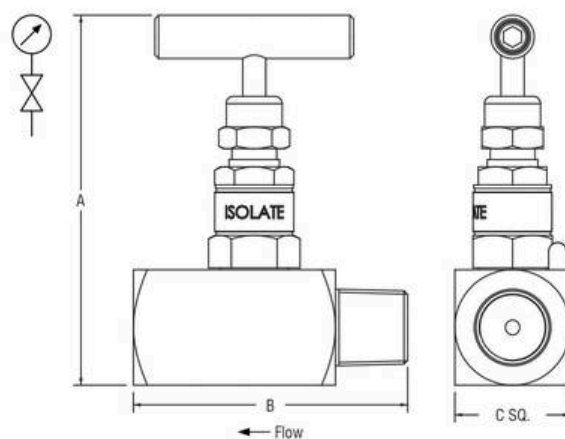
Maximum working Pressure: 690 bar at 38°C (10000 psi at 100°F) (see Pressure & Temperature Chart on page No. 07 for higher temperature applications). MOC in accordance with NACE MR0175 is available on request

ANV 4 FF

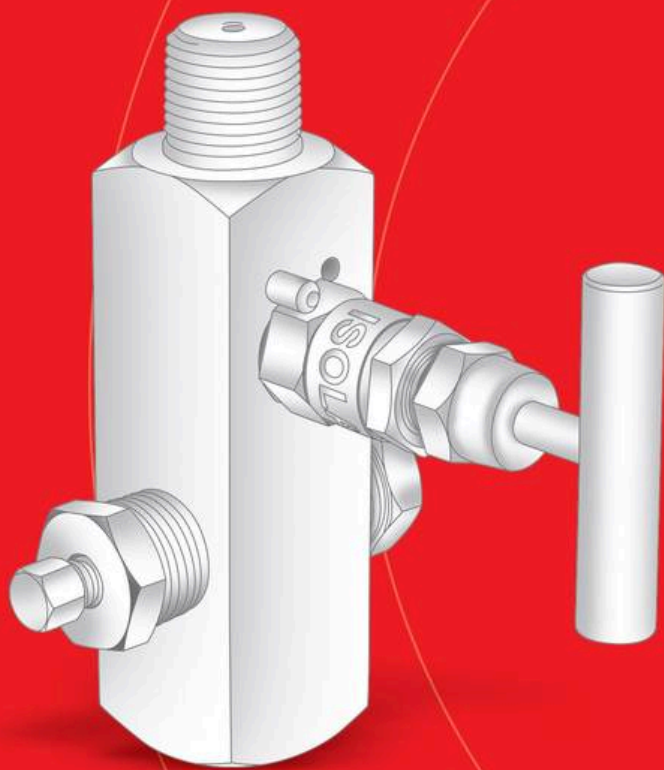


SIZE	A	B	C(SQ.)	WEIGHT (kg)
1/4"	93	55	28	0.44
3/8"	93	55	28	0.44
1/2"	103	67	32	0.65
3/4"	109	70	38	0.80
1"	117	80	45	1.30

ANV 4 MF



SIZE	A	B	C(SQ.)	WEIGHT (kg)
1/4"	93	65	28	0.45
3/8"	93	65	28	0.45
1/2"	103	76	32	0.88
3/4"	109	80	38	0.90
1"	117	95	45	1.40



Gauge Valves

Gauge Valve Selection Chart

A	GV	1	FF	N	12	H	SS	T	OPTION
A = Aptek	GV = Gauge Valve	1 = Bleed Valve 2 = Vent Valve 3 = Multi Port 4 = Forged Body (DIN Standard)	FF = Female X Female MF = Male X Female MM = Male X Male CF = Compression fitting (tube ends) SW = Socket Weld BW = Butt Weld	N = NPT G = BSPP GT = BSPT M = Metric Blank = for CF / SW / BW end	14 = 1/4" 38 = 3/8" 12 = 1/2" 34 = 3/4" 01 = 1" 06M = 6 mm OD tube end 08M = 8 mm OD tube end 10M = 10 mm OD tube end 12M = 12 mm OD tube end	Blank = 6000 psi H = 10,000 psi	SS = SS 316 SSL = SS 316L SSD = SS 316/316L Dual SS4 = SS 304 MO = Monel 400 HC = Hastelloy C 276 DU = Duplex SS SD = Super Duplex IN6 = Inconel-625 IN8 = Inconel-825 CS = Carbon Steel BR = Brass	T = PTFE Packing G = Graphite Packing	NA = NACE MR 01-75 (Latest Edition) OXY = Cleaned for Oxygen Service Ext = Extension Lagging Length TF1 = With Test Flange Ø40 X 5 (Only for DIN Valve) TF2 = With Test Flange 60 X 25 X 10 (Only for DIN Valve) TC = With Test Connection M20 X 1.5 (Only for DIN Valve) AT = Anti Temper LH = Lockable Handle

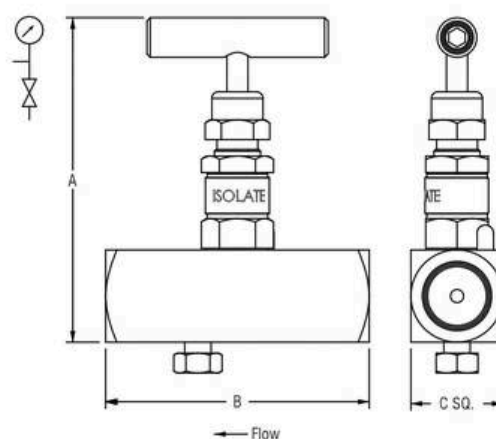
These valves are designed for safe installations of Pressure gauges and Transmitter. The suitable model can be selected for isolation, calibration and drain purpose.

Standard Specifications:

Maximum working Pressure: 413 bar at 38°C (6000 psi at 100°F) (see Pressure & Temperature Chart on page No.07 for higher temperature applications). MOC in accordance with NACE MR0175 is available on request

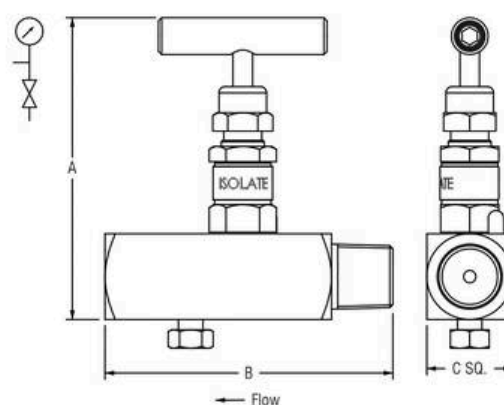
Valves for 10000 PSI working pressure are also available

AGV 1 FF



SIZE	A	B	CSQ	WEIGHT (kg)
1/4"	93	70	25	0.50
3/8"	93	70	25	0.50
1/2"	93	80	28	0.53

AGV 1 MF



SIZE	A	B	CSQ	WEIGHT (kg)
1/4"	93	75	25	0.43
3/8"	93	75	25	0.43
1/2"	93	94	28	0.60

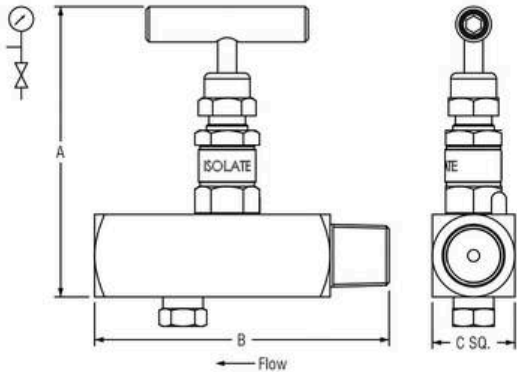
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Gauge Bleed Valve

AGV 2 Gauge Vent Valve

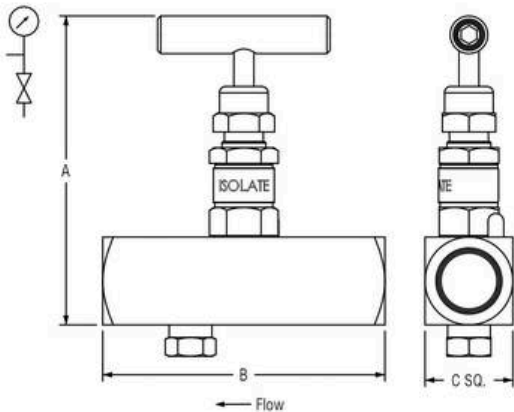
Standard Specifications:
 Maximum working Pressure: 413 bar at 38°C (6000 psi at 100°F)
 (see Pressure & Temperature Chart on page No. 07 for higher temperature applications).
 MOC in accordance with NACE MR0175 is available on request
Valves for 10000 PSI working pressure are also available

AGV 2 MF



SIZE	A	B	CSQ	WEIGHT (kg)
1/4"	93	80	25	0.50
3/8"	93	80	25	0.50
1/2"	95	100	28	0.60

AGV 2 FF



SIZE	A	B	CSQ	WEIGHT (kg)
1/4"	93	80	25	0.45
3/8"	93	80	25	0.45
1/2"	95	90	28	0.55

Multiport or gauge root valves facilitate different positioning of Pressure gauge or Pressure switch without disturbing the main line. The blind plug & Bleed Valve are supplied with the valve.

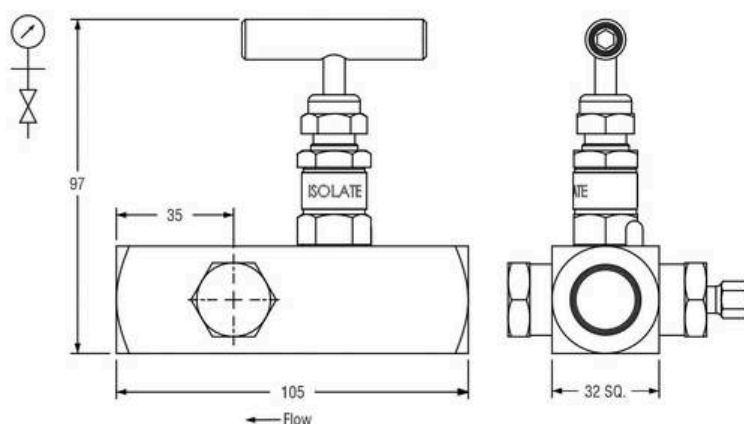
Standard Specifications:

Maximum working Pressure: 413 bar at 38°C (6000 psi at 100°F) (see Pressure & Temperature Chart on page No.07 for higher temperature applications).

MOC in accordance with NACE MR0175 is available on request

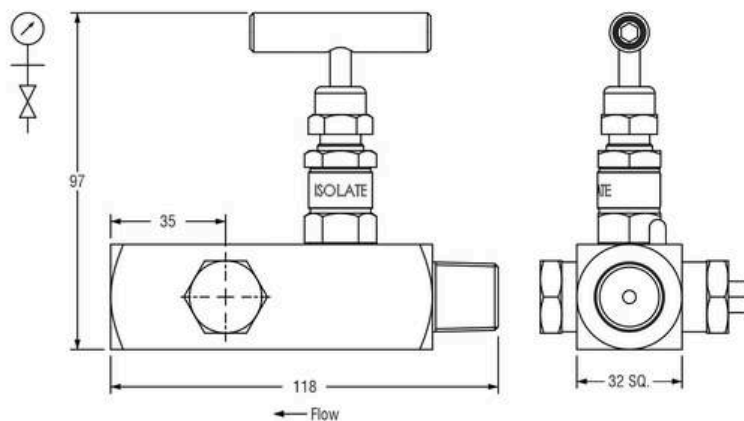
Valves for 10000 PSI working pressure are also available

AGV 3 FF



Weight: 0.95 kg

AGV 3 MF



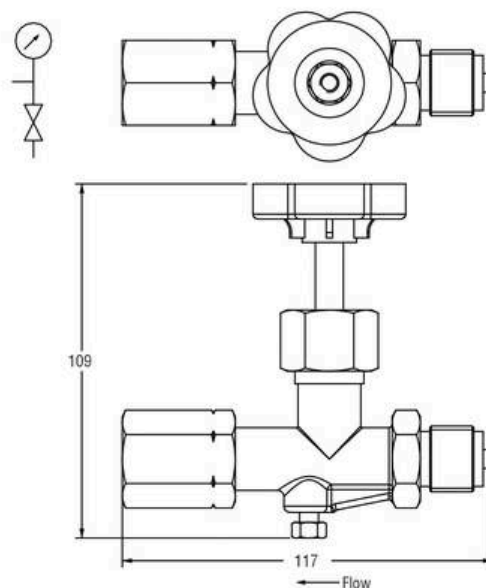
Weight: 1.00 kg

These valves are specially designed in accordance with DIN 16270/16271. The body is forged in union bonnet pattern. The G 1/4 connection provides swivel female outlet for suitable positioning of Pressure gauge. Bleed screw is provided for Vent.

Standard Specifications :

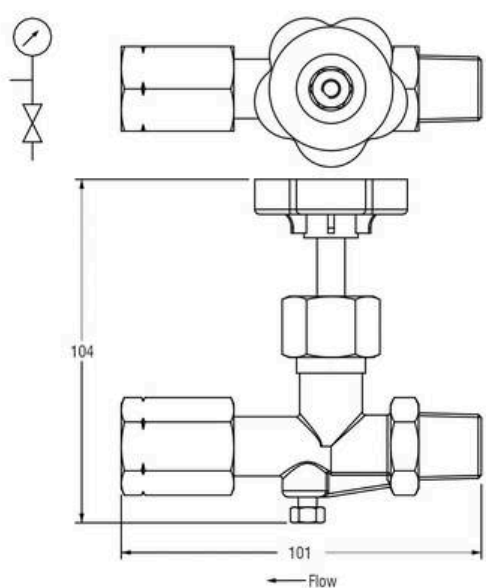
Maximum working pressure : 413 bar at 38°C (6000 psi at 100°F) (see pressure temperature chart on Page No. 07 for higher temperature applications)

AGV 4 G12



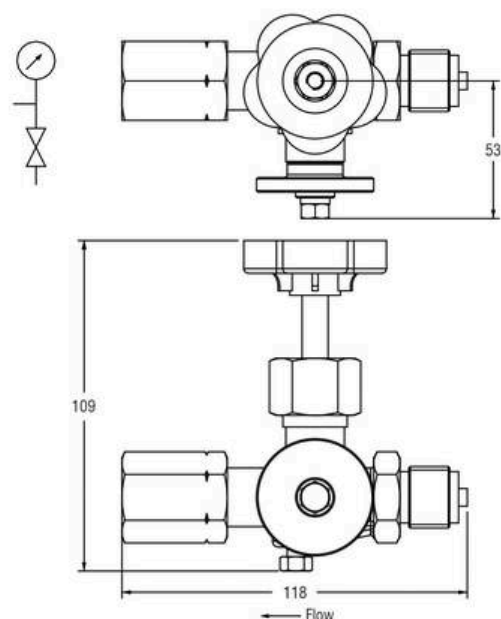
Weight: 0.50 kg

AGV 4 N12



Weight: 0.50 kg

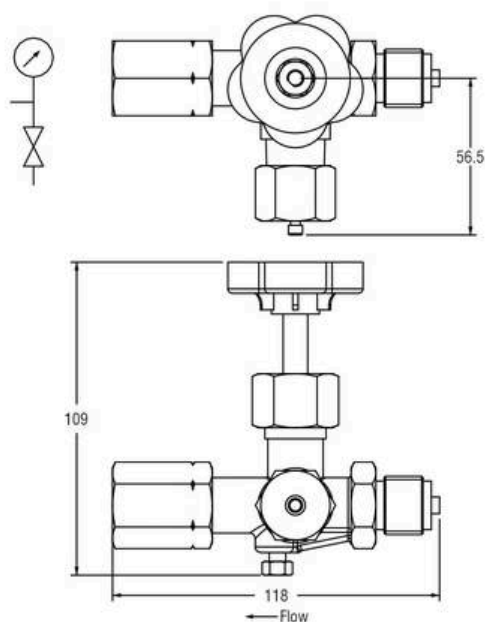
AGV 4 G12 - TF



Weight: 0.50 kg

AGV 4 Test Port

AGV 4 G12 - TP



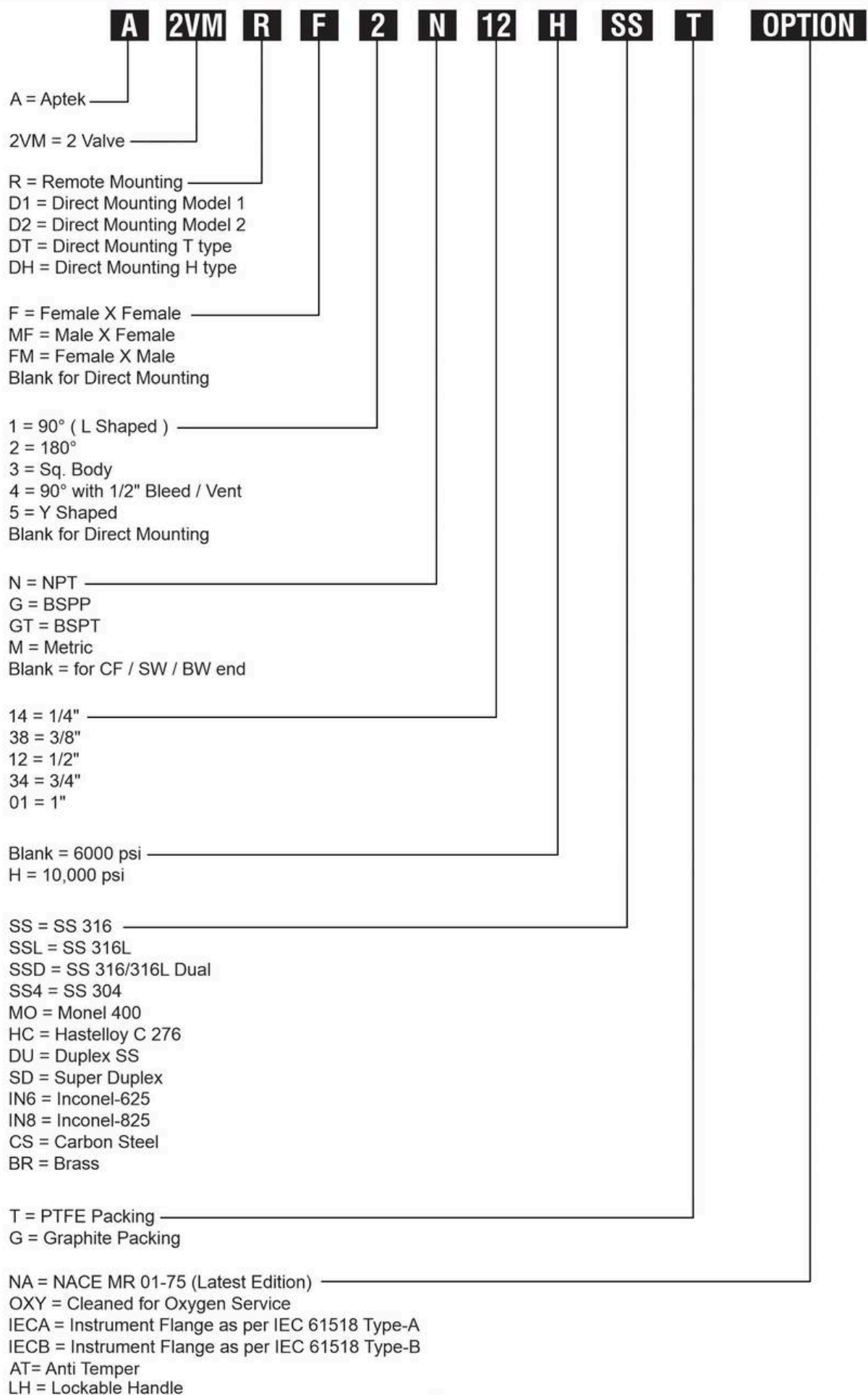
Weight: 0.50 kg

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Manifold Valves

2 Valve Manifold Section Chart

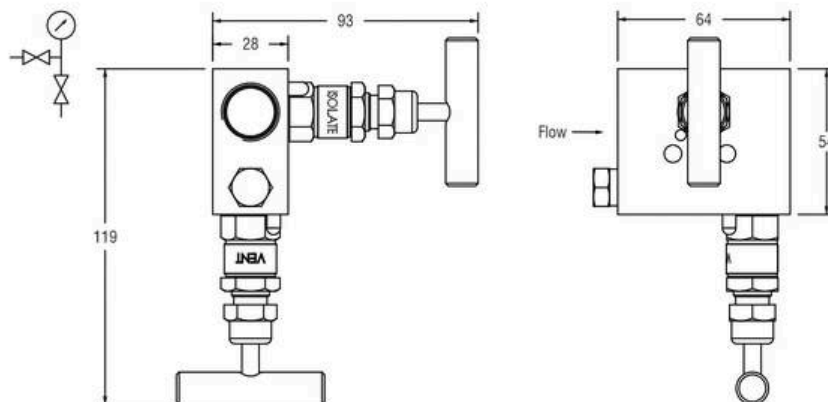


These manifolds are having one Isolation valve of 1/4" connection & one drain valve of 1/2" or 1/4" connection. The outlet of drain valve can be used for calibration. Compact unit saves space, reduces number of joints resulting in cost reduction. These manifolds are used with Pressure gauges, switches or transmitters. Different models are available as shown below.

Standard Specifications:

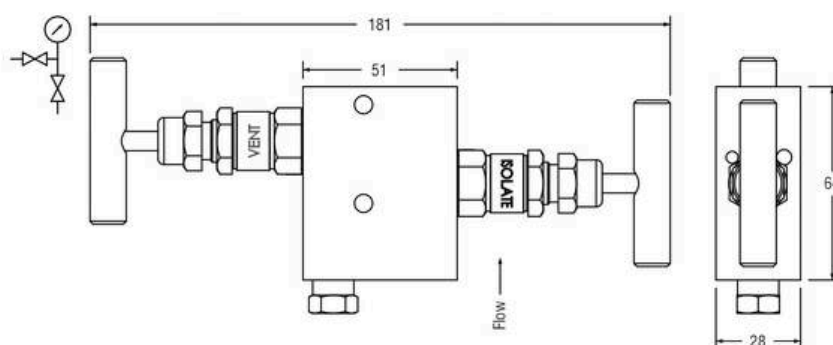
Maximum working Pressure: 413 bar at 38°C (6000 psi at 100°F) (see Pressure & Temperature Chart on page No.07 for higher temperature applications). MOC in accordance with NACE MR0175 is available on request. **Valves for 10000 PSI working pressure are also available**

A 2VM RF 1



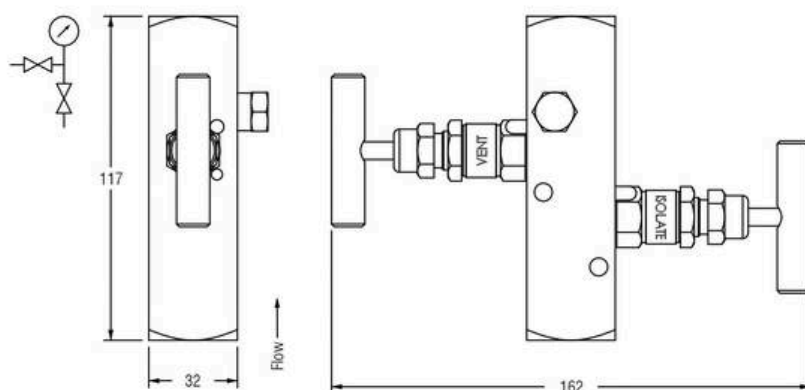
Weight: 1.00 kg

A 2VM RF 2



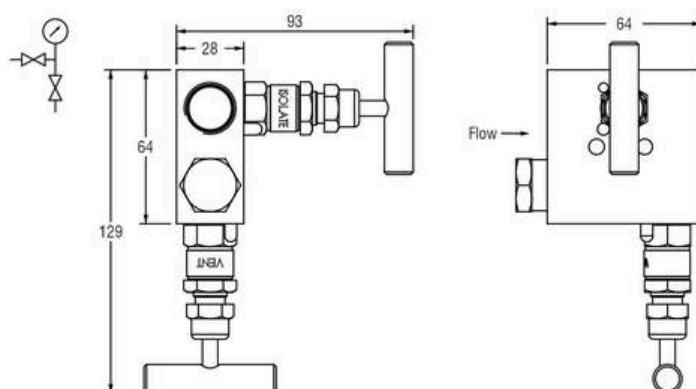
Weight: 1.00 kg

A 2VM RF 3



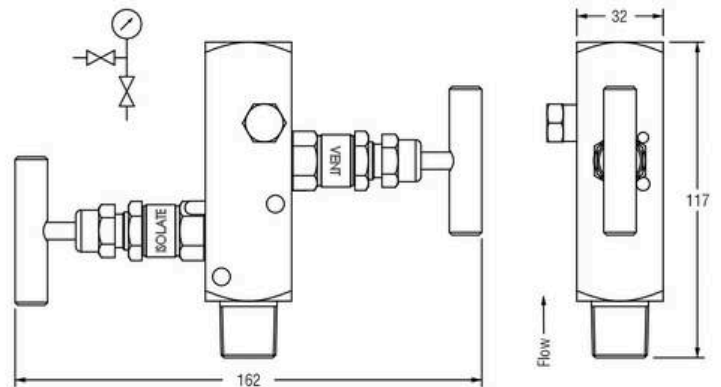
Weight: 1.10 kg

A 2VM RF 4



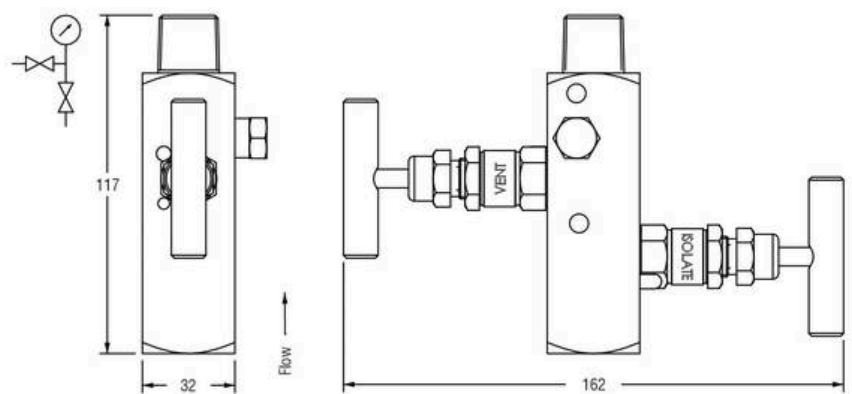
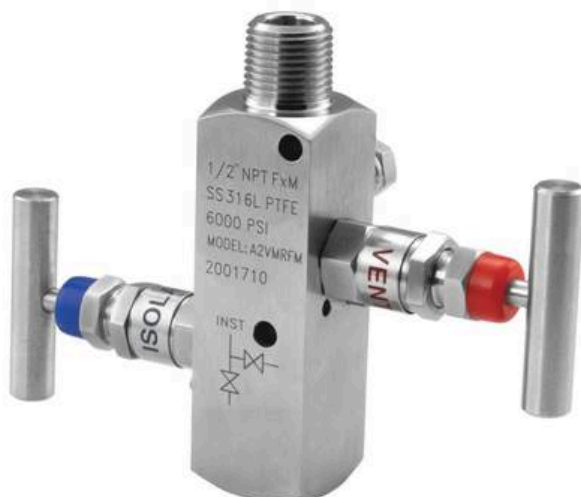
Weight: 1.00 kg

A 2VM RMF



Weight: 1.10 kg

A 2VM RFM



Weight: 1.10 kg

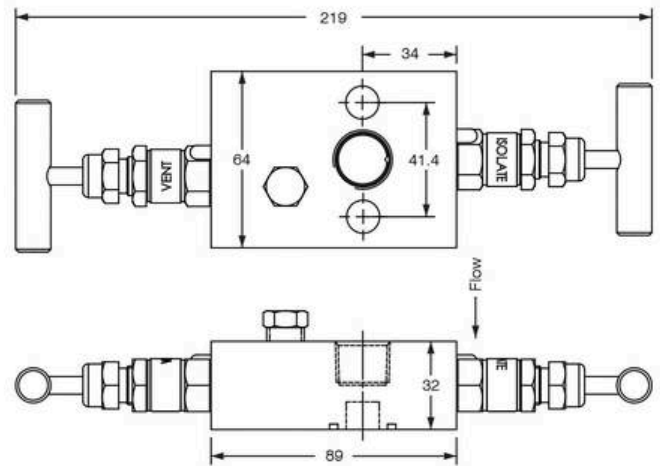
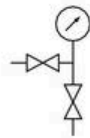
These manifolds provide a direct mounting feature with isolation & drain/calibration in a single unit. These type of manifolds can be directly mounted on the static pressure transmitter to eliminate several parts & joints. Different models available as shown below.

Standard Specifications :

Maximum working Pressure: 413 bar at 38°C (6000 psi at 100°F) (see Pressure & Temperature Chart on page No.07 for higher temperature applications).

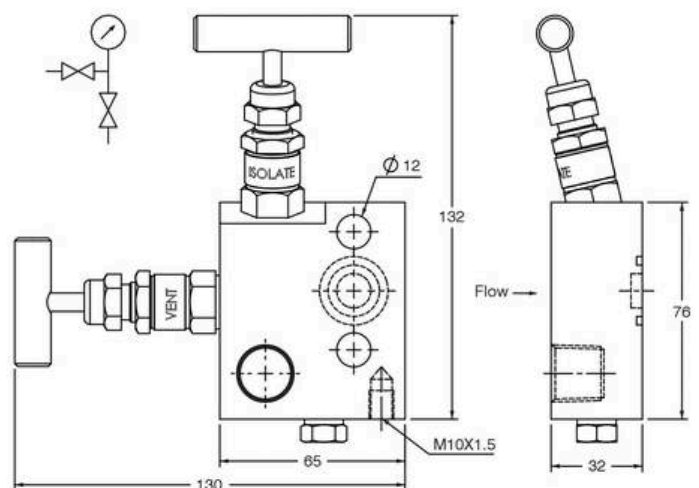
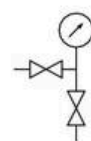
MOC in accordance with NACE MR0175 is available on request. **Valves for 10000 PSI working pressure are also available**

A 2VM D1



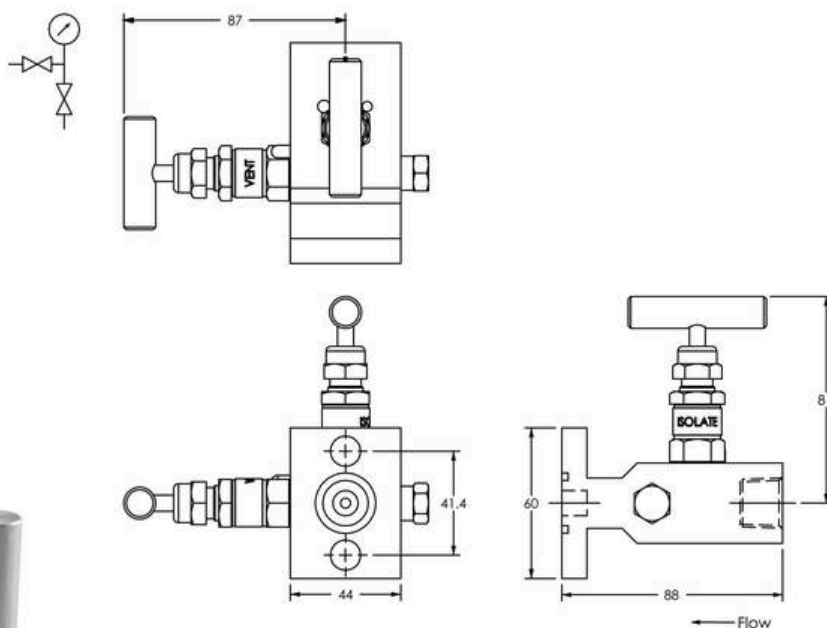
Weight: 1.60 kg

A 2VM D2



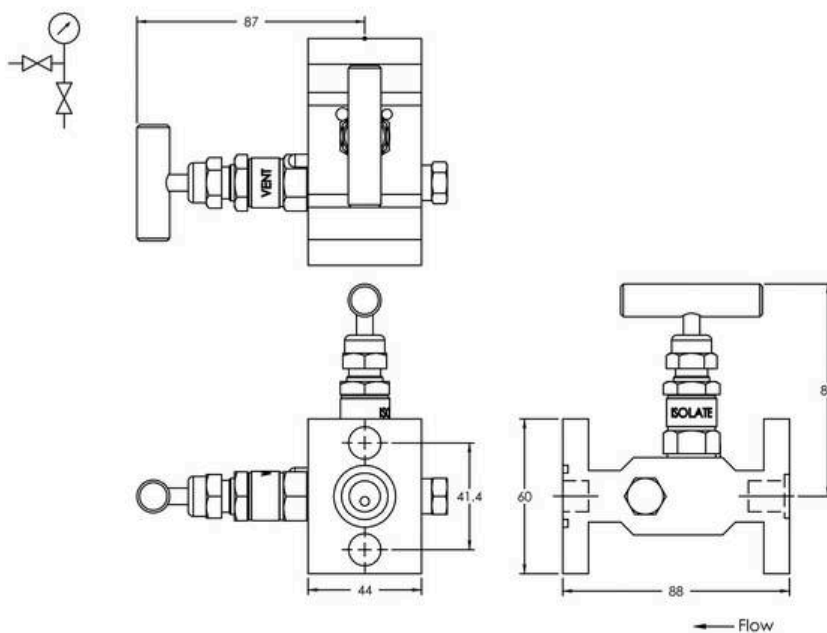
Weight: 1.30 kg

A 2VM DT



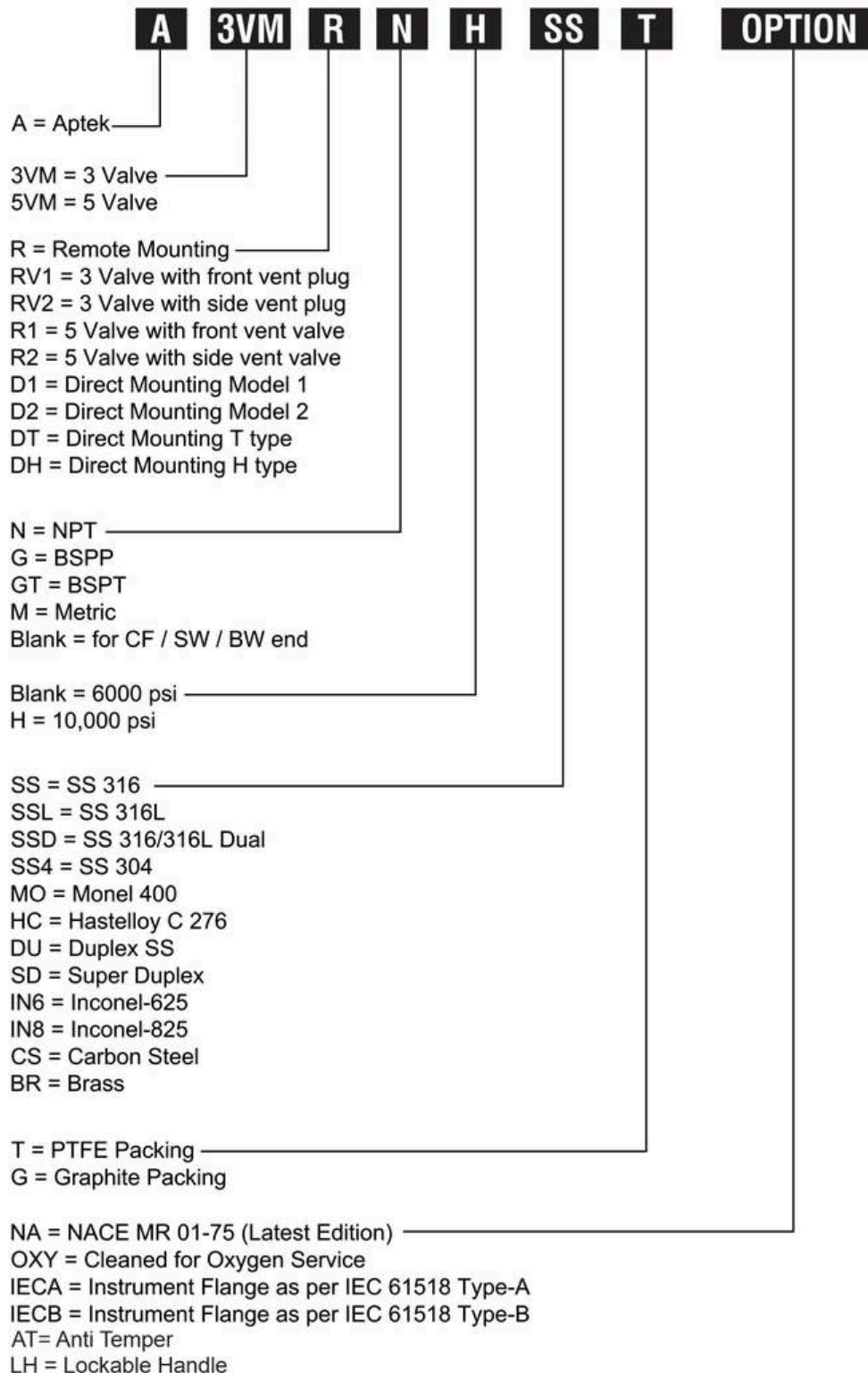
Weight: 1.40 kg

A 2VM DH



Weight: 2.0 kg

3 & 5 Valve Manifold Selection Chart

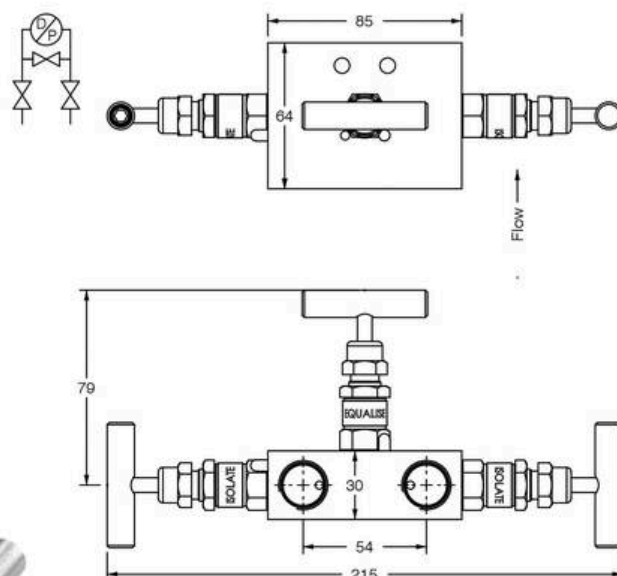


These manifolds are generally used in differential pressure application with Pressure gauges, Pressure switches & pressure transmitters. Two isolation valves & one equilibrating valve is provided. The standard connections are 1/4" NPTF. The drain connection of 1/2" NPT is provided as an option. The different models are available as mentioned below.

Standard Specifications:

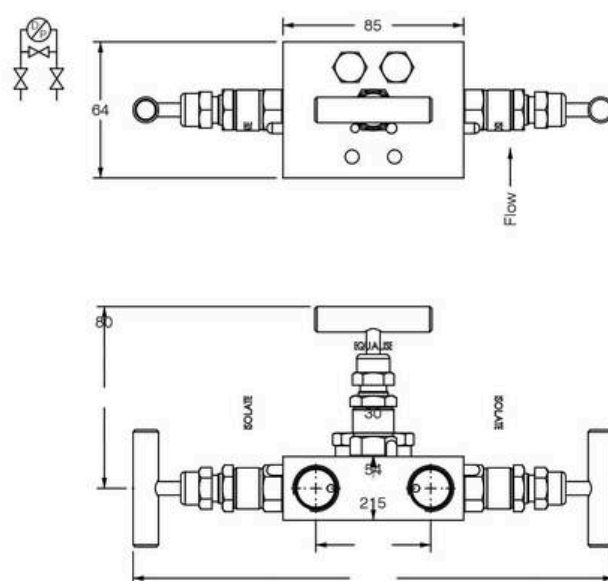
Maximum working pressure : 413 bar at 38°C (6000 psi at 100°F) (see pressure temperature chart on Page No. 07 for higher temperature applications)
MOC in accordance with NACE MR0175 is available on request. **Valves for 10000 PSI working pressure are also available**

A 3VM R



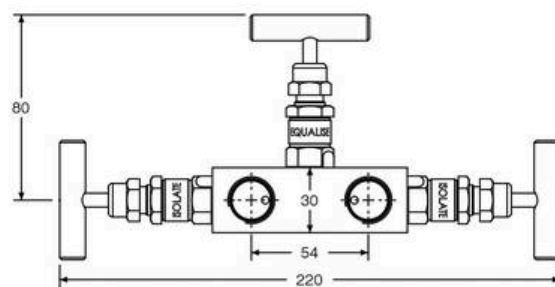
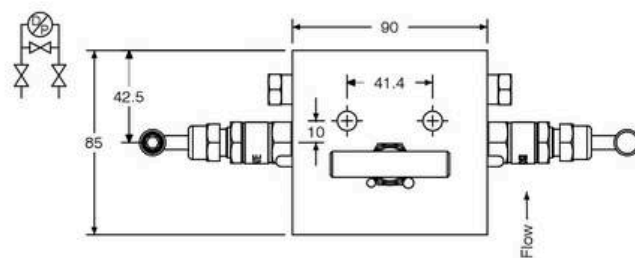
Weight: 1.50 kg

A 3VM RV 1



Weight: 1.56 kg

A 3VM RV 2



Weight: 2.0 kg

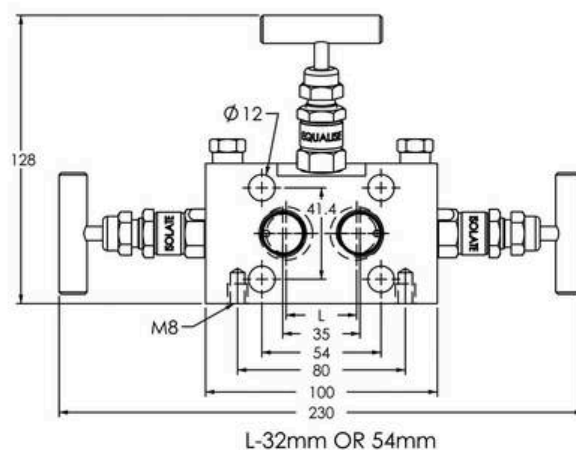
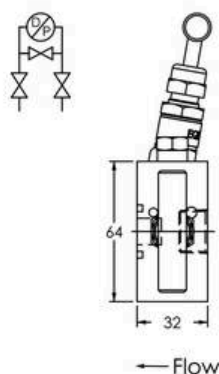
3 Valve Manifold - Direct Mounting

These manifold can be directly mounted on the transmitter having 54 mm or 32 mm center to center connections. The inlet is either 1/4" NPT F or flanged. Two isolation valves & one equalizer valve is provided along with 1/4" NPT F drain as an option. These valves are mainly used with Dp transmitters. Different options are available as shown.

Standard Specifications :

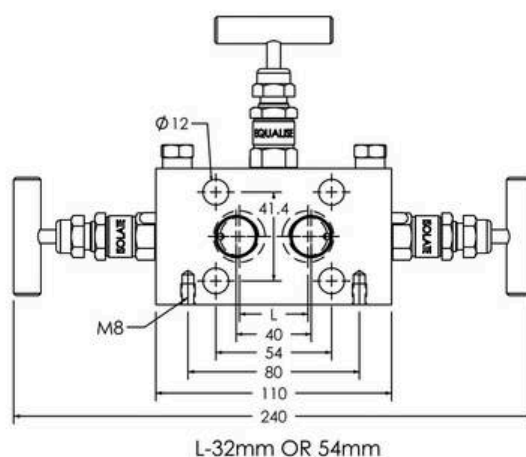
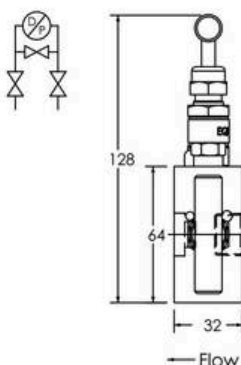
Maximum working Pressure: 413 bar at 38°C (6000 psi at 100°F) (see Pressure & Temperature Chart on page No. 07 for higher temperature applications).
MOC in accordance with NACE MR0175 is available on request. **Valves for 10000 PSI working pressure are also available**

A 3VM D1



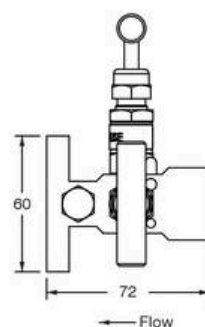
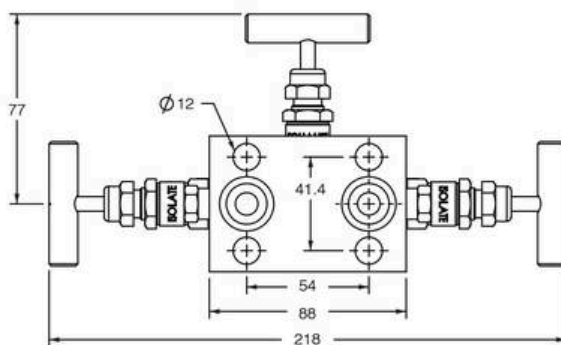
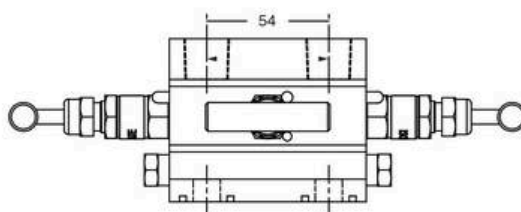
Weight: 2.0 kg

A 3VM D2 (Coplaner)



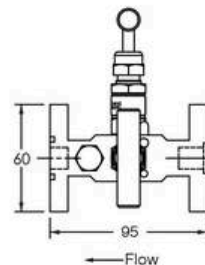
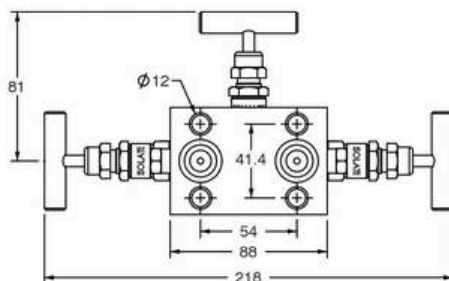
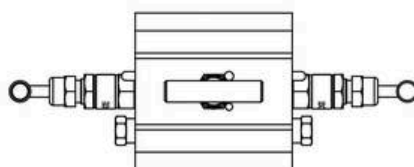
Weight: 2.0 kg

A 3VM DT



Weight: 1.90 kg

A 3VM DH



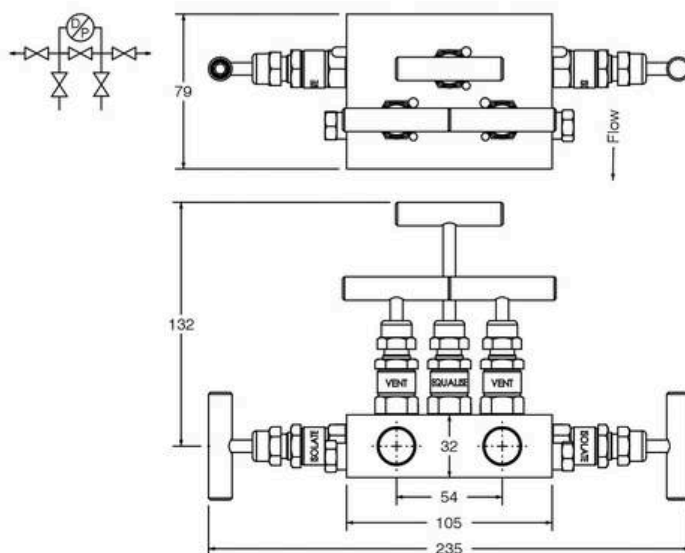
Weight: 2.56 kg

These manifolds are designed for remote mounting on instruments using threaded instrument & process connections. These valves are having two isolation, one equalizer & two drain/calibration valves. The standard process & instrument connections are 1/4" NPT F & the drain connections are 1/2" NPT F. The blank plugs are provided. The instruments connections are positioned at 54 mm center distance but can also be used for other center distances with the help of screwed fittings. Different models are available as shown below.

Standard Specifications:

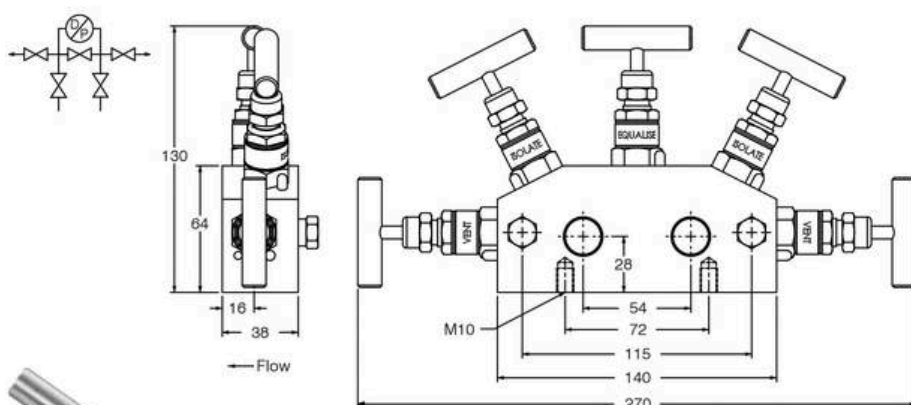
Maximum working Pressure: 413 bar at 38°C (6000 psi at 100°F) (see Pressure & Temperature Chart on page No. 07 for higher temperature applications). MOC in accordance with NACE MR0175 is available on request. **Valves for 10000 PSI working pressure are also available**

A 5VM R1



Weight: 2.70 kg

A 5VM R2



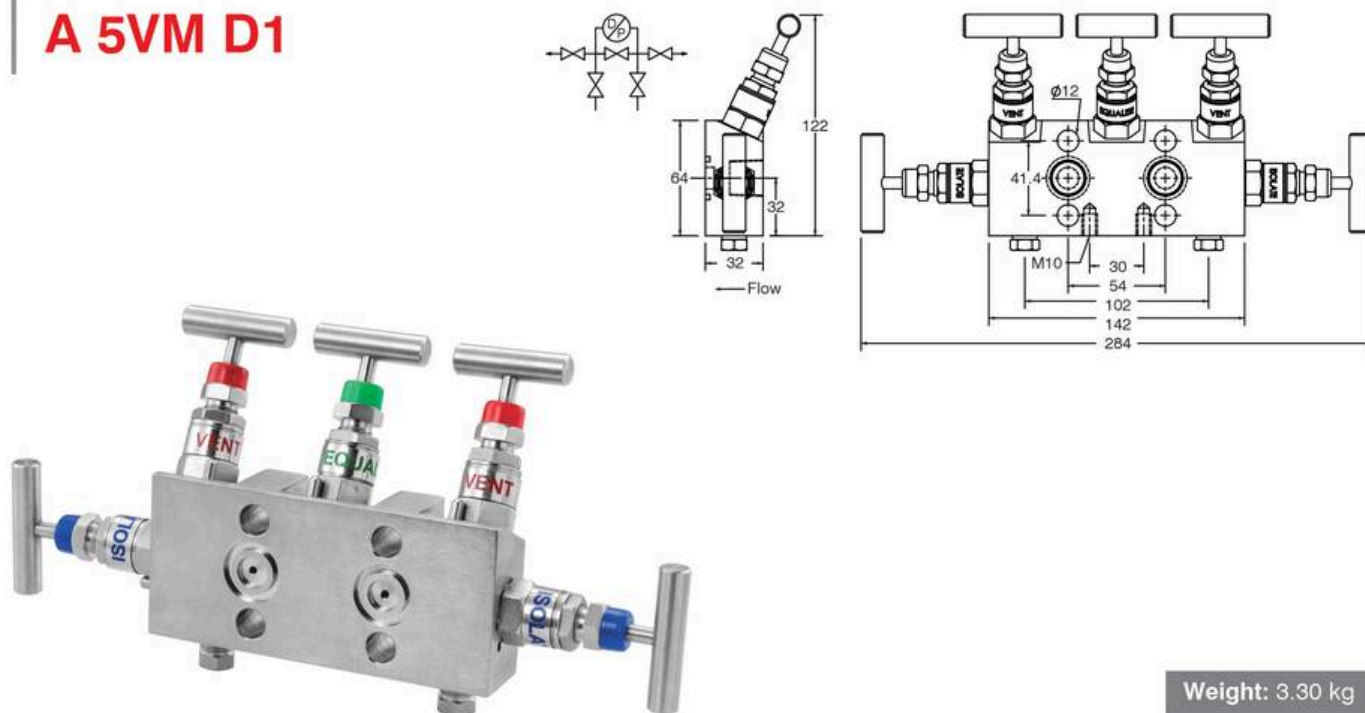
Weight: 3.30 kg

These valves are designed for direct mounting to differential pressure transmitter. The instrument side is provided with a standard flange (both option IEC & MSS available) of 54mm or 32 mm center distance. These valves are having two isolation, one equalizer & two vent / calibration valves. The standard process & instruments connections are 1/4" NPT. Two vent connections are 1/2" NPT in all manifolds & provided with blanking plugs. Different models are available as shown.

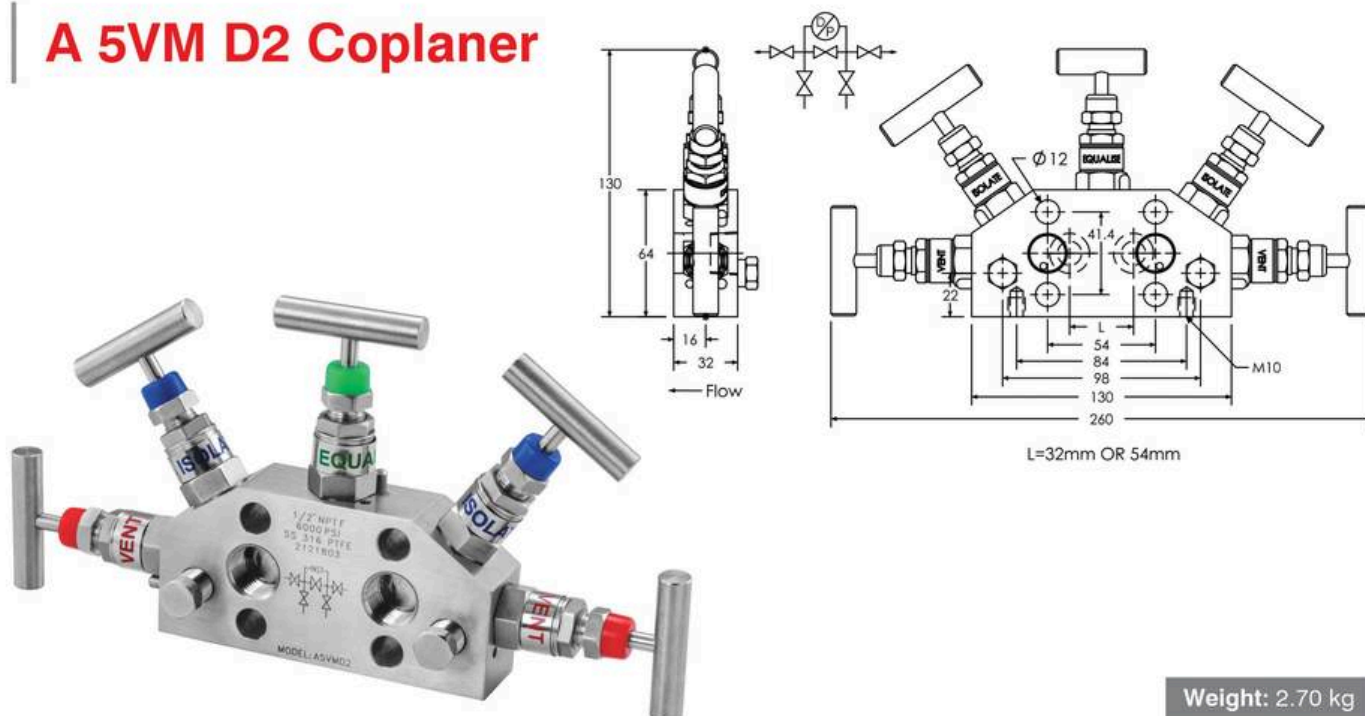
Standard Specifications:

Maximum working Pressure: 413 bar at 38°C (6000 psi at 100°F) (see Pressure & Temperature Chart on page No. 07 for higher temperature applications). MOC in accordance with NACE MR0175 is available on request. **Valves for 10000 PSI working pressure are also available**

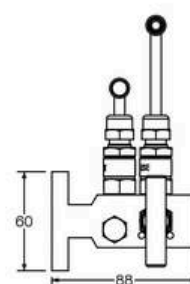
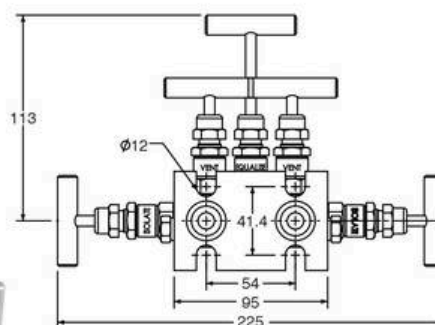
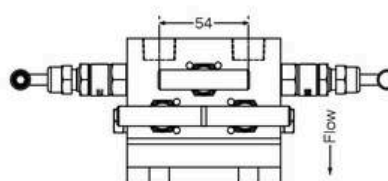
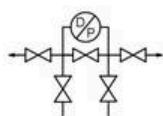
A 5VM D1



A 5VM D2 Coplaner

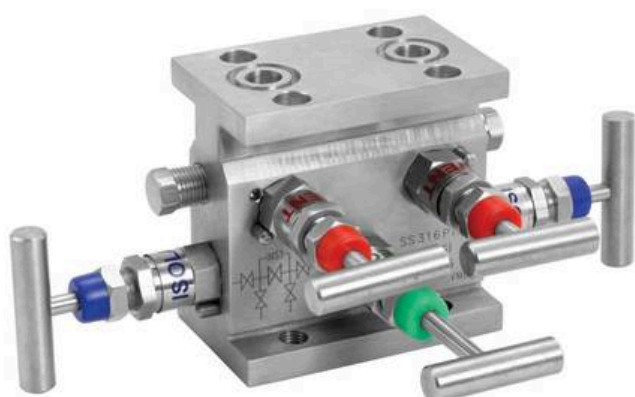
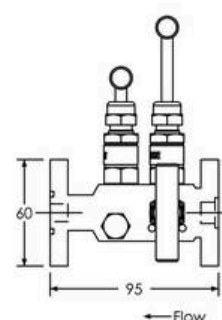
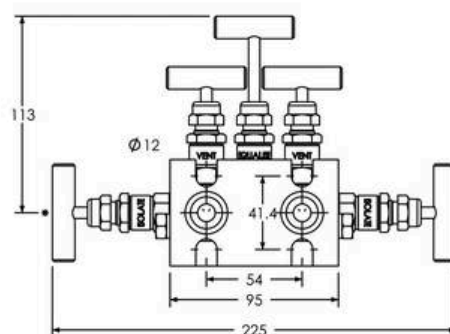
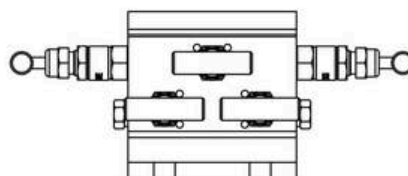
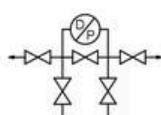


A 5VM DT



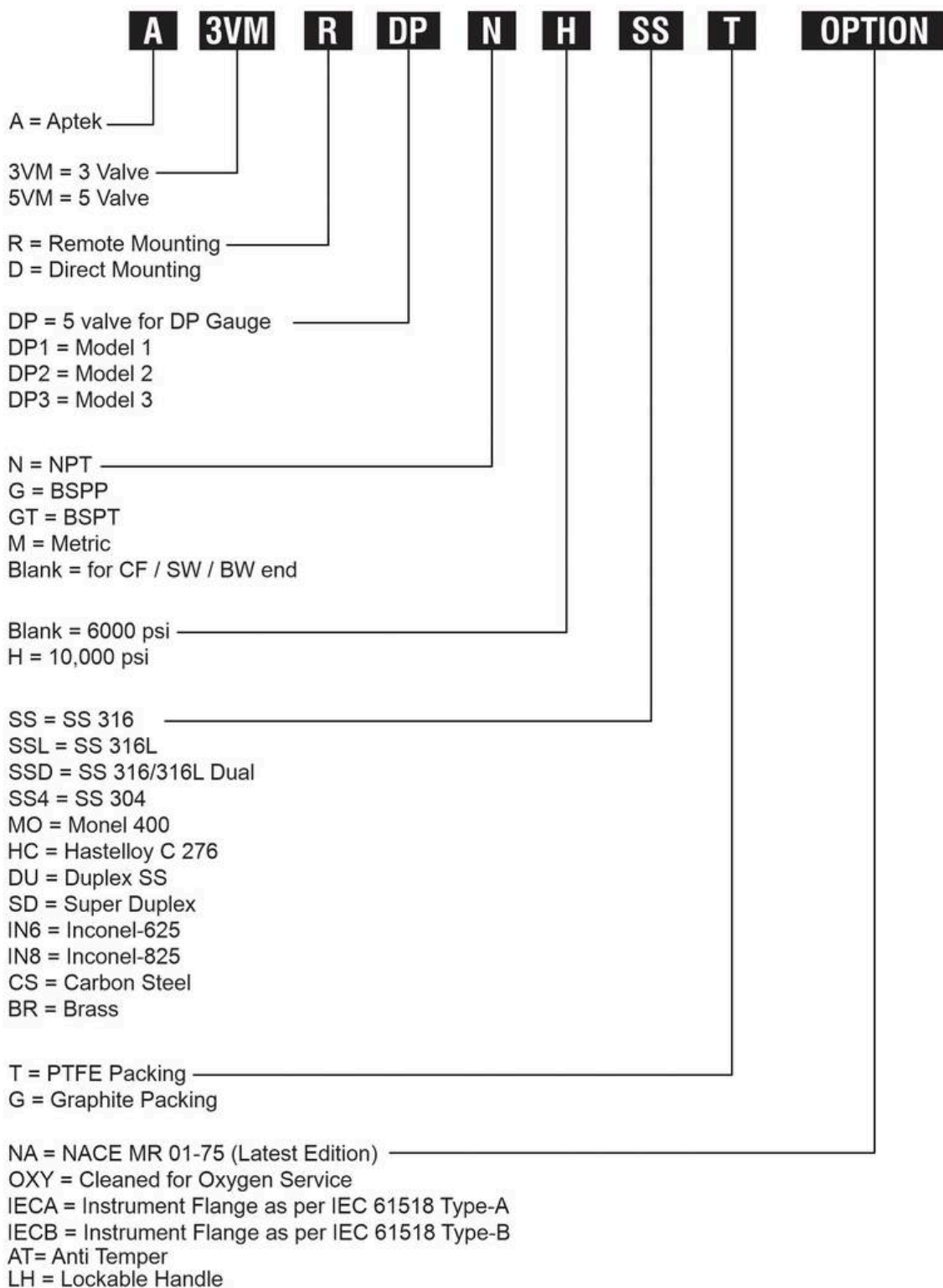
Weight: 2.90 kg

A 5VM DH



Weight: 2.70 kg

3 & 5 Valve Manifold for DP Gauge Selection Chart



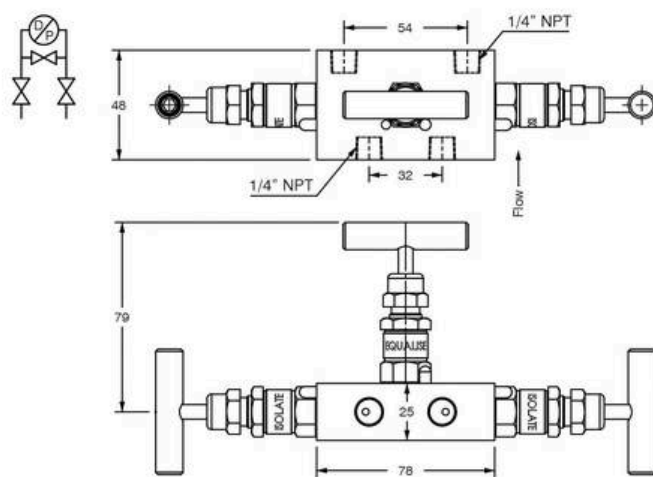
Standard Specifications:

Maximum working Pressure: 413 bar at 38°C (6000 psi at 100°F) (see Pressure & Temperature Chart on page No. 07 for higher temperature applications).

MOC in accordance with NACE MR0175 is available on request.

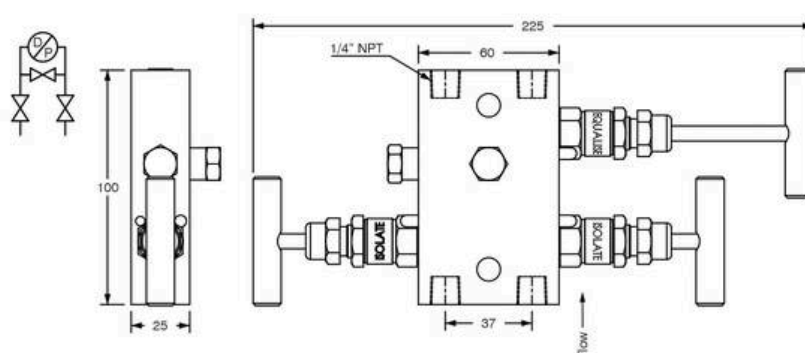
Valves for 10000 PSI working pressure are also available

A 3VM R DP 1



Weight: 1.0 kg

A 3VM R DP 2



Weight: 1.40 kg

Standard Specifications:

Maximum working Pressure: 413 bar at 38°C (6000 psi at 100°F)

(see Pressure & Temperature Chart on page No. 07 for higher temperature applications).

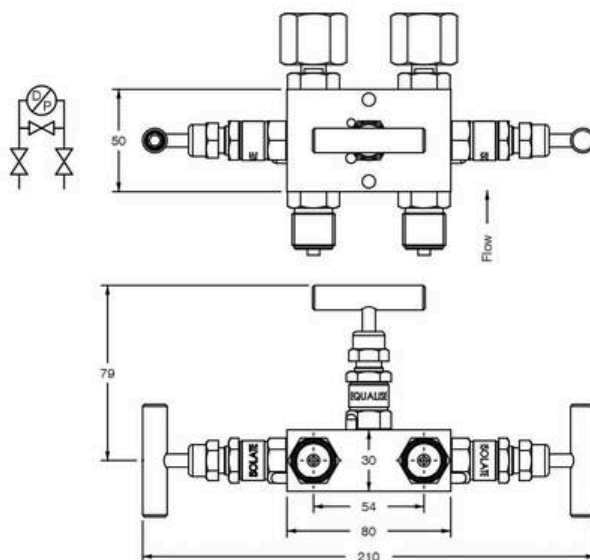
MOC in accordance with NACE MR0175 is available on request.

Standard Process Connection : G 1/2" M or 1/2" NPT M

Standard Instrument Connection : G 1/2" F Swivel

Valves for 10000 PSI working pressure are also available

A 3VM R DP 3



Weight: 1.90 kg

Standard Specifications:

Maximum working Pressure: 413 bar at 38°C (6000 psi at 100°F)

(see Pressure & Temperature Chart on page No. 07 for higher temperature applications).

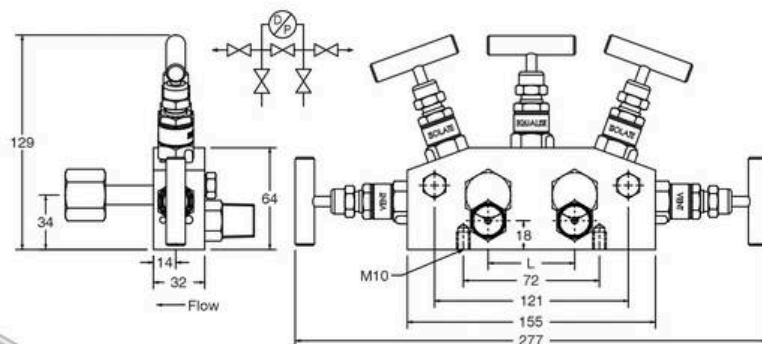
MOC in accordance with NACE MR0175 is available on request

Standard Process Connection : G 1/2" M or 1/2" NPT M

Standard Instrument Connection : G 1/2" F Swivel

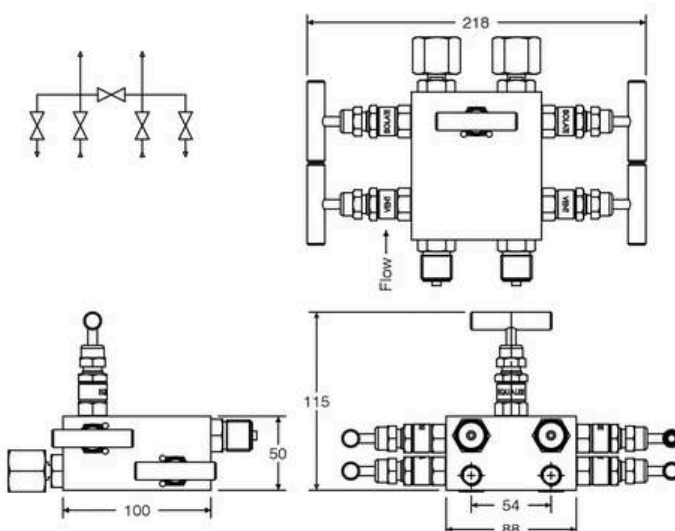
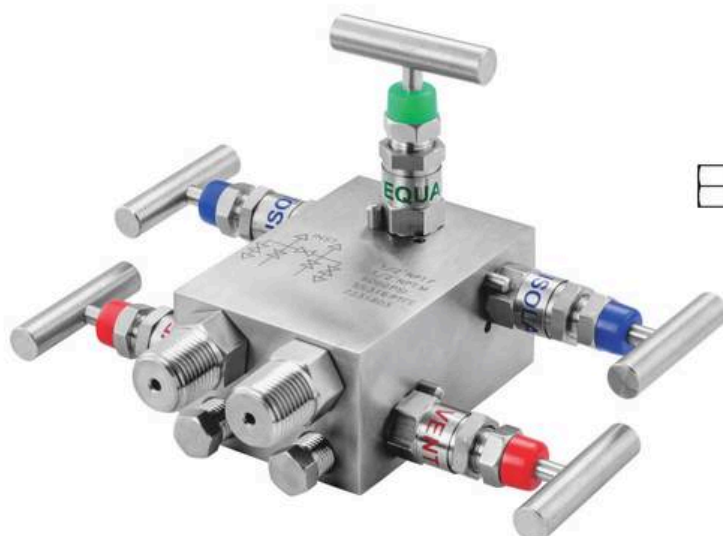
Valves for 10000 PSI working pressure are also available

A 5VM D DP1



Weight: 3.70 kg

A 5VM D DP2



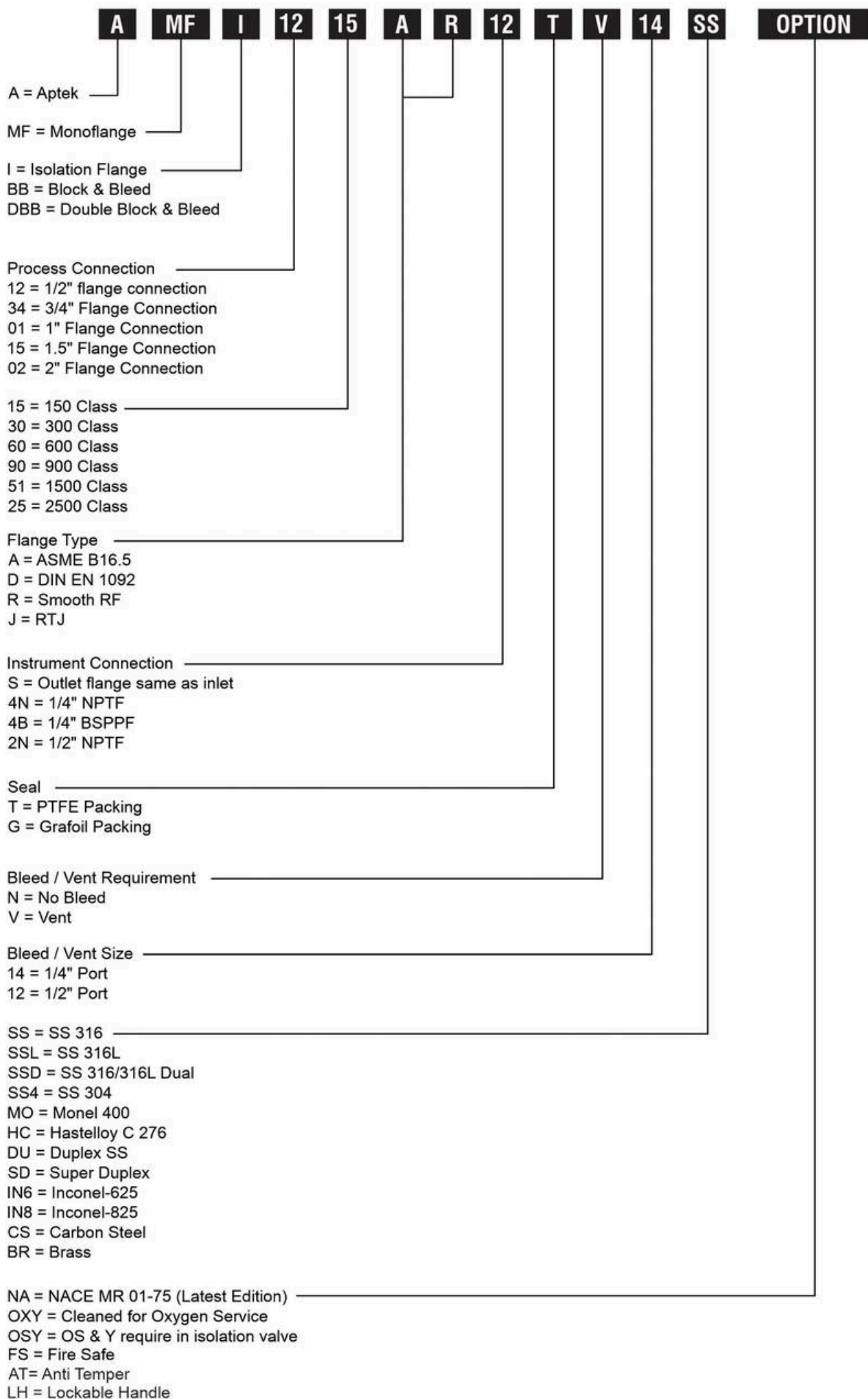
Weight: 3.340 kg

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Monoflanges

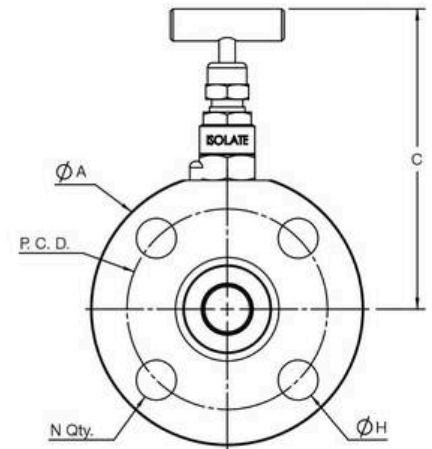
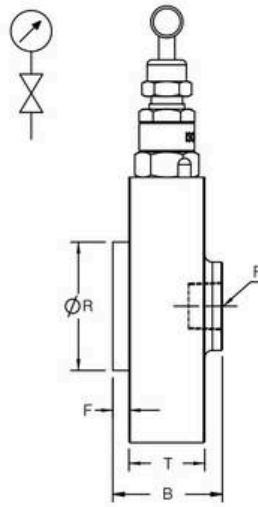


Monoflange Selection Chart

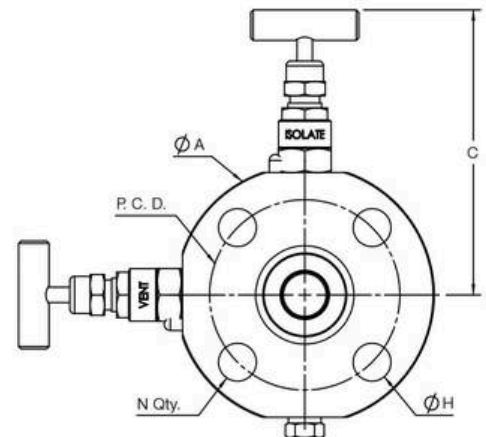
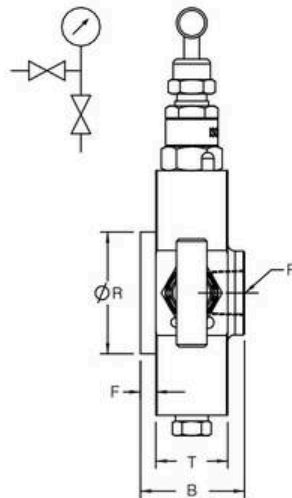


Monoflanges are design for Shut-off and venting of pressure measuring instruments.

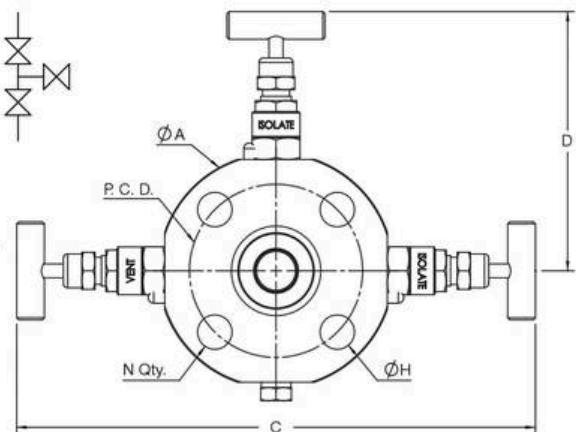
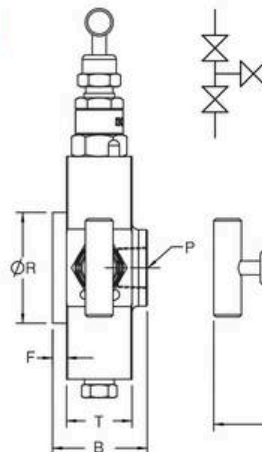
Isolation



Block & Bleed



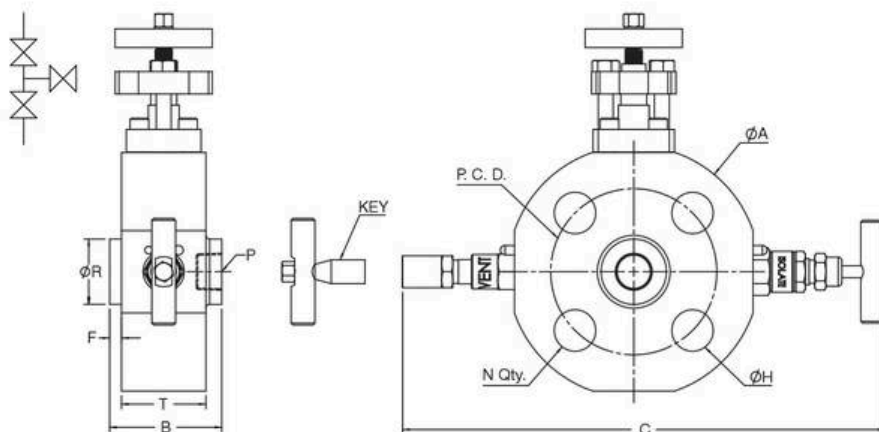
Double Block & Bleed



SIZE	CLASS	ØA	B	P	C	D	F	T	ØR	ØH	N Qty.	P. C. D.
1/2"	150	88.9	38.6	1/2" NPT	220	110	1.6	30	35.05	15.8	4	60.45
	300	95.25	38.6		226	113	1.6	30		15.8	4	66.55
	600	95.25	43.4		226	113	6.4	30		15.8	4	66.55
	900 & 1500	120.65	43.4		250	125	6.4	30		22.2	4	82.55
	2500	133.35	43.4		266	133	6.4	30		22.2	4	88.9
3/4"	150	98.55	38.6		230	115	1.6	30	42.93	15.8	4	69.85
	300	117.35	38.6		250	125	1.6	30		19.05	4	82.55
	600	117.35	43.4		250	125	6.4	30		19.05	4	82.55
	900 & 1500	130.05	43.4		260	130	6.4	30		22.2	4	88.9
	2500	139.7	45.15			135	6.4	31.75		22.2		95.25
1"	150	107.95	38.6		240	120	1.6	30	50.8	15.8	4	79.25
	300	123.95	38.6		256	128	1.6	30		19.05	4	88.9
	600	123.95	43.4		256	128	6.4	30		19.05	4	88.9
	900 & 1500	149.35	43.4		280	140	6.4	30		25.4	4	101.6
	2500	158.75	48.45		290	145	6.4	35		25.4	4	107.95
1-1/2"	150	127	38.6		258	129	1.6	30	73.15	15.8	4	98.55
	300	155.45	38.6		288	144	1.6	30		22.2	4	114.3
	600	155.45	43.4		288	144	6.4	30		22.2	4	114.3
	900 & 1500	177.8	45.15		308	154	6.4	31.75		28.6	4	123.95
	2500	203.2	57.85		334	167	6.4	44.45		31.75	4	146.05
2"	150	152.4	38.6		284	142	1.6	30	91.95	19.05	4	120.65
	300	165.1	38.6		298	149	1.6	30		19.05	8	127
	600	165.1	43.4		298	149	6.4	30		19.05	8	127
	900 & 1500	215.9	51.5		358	179	6.4	38.1		25.4	8	165.1
	2500	234.95	64.2		378	189	6.4	50.8		28.6	8	171.45

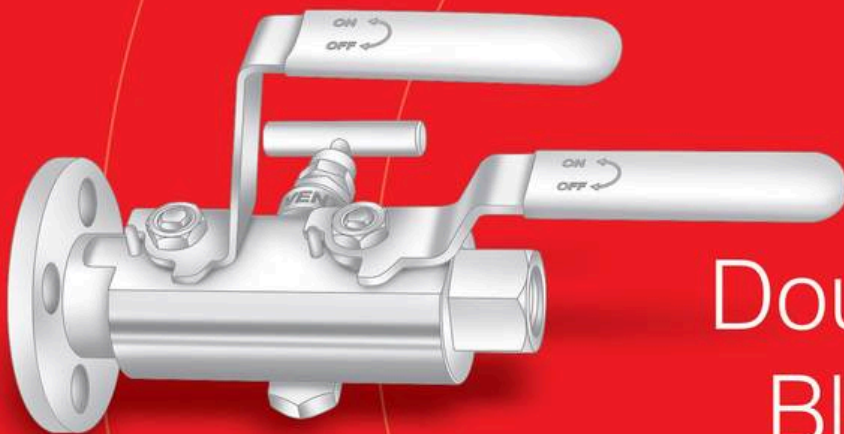
OS & Y Type

Double Block & Bleed



SIZE	CLASS	ØA	B	P	C	F	T	ØR	ØH	N Qty.	P.C.D.
1/2"	150	88.9	54.1	1/2" NPT	210	1.6	45.5	35.05	15.8	4	60.45
	300	95.25	54.1		216	1.6	45.5		15.8	4	66.55
	600	95.25	58.9		216	6.4	45.5		15.8	4	66.55
	900 & 1500	120.65	58.9		240	6.4	45.5		22.2	4	82.55
	2500	133.35	58.9		256	6.4	45.5		22.2	4	88.9
3/4"	150	98.55	54.1		220	1.6	45.5	42.93	15.8	4	69.85
	300	117.35	54.1		240	1.6	45.5		19.05	4	82.55
	600	117.35	58.9		240	6.4	45.5		19.05	4	82.55
	900 & 1500	130.05	58.9		250	6.4	45.5		22.2	4	88.9
	2500	139.7	58.9		260	6.4	45.5		22.2	4	95.25
1"	150	107.95	54.1		230	1.6	45.5	50.8	15.8	4	79.25
	300	123.95	54.1		246	1.6	45.5		19.05	4	88.9
	600	123.95	58.9		246	6.4	45.5		19.05	4	88.9
	900 & 1500	149.35	58.9		270	6.4	45.5		25.4	4	101.6
	2500	158.75	58.9		280	6.4	45.5		25.4	4	107.95
1-1/2"	150	127	54.1		248	1.6	45.5	73.15	15.8	4	98.55
	300	155.45	54.1		278	1.6	45.5		22.2	4	114.3
	600	155.45	58.9		278	6.4	45.5		22.2	4	114.3
	900 & 1500	177.8	58.9		298	6.4	45.5		28.6	4	123.95
	2500	203.2	58.9		324	6.4	45.5		31.75		146.05
2"	150	152.4	54.1		274	1.6	45.5	91.95	19.05	4	120.65
	300	165.1	54.1		288	1.6	45.5		19.05	8	127
	600	165.1	58.9		288	6.4	45.5		19.05	8	127
	900 & 1500	215.9	58.9		348	6.4	45.5		25.4	8	165.1
	2500	234.95	64.2		368	6.4	50.8		28.6	8	171.45

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Double Block & Bleed Valves

Double Block & Bleed Valves

Selection Chart

	A	DBB	AN	12	15	A	R	4NF	T	V	14	SS	OPTION
A = Aptek													
DBB = Double Block & Bleed Valve													
AN = All Needle Valve													
BNB = Block - Ball & Bleed - Needle													
Process Connection													
12 = 1/2" flange Connection													
34 = 3/4" Flange Connection													
01 = 1" Flange Connection													
15 = 1.5" Flange Connection													
02 = 2" Flange Connection													
15 = 150 Class													
30 = 300 Class													
60 = 600 Class													
90 = 900 Class													
51 = 1500 Class													
25 = 2500 Class													
4NF = 1/4" NPTF													
4BF = 1/4" BSPPF													
2NF = 1/2" NPTF													
4NM = 1/4" NPTM													
4BM = 1/4" BSPPM													
2NM = 1/2" NPTM													
Flange Type													
A = ASME B16.5													
D = DIN EN 1092													
R = Smooth RF													
J = RTJ													
Instrument Connection													
S = Outlet flange same as inlet													
4NF = 1/4" NPTF													
4BF = 1/4" BSPPF													
2NF = 1/2" NPTF													
4NM = 1/4" NPTM													
4BM = 1/4" BSPPM													
2NM = 1/2" NPTM													
Seal													
T = PTFE Packing													
G = Grafoil Packing													
Bleed / Vent Requirement													
N = No Bleed													
V = Vent													
Bleed / Vent Size													
14 = 1/4" Port													
12 = 1/2" Port													
SS = SS 316													
SSL = SS 316L													
SSD = SS 316/316L Dual													
SS4 = SS 304													
MO = Monel 400													
HC = Hastelloy C 276													
DU = Duplex SS													
SD = Super Duplex													
IN6 = Inconel-625													
IN8 = Inconel-825													
CS = Carbon Steel													
BR = Brass													
NA = NACE MR 01-75 (Latest Edition)													
OXY = Cleaned for Oxygen Service													
OSY = OS & Y require in isolation valve													
FS = Fire Safe													
LH = Lockable Handle													

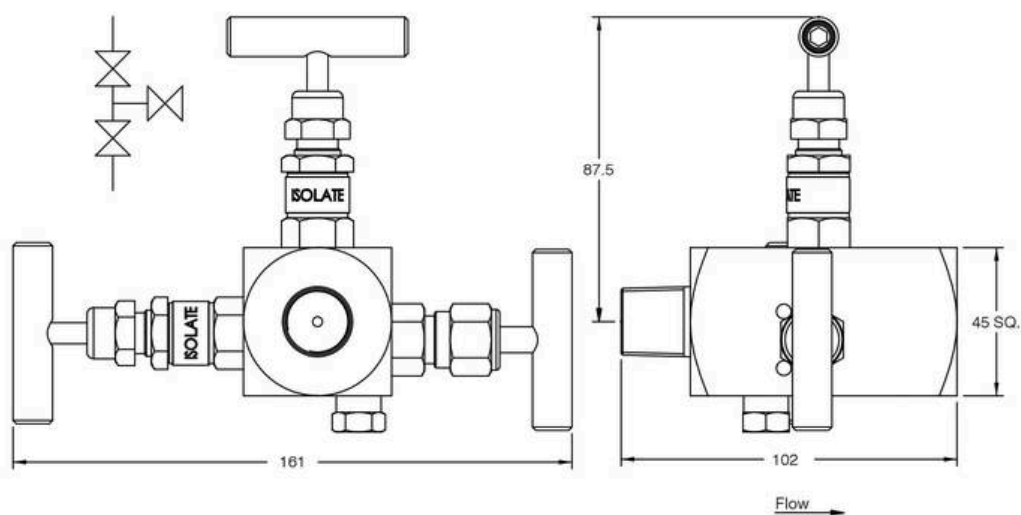
DBB instrument valves offer improved safety with 2 isolation valves & a vent valve between the primary & secondary isolation valves. Available in different options such as all needle, Ball - Needle - Ball, Threaded & Flanged connections.

Standard Specifications:

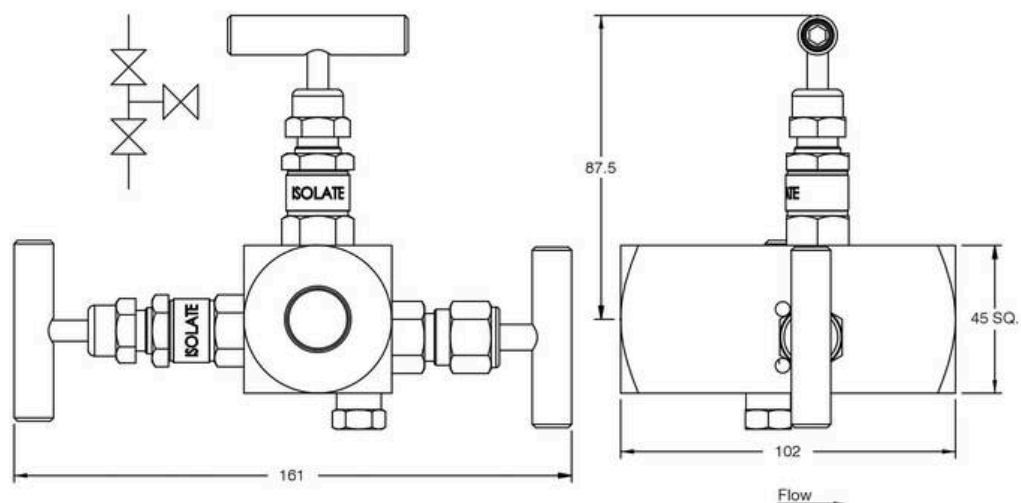
Maximum working Pressure: 6000 & 10000 Psi
(see Pressure & Temperature Chart on page No. 07 for higher temperature applications). MOC in accordance with NACE MR0175 is available on request



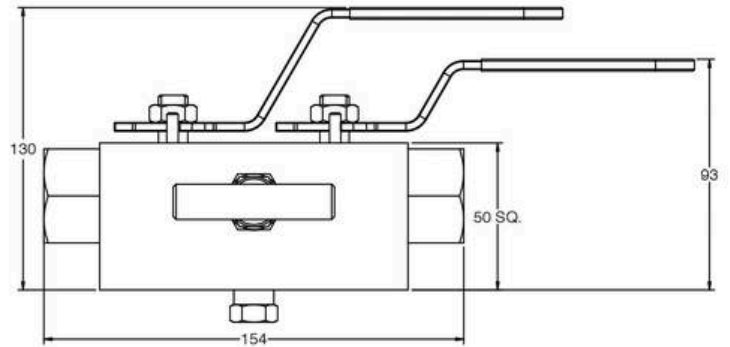
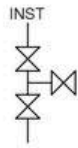
DBB AN - MF



DBB AN - FF

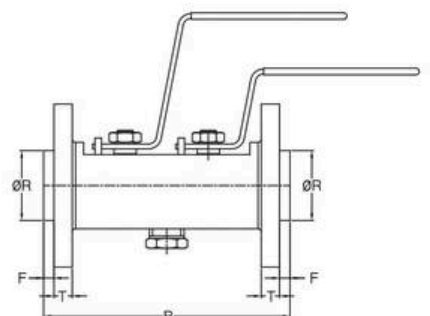
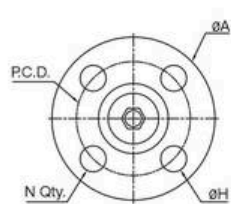
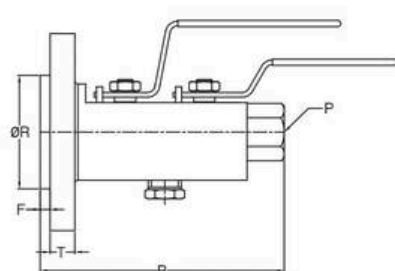
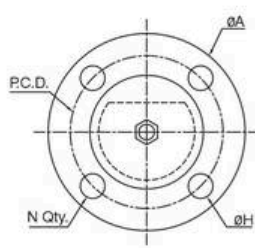
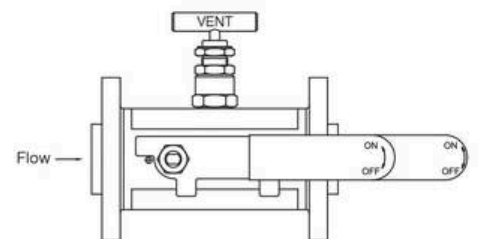
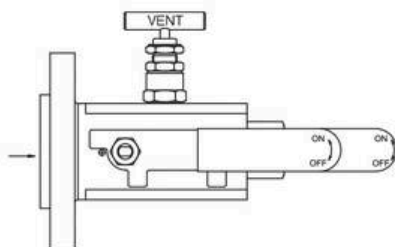
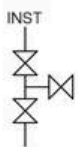


DBB BNB - S



DBB BNB - FS

DBB BNB - FL



DBB BNB - FS

SIZE	CLASS	ØA	B	P	F	T	ØR	ØH	N Qty.	P. C. D.
1/2"	150	88.9	161.5	1/2" NPT	1.6	9.65	35.05	15.8	4	60.45
	300	95.25	164		1.6	12.7		15.8	4	66.55
	600	95.25	170.5		6.4	14.2		15.8	4	66.55
	900 & 1500	120.65	178.5		6.4	22.35		22.2	4	82.55
	2500	133.35	186.5		6.4	30.2		22.2	4	88.9
3/4"	150	98.55	162.5		1.6	11.2	42.93	15.8	4	69.85
	300	117.35	165.5		1.6	14.2		19.05	4	82.55
	600	117.35	172.1		6.4	15.7		19.05	4	82.55
	900 & 1500	130.05	174		6.4	25.4		22.2	4	88.9
	2500	139.7	181.5		6.4	31.75		22.2		95.25
1"	150	107.95	188		1.6	12.7	50.8	15.8	4	79.25
	300	123.95	164		1.6	15.7		19.05	4	88.9
	600	123.95	167.5		6.4	17.5		19.05	4	88.9
	900 & 1500	149.35	185		6.4	28.4		25.4	4	101.6
	2500	158.75	191.5		6.4	35.05		25.4	4	107.95
1-1/2"	150	127	167.5		1.6	15.7	73.15	15.8	4	98.55
	300	155.45	170.5		1.6	19.05		22.2	4	114.3
	600	155.45	178.5		6.4	22.35		22.2	4	114.3
	900 & 1500	177.8	188.5		6.4	31.75		28.6	4	123.95
	2500	203.2	201		6.4	44.45		31.75	4	146.05
2"	150	152.4	169		1.6	17.5	91.95	19.05	4	120.65
	300	165.1	172.5		1.6	20.6		19.05	8	127
	600	165.1	181.5		6.4	25.4		19.05	8	127
	900 & 1500	215.9	194.5		6.4	38.1		25.4	8	165.1
	2500	234.95	207		6.4	50.8		28.6	8	171.45

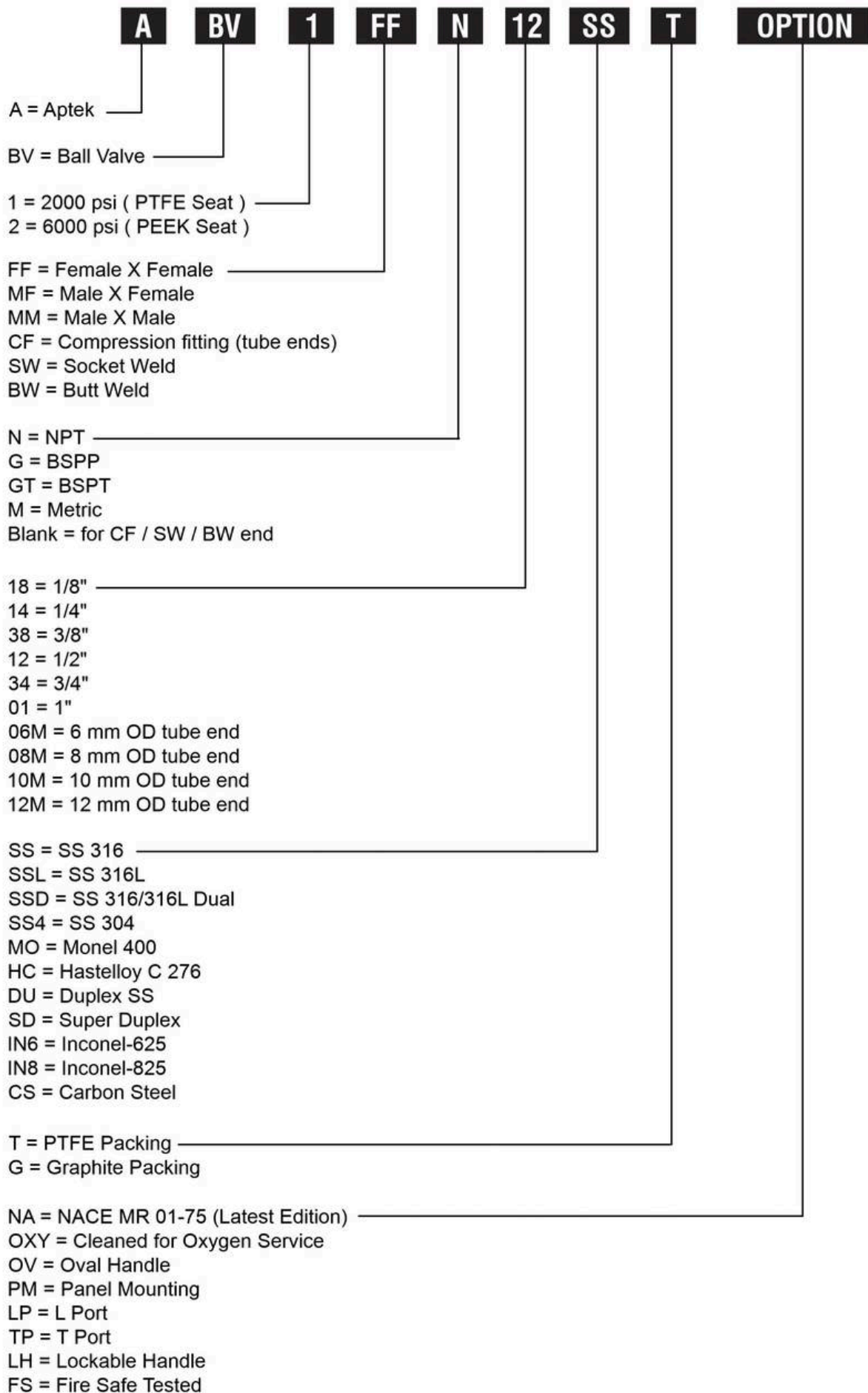
DBB BNB - FL

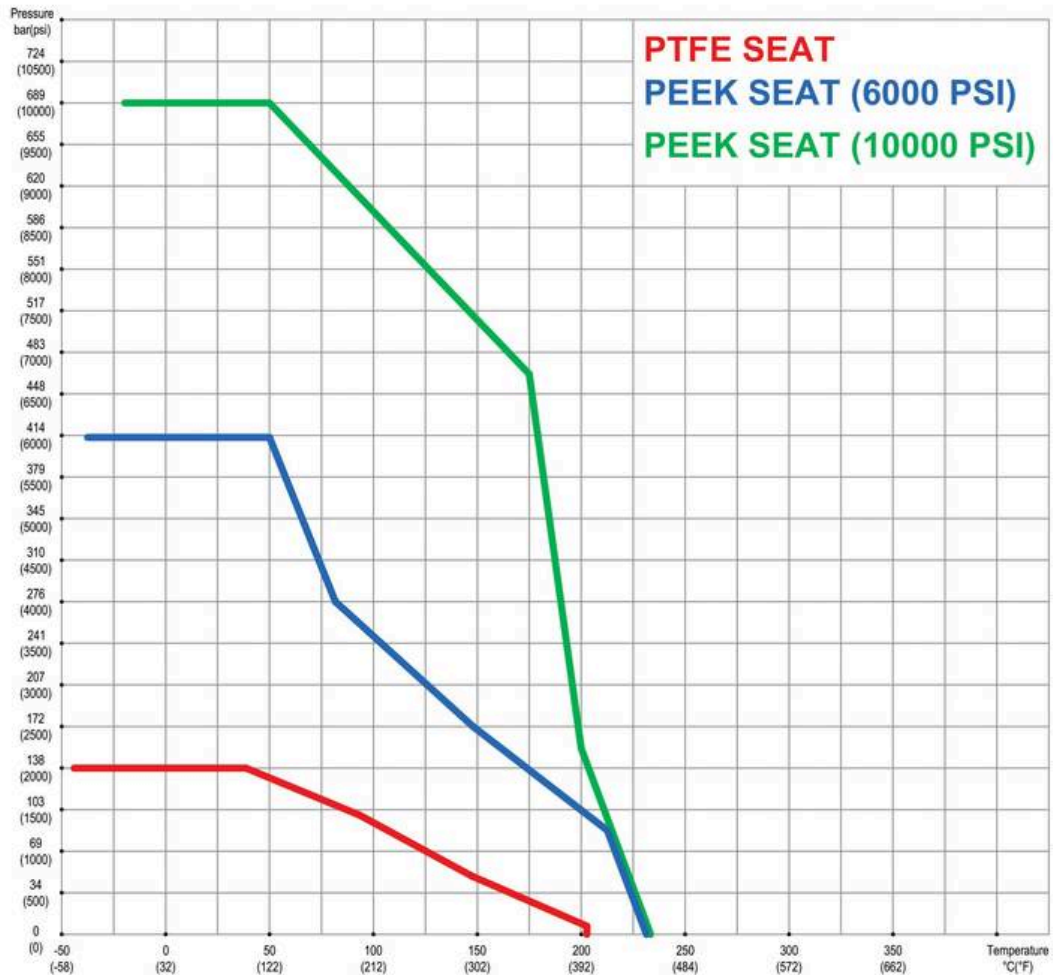
SIZE	CLASS	ØA	B	F	T	ØR	ØH	N Qty.	P. C. D.
1/2"	150	88.9	175.5	1.6	9.65	35.05	15.8	4	60.45
	300	95.25	181.5	1.6	12.7		15.8	4	66.55
	600	95.25	194	6.4	14.2		15.8	4	66.55
	900 & 1500	120.65	210.5	6.4	22.35		22.2	4	82.55
	2500	133.35	226	6.4	30.2		22.2	4	88.9
3/4"	150	98.55	178.6	1.6	11.2	42.93	15.8	4	69.85
	300	117.35	184.5	1.6	14.2		19.05	4	82.55
	600	117.35	197	6.4	15.7		19.05	4	82.55
	900 & 1500	130.05	216.5	6.4	25.4		22.2	4	88.9
	2500	139.7	229	6.4	31.75		22.2		95.25
1"	150	107.95	181.5	1.6	12.7	50.8	15.8	4	79.25
	300	123.95	187.5	1.6	15.7		19.05	4	88.9
	600	123.95	201	6.4	17.5		19.05	4	88.9
	900 & 1500	149.35	222.5	6.4	28.4		25.4	4	101.6
	2500	158.75	236	6.4	35.05		25.4	4	107.95
1-1/2"	150	127	187.5	1.6	15.7	73.15	15.8	4	98.55
	300	155.45	194	1.6	19.05		22.2	4	114.3
	600	155.45	210.5	6.4	22.35		22.2	4	114.3
	900 & 1500	177.8	229	6.4	31.75		28.6	4	123.95
	2500	203.2	254.5	6.4	44.45		31.75	4	146.05
2"	150	152.4	191	1.6	17.5	91.95	19.05	4	120.65
	300	165.1	197	1.6	20.6		19.05	8	127
	600	165.1	216.5	6.4	25.4		19.05	8	127
	900 & 1500	215.9	242	6.4	38.1		25.4	8	165.1
	2500	234.95	267	6.4	50.8		28.6	8	171.45



Instrumentation Ball Valve

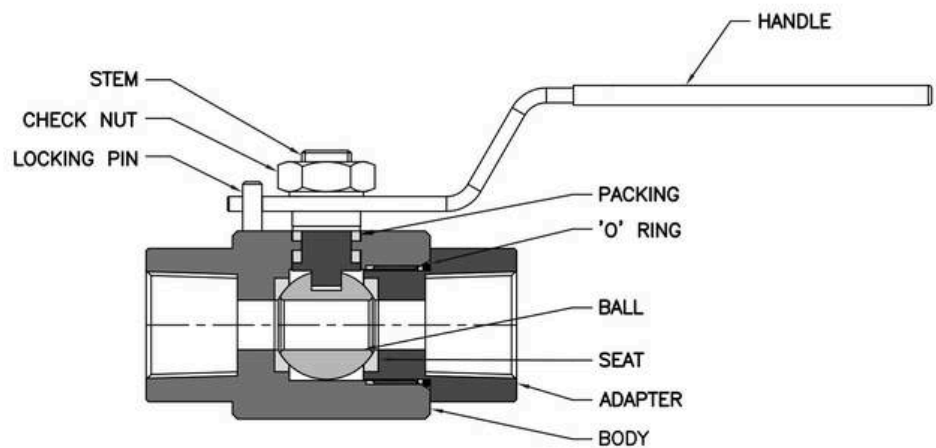
Ball Valve Selection Chart



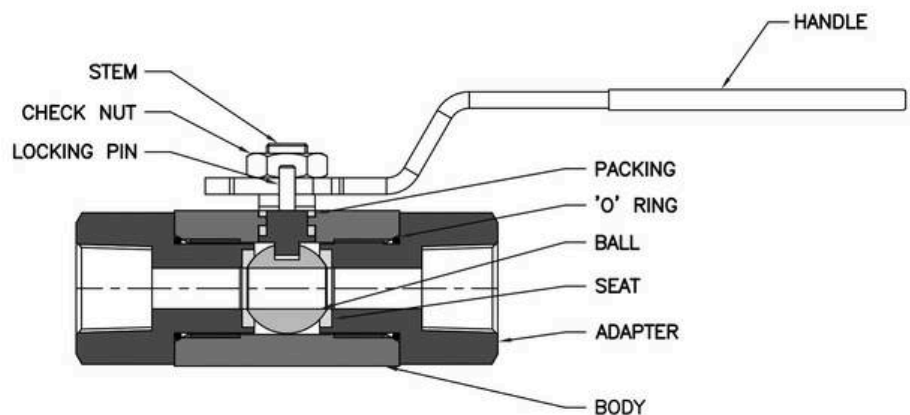


P-T CHART FOR BALL VALVE SEAT

ABV 1 Low Pressure Ball Valve



ABV 2 High Pressure Ball Valve



This valves are safe, reliable & economical option for low pressure application

Standard Specifications:

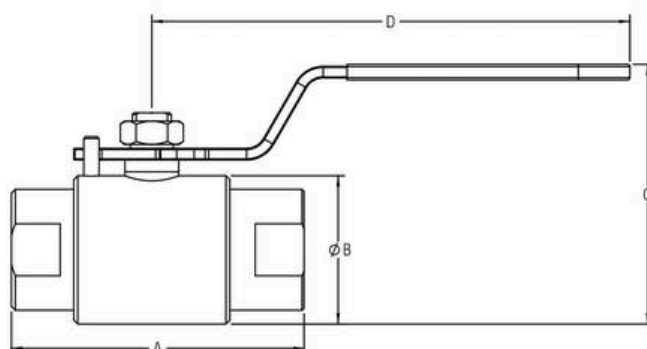
Maximum working pressure : 138 bar at 38°C (2000 PSI at 100°F)

See pressure temperature chart for high temperature applications

ABV 1 FF



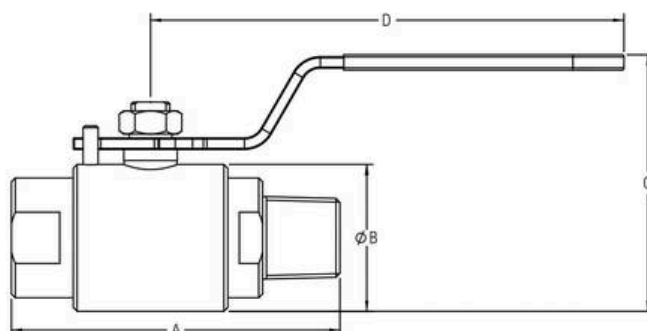
SIZE	A	ØB	C	D	WEIGHT (kg)
1/4"	60	32	60	112	0.30
3/8"	64	32	60	112	0.40
1/2"	75	38	68	112	0.50
3/4"	86	50	80	136	0.75
1"	94	50	80	136	0.90



ABV 1 MF



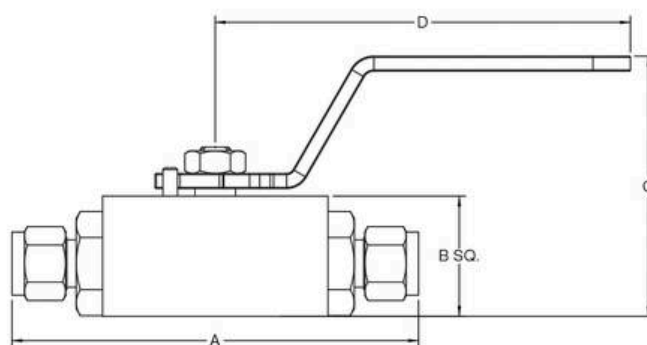
SIZE	A	ØB	C	D	WEIGHT (kg)
1/4"	70	32	60	112	0.30
3/8"	75	32	60	112	0.40
1/2"	86	38	68	122	0.50
3/4"	97	50	80	136	0.75
1"	105	50	80	136	0.90



ABV 1 CF



SIZE	A	ØB	C	D	WEIGHT (kg)
1/4"	80	32	60	112	0.30
3/8"	87	32	60	112	0.40
1/2"	102	38	68	122	0.50
3/4"	110	50	80	136	0.75
1"	117	50	80	136	0.90



High Pressure Ball Valve

This range offers reliable option for the high pressure application. These valves are recommended for the use in Oil, Gas, Petrochemicals where the long life is required.

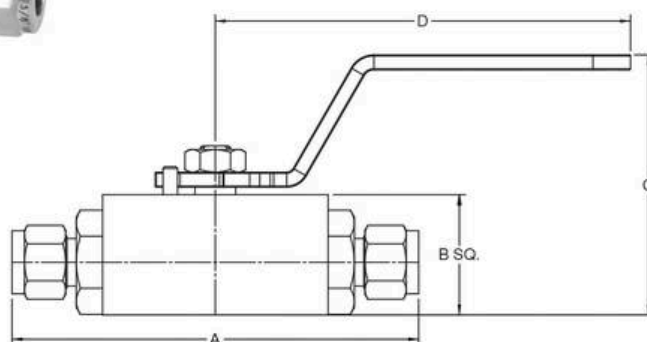
Standard Specifications:

Maximum working pressure : 413 bar at 38°C (6000 psi at 100°F), 690 bar at 38°C (10000 psi at 100°F) (See pressure temperature chart for high temperature applications)

ABV 2 CF



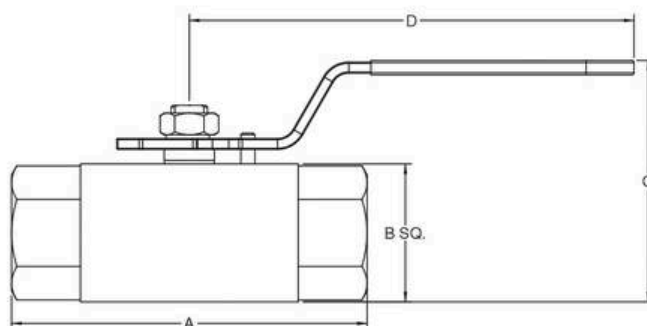
SIZE	A	B SQ.	C	D	WEIGHT (kg)
1/4"	97	32	61	85	0.6
3/8"	100	32	61	85	0.7
1/2"	132	38	67	120	1.5
3/4"	155	50	85	135	2.4
1"	160	50	85	135	3.0



ABV 2 FF



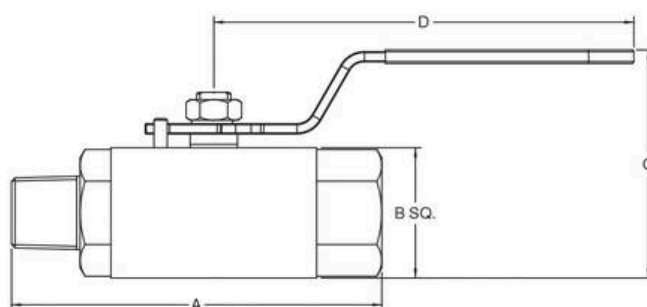
SIZE	A	B SQ	C	D	WEIGHT (kg)
1/4"	60	32	61	85	0.4
3/8"	60	32	61	85	0.4
1/2"	103	38	67	120	1.2
3/4"	123	50	85	135	2.0
1"	123	50	85	135	2.4



ABV 2 MF



SIZE	A	B SQ	C	D	WEIGHT (kg)
1/4"	75	32	61	85	0.5
3/8"	75	32	61	85	0.6
1/2"	114	38	67	120	1.3
3/4"	141	50	85	135	2.2
1"	144	50	85	135	2.6



Standard Specifications:

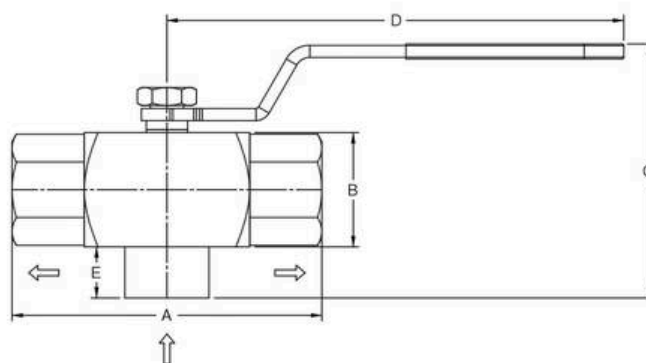
Maximum working pressure : 138 bar at 38°C (2000 PSI at 100°F), 413 bar at 38°C (6000 PSI at 100°F)

See pressure temperature chart for high temperature applications

3 Way Ball Valve



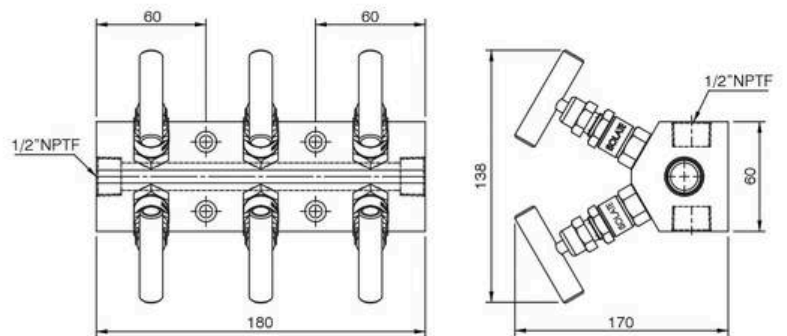
SIZE	A	B SQ	C	D	E	WEIGHT (kg)
1/4"	60	32	75	85	14	0.45
3/8"	60	32	75	85	14	0.45
1/2"	103	38	84	120	17	1.25
3/4"	123	50	105	135	20	2.05
1"	123	50	105	135	20	2.45



Standard Specifications:

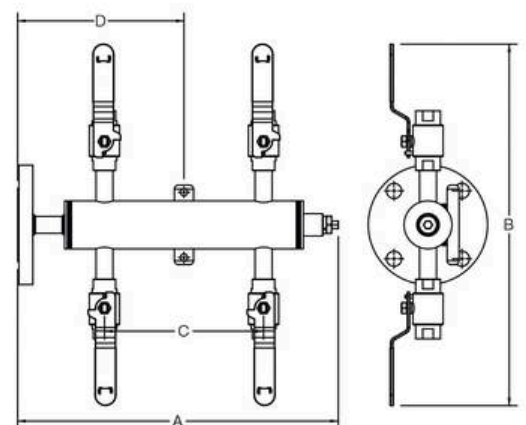
Maximum working Pressure: 2000/6000/10000 Psi (see Pressure & Temperature Chart on page No.07 for higher temperature applications).
MOC in accordance with NACE MR0175 is available on request Also Available in 4,6,8,10 Valves

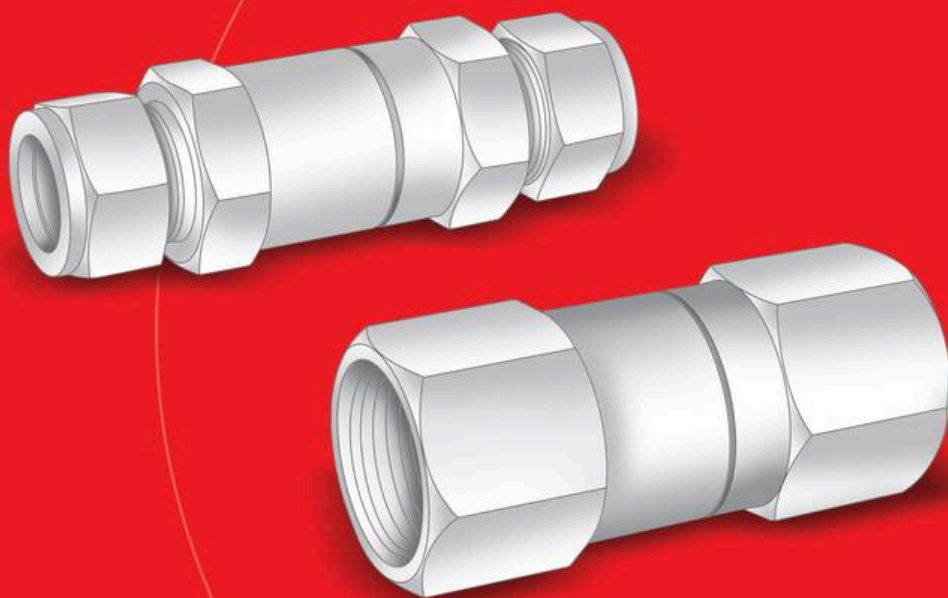
Needle Valve Air Header



Distribution Header

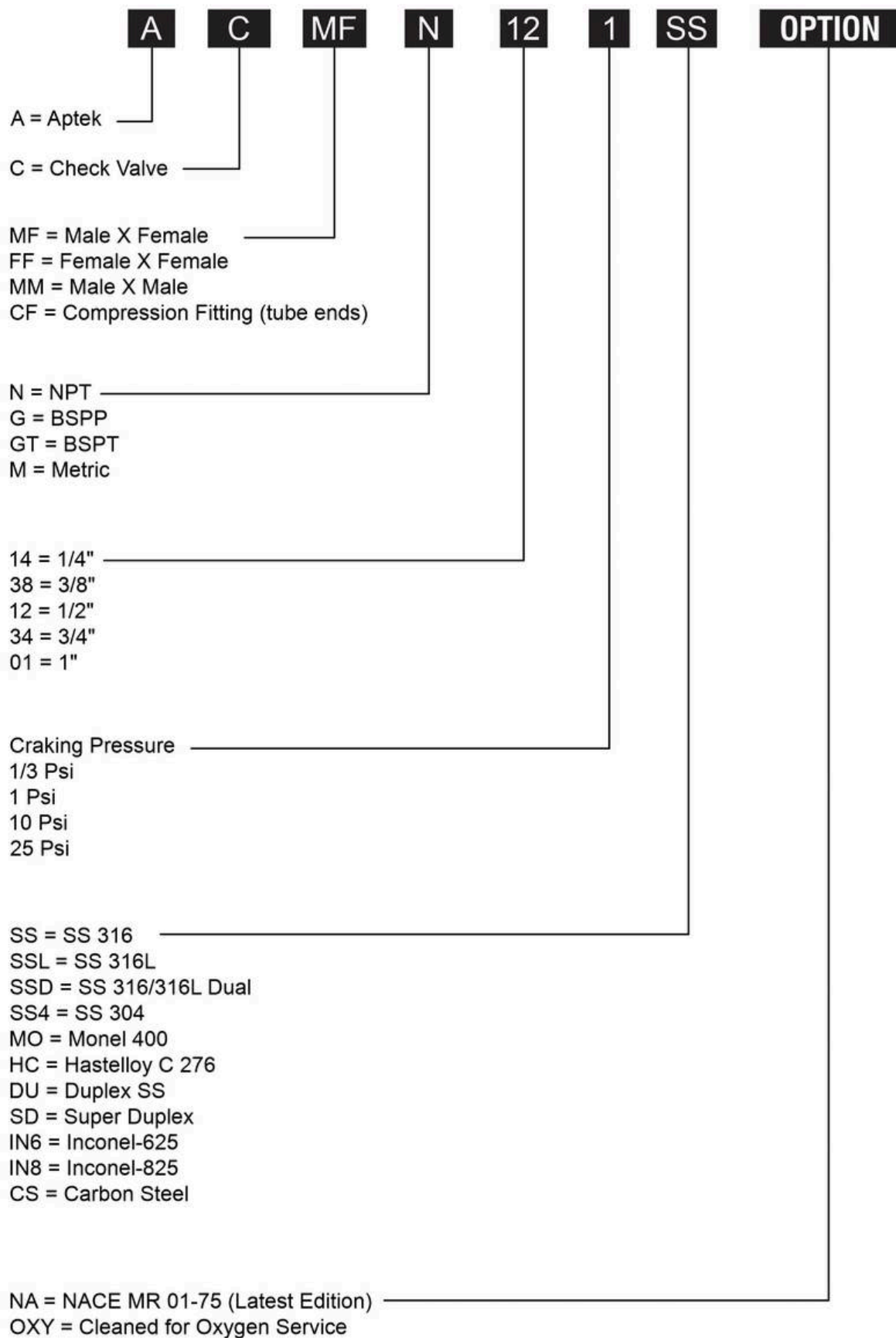
Ball Valve Air Header





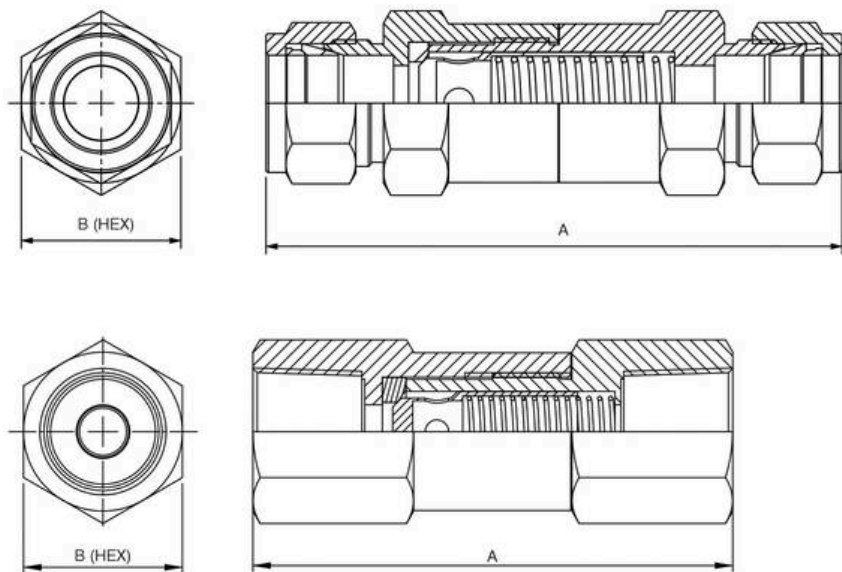
Check Valves

Check Valve Selection Chart



A check valve is a one-way valve, in which the flow can run freely one way, but if the flow returns the valve will close to protect the piping, other valves, pumps etc.

Available Cracking Pressure, psi (bar) 1/3 (0.03), 1 (0.07), 10 (0.69), 25 (1.80)

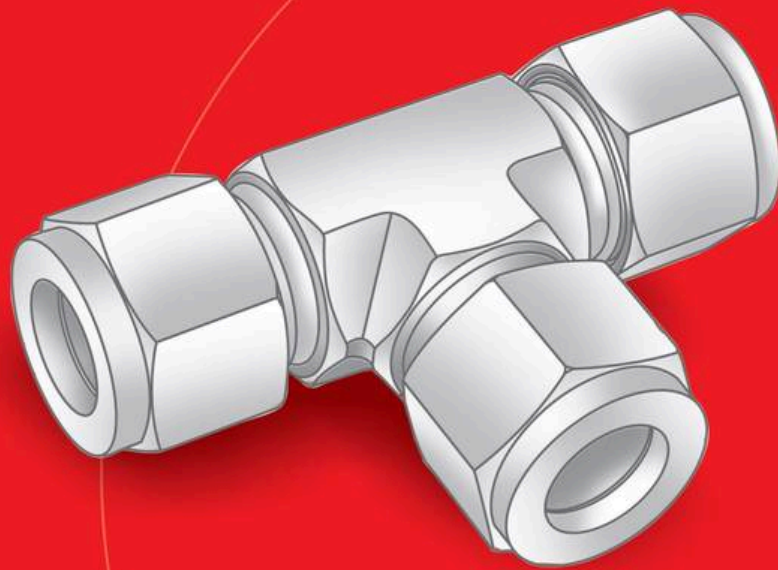


INLET / OUTLET	Size	A (LENGTH)	B (HEX)
NPT FXF	1/4"	58.00	19
	3/8"	79.00	22
	1/2"	82.00	27
	3/4"	103.00	32
	1"	109.00	41
TUBE FITTING END	1/4"	71.00	19
	3/8"	83.00	22
	1/2"	95.00	27
	3/4"	116.00	32
	1"	134.00	41

Available Cracking Pressure

Psi	Bar
1/3	0.03
1	0.07
10	0.69
25	1.80

aptek

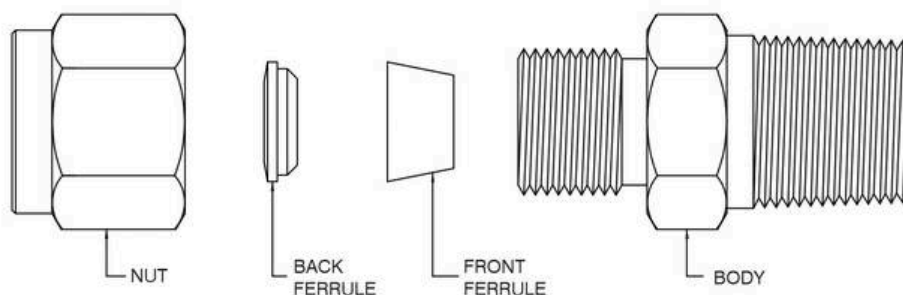


Instrumentation Tube Fittings

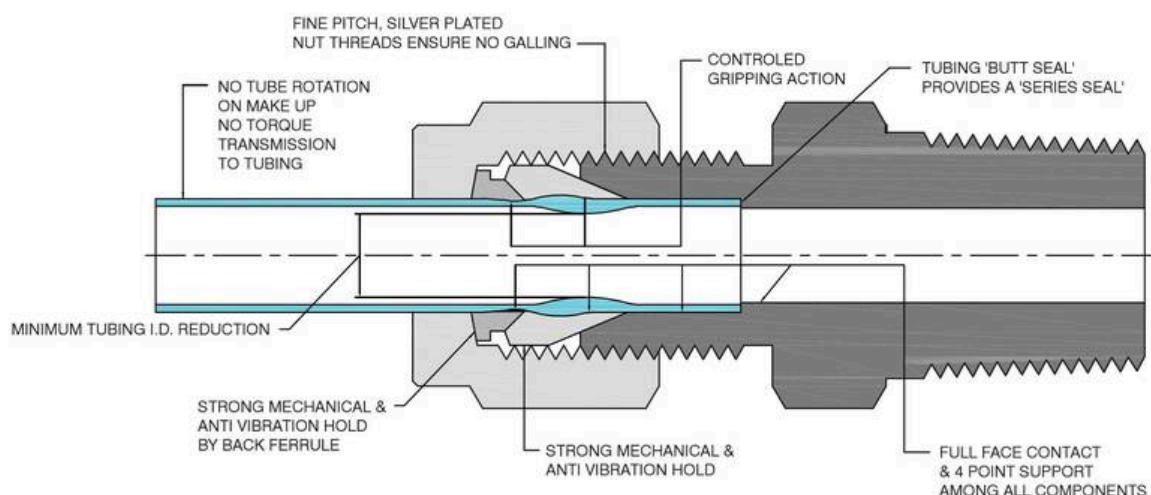
Tube Fitting Selection Chart

A	MC	I	12	12	N	SS	OPTION
A = Aptek	MC = Male Connector , FC = Female Connector EU = Equal Union , UU = Unequal Union BMC = Bulkhead Male Connector , C = Cap BFC = Bulkhead Female Connector , P = Plug BU = Bulkhead Union , UE = Union Elbow ME = Male Elbow , FE = Female Elbow UC = Equal Union Cross , UT = Union Tee MBT = Male Branch Tee , MRT = Male Run Tee FBT = Female Branch Tee , FRT = Female Run Tee N = Nut , F = Ferrule , FTA = Female Tube Adaptor MTA = Male Tube Adaptor , WC = Weld Connector VP = Vent Plug	I = Imperial Range M = Metric Range	14 = 1/4" , 38 = 3/8" 12 = 1/2" for Imperial & 12mm OD for Metric 34 = 3/4" , 01 = 1" 06 = 6mm OD , 08 = 8mm OD 10 = 10mm OD , 20 = 20mm OD 25 = 25mm OD	14 = 1/4" , 38 = 3/8" 12 = 1/2" for Imperial & 12mm OD for Metric 34 = 3/4" , 01 = 1" 06 = 6mm OD , 08 = 8mm OD 10 = 10mm OD , 20 = 20mm OD 25 = 25mm OD , Blank for Vent Plug , Nut & Ferrule	N = NPT G = BSPP GT = BSPT M = Metric S = SAE U = UNF	SS = SS 316 SSL = SS 316L SSD = SS 316/316L Dual SS4 = SS 304 MO = Monel 400 HC = Hastelloy C 276 DU = Duplex SS SD = Super Duplex IN6 = Inconel-625 IN8 = Inconel-825 CS = Carbon Steel BR = Brass	NA = NACE MR 01-75 (Latest Edition) OXY = Cleaned for Oxygen Service BR = Bored Thru SPF = Silver Plated Ferrule VO = Viton 'O' Ring

Aptek Instrument tube fittings are used for leak proof, torque free seal in different instrumentation & process tubing connections. These products eliminate costly & hazardous leakages.



As shown in the figure the normal tube fitting is made of four parts. The parts are machined as per international standards to achieve the high quality. The ferrules are hardened & burnished for better swaging. The nuts are silver plated for preventing the Galling up on the threads. The fitting consists of four parts but when it is fixed to the tube as shown; it becomes a five parts leak proof connection. No special tool is required to install the tube fittings.

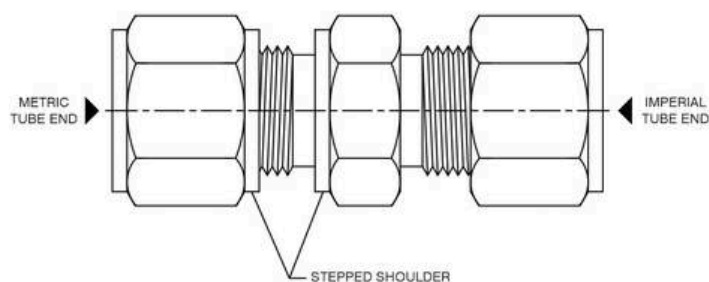


Materials Standards

Aptek fittings are available in different materials as shown in the below mentioned table.

Materials	Barstock	Forgings
SS 316/316L	ASTM A276, EN 1.4401	ASTM A182, EN 1.4401
6- Moly	ASTM A479	ASTM A182
Alloy 400	ASTM B164	ASTM B564
Alloy 600	ASTM B166	ASTM B564
Alloy 625	ASTM B446	ASTM B564
Alloy 825	ASTM B425	ASTM B564
Alloy C-276	ASTM B574	ASTM B564
Brass	ASTM B16, ASTM B453	ASTM B283
Carbon Steel	ASTM A108	-
Alloy 2507-Super Duplex	ASTM A479	ASTM A182
Titanium (Grade 4)	ASTM B348	ASTM B381

Identification of metric & fractional



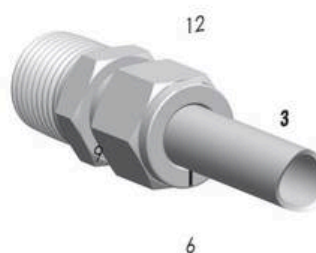
Installation Instruction for Tube Fittings

Assembly Instructions:

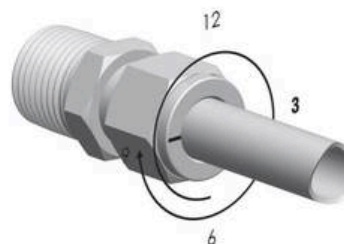
- 1 Insert the tube in APTEK tube fitting till the face of tube rest in the fitting body as shown.



- 2 Tighten nut finger tight and scribe the nut at the 6 o'clock position as shown.



- 3 Hold Fitting body steady, tighten the nut 1-1/4 turn from finger tight for fitting size in fraction 1/4" to 1", in metric 6 mm to 25 mm and 3/4" turn from finger tight for fitting size in fraction 1/16" to 3/16", in metric 2 mm to 4 mm



- 4 Never turn the fitting body. Instead, hold the fitting body and turn the nut.

Reassembly Instructions:

- A Prior to dis-assembly mark a line along the nut and fitting body flats as shown. Make sure that the fitting body and tube assembly are free from contaminants or damages.



- B Insert the tube assembly into the fitting body until the front ferrule seats firmly. Assemble the tube and nut to the finger tight position.



- C Rotate the nut till previously marked position is reached. Further the nut should be rotated until significant rise in torque is felt to ensure the assembly is positively seal.



TABLE - 1

NPT / ISO PIPE SIZE IN.	SS 316 SS & Carbon Steel	
	Male Threads (PSI)	Female Threads (PSI)
1/8" NPT	10000	6400
1/4" NPT	8300	6500
3/8" NPT	8000	5200
1/2" NPT	7800	4800
3/4" NPT	7500	4600
1" NPT	5300	4400

Compression fittings are rated for working pressure higher than the tubing. Therefore pressure rating depends on choice of tubing which should be based on material. Wall thickness and temperature (see table below). De rating factors must be used to determinate tubing working pressure at high temperatures.

Maximum Allowable Working Pressure For Fractional Size Tubing (Value in PSI)

TABLE - 2

Tubing O.D. (inch)	Tubing wall thickness (inch), SS 316 & SS 304												
	0.010	0.012	0.014	0.016	0.020	0.028	0.035	0.049	0.065	0.083	0.095	0.109	0.120
1/16"	5600	6800	8100	9400	12000								
1/8"						8500	10900						
3/16"						5400	7000	10200					
1/4"						4000	5100	7500	10200				
5/16"							4000	5800	8000				
3/8"							3300	4800	6500				
1/2"							2600	3700	5100	6700			
5/8"								2900	4000	5200	6000		
3/4"								2400	3300	4200	4900	5800	
7/8"								2000	2800	3600	4200	4800	
1"									2400	3100	3600	4200	4700

Note: For Gas service, use the tubing wall listed on un-shadowed section in Table 2 & 3.

Maximum Allowable Working Pressure For Metric Size Tubing (value In BAR)

TABLE - 3

Tubing O.D. (mm)	Tubing wall thickness (mm), SS 316 & SS 304									
	0.8	1.0	1.2	1.5	1.8	2.0	2.2	2.5	2.8	3.0
3	680									
6	320	430	550	720						
8		310	390	530						
10		240	310	410	510	580				
12		200	250	330	420	480				
14		160	200	270	340	390	430			
16			180	230	290	330	370			
18			150	210	260	290	330	380		
20			140	180	230	260	290	330	380	
22			120	170	210	240	260	300	340	
25					180	200	230	260	300	320

- AISI 304 and AISI 316 stainless steel annealed seamless tubing ASTM A-269 or A-213 or equivalent
- Above mentioned working pressure is applicable only for -29 °C to 38 °C for AISI 304 & AISI 316/316 L stainless steel. For other temperature range use table-4 (Pressure Rating at Elevated Temperature) for derating factor

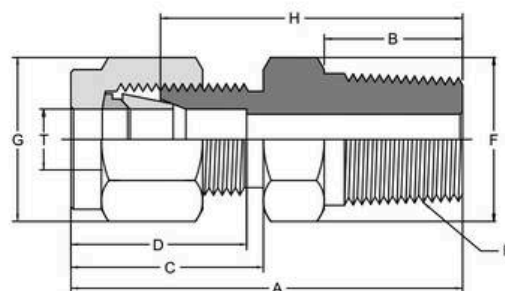
Pressure Ratings at Elevated Temperatures

Elevated Temperature Factor

TABLE - 4

Temperature		Tubing Material	
F	C	SS 304/304L	SS 316/316L
200	93	1.00	1.00
400	204	0.93	0.96
600	315	0.82	0.85
800	426	0.76	0.8
1000	537	0.69	0.76

Male Connector



Imperial Range

Dimensions in Inches

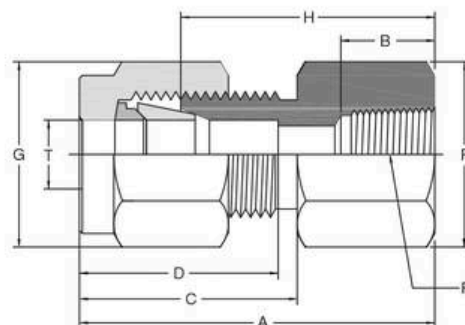
PART. NO.	T O/D	P	A LENGTH	B	C	D	F BODY	G NUT	H BODY
MCI-1616	1/16	1/16	0.94	0.38	0.44	0.34	5/16	5/16	0.78
MCI-1618	1/16	1/8	1.00	0.38	0.44	0.34	7/16	5/16	0.84
MCI-1816	1/8	1/16	1.19	0.38	0.63	0.53	7/16	7/16	0.91
MCI-1818	1/8	1/8	1.19	0.38	0.63	0.53	7/16	7/16	0.91
MCI-1814	1/8	1/4	1.41	0.56	0.63	0.53	9/16	7/16	1.13
MCI-3618	3/16	1/8	1.21	0.38	0.63	0.54	7/16	1/2	0.95
MCI-3614	3/16	1/4	1.42	0.56	0.63	0.54	9/16	1/2	1.16
MCI-1416	1/4	1/16	1.29	0.38	0.70	0.60	1/2	9/16	1.00
MCI-1418	1/4	1/8	1.29	0.38	0.70	0.60	1/2	9/16	1.00
MCI-1414	1/4	1/4	1.50	0.56	0.72	0.63	9/16	9/16	1.19
MCI-1438	1/4	3/8	1.51	0.56	0.70	0.60	11/16	9/16	1.19
MCI-1412	1/4	1/2	1.73	0.75	0.70	0.60	7/8	9/16	1.44
MCI-5618	5/16	1/8	1.34	0.38	0.73	0.64	9/16	5/8	1.05
MCI-5614	5/16	1/4	1.52	0.56	0.73	0.64	9/16	5/8	1.23
MCI-3818	3/8	1/8	1.38	0.38	0.76	0.66	5/8	11/16	1.09
MCI-3814	3/8	1/4	1.57	0.56	0.76	0.66	5/8	11/16	1.28
MCI-3838	3/8	3/8	1.57	0.56	0.76	0.66	11/16	11/16	1.28
MCI-3812	3/8	1/2	1.79	0.75	0.76	0.66	7/8	11/16	1.50
MCI-3834	3/8	3/4	1.82	0.75	0.76	0.66	1-1/16	11/16	1.53
MCI-1214	1/2	1/4	1.71	0.56	0.86	0.90	13/16	7/8	1.31
MCI-1238	1/2	3/8	1.71	0.56	0.86	0.90	13/16	7/8	1.31
MCI-1212	1/2	1/2	1.90	0.75	0.86	0.90	7/8	7/8	1.50
MCI-1234	1/2	3/4	1.93	0.75	0.86	0.90	1-1/16	7/8	1.53
MCI-5838	5/8	3/8	1.74	0.56	0.86	0.96	15/16	1	1.34
MCI-5812	5/8	1/2	1.93	0.75	0.86	0.96	15/16	1	1.53
MCI-5834	5/8	3/4	1.93	0.75	0.86	0.96	1-1/16	1	1.53
MCI-3412	3/4	1/2	1.99	0.75	0.86	0.96	1-1/16	1-1/8	1.59
MCI-3434	3/4	3/4	1.99	0.75	0.86	0.96	1-1/16	1-1/8	1.59
MCI-3401	3/4	1	2.21	0.94	0.86	0.96	1-3/16	1-1/8	1.81
MCI-0134	1	3/4	2.26	0.75	1.04	1.23	1-3/8	1-1/2	1.78
MCI-0101	1	1	2.45	0.94	1.04	1.23	1-3/8	1-1/2	1.97

Matric Range

Dimensions in mm

PART. NO.	T O/D	P	A LENGTH	B	C	D	F BODY	G NUT	H BODY
MCM-0418	4	1/8"	31.35	9.53	16.67	14.29	12	12	24.21
MCM-0414	4	1/4"	35.72	14.29	16.67	14.29	14	12	29.37
MCM-0618	6	1/8"	33.34	9.53	18.26	15.88	14	14	25.40
MCM-0614	6	1/4"	33.34	14.29	18.26	15.88	14	14	30.16
MCM-0638	6	3/8"	38.89	14.29	18.26	15.88	18	14	30.96
MCM-0612	6	1/2"	44.45	19.05	18.26	15.88	22	14	36.51
MCM-0818	8	1/8"	34.53	9.53	19.05	16.67	15	17	26.59
MCM-0814	8	1/4"	39.29	14.29	19.05	16.67	15	17	31.35
MCM-1018	10	1/8"	35.72	9.53	19.84	17.46	18	19	27.78
MCM-1014	10	1/4"	40.48	14.29	19.84	17.46	18	19	32.54
MCM-1038	10	3/8"	40.48	14.29	19.84	17.46	18	19	32.54
MCM-1012	10	1/2"	46.04	19.05	19.84	17.46	22	19	38.10
MCM-1034	10	3/4"	46.83	19.05	19.84	17.46	27	19	38.89
MCM-1214	12	1/4"	43.66	14.29	22.23	23.02	22	22	33.34
MCM-1238	12	3/8"	43.66	14.29	22.23	23.02	22	22	33.34
MCM-1212	12	1/2"	48.42	19.05	22.23	23.02	22	22	38.10
MCM-1234	12	3/4"	49.21	19.05	22.23	23.02	27	22	38.89
MCM-1638	16	3/8"	44.45	14.29	22.23	24.61	24	27	34.13
MCM-1612	16	1/2"	49.21	19.05	22.23	24.61	24	27	38.89
MCM-1634	16	3/4"	49.21	19.05	22.23	24.61	27	27	38.89
MCM-1812	18	1/2"	50.80	19.05	22.23	24.61	27	30	40.48
MCM-1834	18	3/4"	50.80	19.05	22.23	24.61	27	30	40.48
MCM-1801	18	1"	56.36	23.81	22.23	24.61	36	30	46.40
MCM-2012	20	1/2"	50.80	19.05	22.23	24.61	30	30	40.48
MCM-2034	20	3/4"	50.80	19.05	22.23	24.61	30	30	40.48
MCM-2001	20	1"	56.36	23.81	22.23	24.61	35	30	46.04
MCM-2534	25	3/4"	57.15	19.05	26.19	30.96	35	41	45.24
MCM-2501	25	1"	61.91	23.81	26.19	30.96	35	41	50.01

Female Connector



Imperial Range

Dimensions in Inches

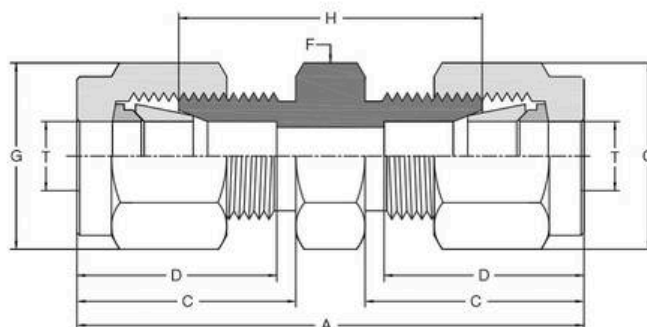
PART. NO.	T O/D	P	A LENGTH	B	C	D	F BODY	G NUT	H BODY
FCI-1616	1/16	1/16	0.94	0.36	0.44	0.34	7/16	5/16	0.78
FCI-1818	1/8	1/8	1.16	0.39	0.63	0.53	9/16	7/16	0.88
FCI-1814	1/8	1/4	1.34	0.59	0.63	0.53	3/4	7/16	1.06
FCI-3618	3/16	1/8	1.17	0.39	0.63	0.54	9/16	9/16	0.91
FCI-1418	1/4	1/8	1.23	0.39	0.70	0.60	9/16	9/16	0.94
FCI-1414	1/4	1/4	1.41	0.59	0.70	0.60	3/4	9/16	1.12
FCI-1438	1/4	3/8	1.48	0.59	0.70	0.60	7/8	9/16	1.19
FCI-1412	1/4	1/2	1.7	0.78	0.70	0.60	1-1/16	9/16	1.41
FCI-5618	5/16	1/8	1.26	0.39	0.73	0.64	9/16	5/8	0.97
FCI-5614	5/16	1/4	1.45	0.59	0.73	0.64	3/4	5/8	1.16
FCI-3818	3/8	1/8	1.29	0.39	0.76	0.66	5/8	11/16	1.00
FCI-3814	3/8	1/4	1.48	0.59	0.76	0.66	3/4	11/16	1.19
FCI-3838	3/8	3/8	1.54	0.59	0.76	0.66	7/8	11/16	1.25
FCI-3812	3/8	1/2	1.73	0.78	0.76	0.66	1-1/16	11/16	1.44
FCI-1214	1/2	1/4	1.65	0.59	0.86	0.90	13/16	7/8	1.25
FCI-1238	1/2	3/8	1.65	0.59	0.86	0.90	7/8	7/8	1.25
FCI-1212	1/2	1/2	1.84	0.78	0.86	0.90	1-1/16	7/8	1.44
FCI-5838	5/8	3/8	1.65	0.59	0.86	0.96	15/16	1	1.25
FCI-5812	5/8	1/2	1.84	0.78	0.86	0.96	1-1/16	1	1.44
FCI-3412	3/4	1/2	1.84	0.78	0.86	0.96	1-1/16	1-1/8	1.44
FCI-3434	3/4	3/4	1.9	0.81	0.86	0.96	1-5/16	1-1/8	1.50
FCI-0134	1	3/4	2.1	0.81	1.04	1.23	1-3/8	1-1/2	1.62
FCI-0101	1	1	2.45	1.00	1.04	1.23	1-5/8	1-1/2	1.97

Matric Range

Dimensions in mm

PART. NO.	T O/D	P	A LENGTH	B	C	D	F BODY	G NUT	H BODY
FCM-0418	4	1/8"	30.16	10	16.67	14.29	14	12	23.02
FCM-0618	6	1/8"	31.75	10	18.26	15.88	14	14	23.81
FCM-0614	6	1/4"	36.51	14	18.26	15.88	19	14	28.58
FCM-0638	6	3/8"	38.07	14	18.26	15.88	22	14	30.16
FCM-0612	6	1/2"	43.66	19	18.26	15.88	27	14	35.72
FCM-0818	8	1/8"	32.54	10	19.05	16.67	15	17	24.61
FCM-0814	8	1/4"	37.31	14	19.05	16.67	19	17	29.37
FCM-1018	10	1/8"	33.34	10	19.84	17.46	18	19	25.40
FCM-1014	10	1/4"	38.10	14	19.84	17.46	19	19	30.16
FCM-1038	10	3/8"	39.69	14	19.84	17.46	22	19	31.75
FCM-1012	10	1/2"	44.45	19	19.84	17.46	27	19	36.51
FCM-1214	12	1/4"	42.07	14	22.23	23.02	22	22	31.75
FCM-1238	12	3/8"	42.07	14	22.23	23.02	22	22	31.75
FCM-1212	12	1/2"	46.83	19	22.23	23.02	27	22	36.51
FCM-1612	16	1/2"	46.83	19	22.23	24.61	27	27	36.51
FCM-1812	18	1/2"	46.83	19	22.23	24.61	27	30	36.51
FCM-1834	18	3/4"	48.42	19	22.23	24.61	35	30	38.10
FCM-2012	20	1/2"	46.83	19	22.23	24.61	30	30	36.51
FCM-2034	20	3/4"	48.42	19	22.23	24.61	35	30	38.10
FCM-2534	25	3/4"	53.18	19	26.19	31.75	35	41	41.28
FCM-2501	25	1"	61.91	24	26.19	30.96	41	41	50.01

Equal Union



Imperial Range

Dimensions in Inches

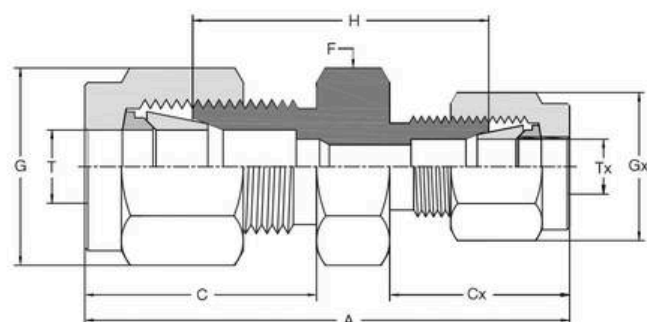
PART NO.	T O/D	A LENGTH	C	D	F BODY	G NUT	H BODY
EUI-16	1/16	1.00	0.44	0.34	5/16	5/16	0.689
EUI-18	1/8	1.44	0.63	0.53	7/16	7/16	0.881
EUI-36	3/16	1.47	0.63	0.54	7/16	1/2	0.945
EUI-14	1/4	1.61	0.70	0.60	1/2	9/16	1.043
EUI-56	5/16	1.68	0.73	0.64	9/16	5/8	1.102
EUI-38	3/8	1.77	0.76	0.66	5/8	11/16	1.189
EUI-12	1/2	2.02	0.86	0.90	13/16	7/8	1.220
EUI-58	5/8	2.05	0.86	0.96	15/16	1	1.26
EUI-34	3/4	2.11	0.86	0.96	1-1/16	1-1/8	1.319
EUI-01	1	2.55	1.04	1.23	1-3/8	1-1/2	1.594

Metric Range

Dimensions in mm

PART NO.	T O/D	A LENGTH	C	D	F BODY	G NUT	H BODY
EUM-04	4	38.50	16.67	14.29	12	12	24.21
EUM-06	6	42.07	18.26	15.88	14	14	26.19
EUM-08	8	44.04	19.05	16.67	15	17	28.18
EUM-10	10	46.04	19.84	17.46	18	19	31
EUM-12	12	51.59	22.23	23.01	22	22	31
EUM-16	16	52.39	22.23	24.61	24	25	32
EUM-18	18	53.98	22.23	24.61	27	30	33.34
EUM-20	20	55	22.23	24.61	30	32	34
EUM-25	25	64.29	26.19	30.96	35	38	40.48

Unequal Union



Imperial Range

Dimensions in Inches

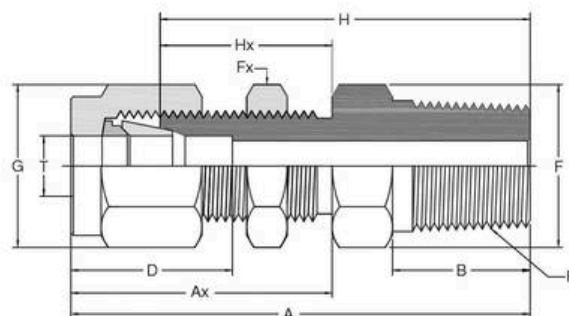
PART NO.	T O/D	TX O/D	A LENGTH	C	CX	F BODY	G NUT	GX NUT	H BODY
UUI-1816	1/8	1/16	1.25	0.63	0.44	17/16	7/16	5/16	0.81
UUI-3618	3/16	1/8	1.48	0.66	0.63	17/16	1/2	7/16	0.92
UUI-1416	1/4	1/16	1.37	0.72	0.44	1/2	9/16	5/16	0.91
UUI-1418	1/4	1/8	1.56	0.72	0.63	1/2	9/16	7/16	0.97
UUI-1436	1/4	3/16	1.55	0.70	0.63	1/2	9/16	1/2	1.00
UUI-5614	5/16	1/4	1.66	0.73	0.70	9/16	5/8	9/16	1.08
UUI-3814	3/8	1/4	1.70	0.76	0.70	5/8	11/16	9/16	1.12
UUI-3856	3/8	5/16	1.74	0.76	0.73	5/8	11/16	5/8	1.16
UUI-1214	1/2	1/4	1.85	0.86	0.70	13/16	7/8	9/16	1.16
UUI-1238	1/2	3/8	1.91	0.86	0.76	13/16	7/8	11/16	1.22
UUI-5638	5/8	3/8	1.94	0.86	0.76	15/16	1	11/16	1.25
UUI-5812	5/8	1/2	2.05	0.86	0.86	15/16	1	7/8	1.25
UUI-3412	3/4	1/2	2.11	0.86	0.86	1-1/16	1-1/8	7/8	1.31
UUI-3458	3/4	5/8	2.11	0.86	0.86	1-1/16	1-1/8	1	1.31
UUI-0134	1	3/4	2.47	1.04	0.86	1-3/8	1-1/2	1-1/8	1.59

Metric Range

Dimensions in mm

PART NO.	T O/D	TX O/D	A LENGTH	C	CX	F BODY	G NUT	GX NUT	H BODY
UUM-0604	6	4	40.48	19.97	19.05	14	14	12	25.40
UUM-0804	8	6	43.26	19.05	19.97	15	17	14	27.38
UUM-1006	10	6	44.45	19.84	19.97	18	19	14	28.58
UUM-1008	10	8	45.24	19.84	19.05	18	19	17	29.37
UUM-1206	12	6	47.63	22.23	19.97	22	22	14	29.37
UUM-1210	12	10	49.21	22.23	19.84	22	22	19	30.96
UUM-1610	16	10	50.01	22.23	19.84	24	27	19	31.75
UUM-1612	16	12	52.39	22.23	22.23	24	27	22	31.75
UUM-2012	20	12	53.98	22.23	22.23	30	30	22	33.34
UUM-2016	20	16	53.98	22.23	22.23	30	30	27	33.34
UUM-2520	25	20	60.32	26.19	22.23	35	41	30	38.10

Bulkhead Male Connector



Imperial Range

Dimensions in Inches

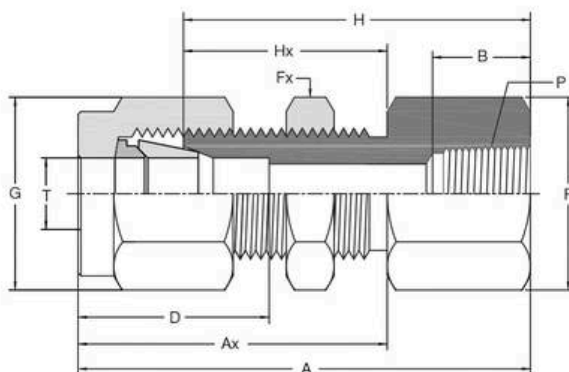
PART NO.	T O/D	P	B	D	F BODY	G NUT	FX	H	AX	HX	A	PANEL HOLE	MAX PANEL
BMCI-1418	1/4	1/8	0.38	0.60	5/8	9/16	5/8	1.66	1.32	1.03	1.95	29/64	0.40
BMCI-1414	1/4	1/4	0.56	0.60	5/8	9/16	5/8	1.81	1.32	1.03	2.10	29/64	0.40
BMCI-3814	3/8	1/4	0.56	0.66	3/4	11/16	3/4	1.97	1.45	1.16	2.26	37/64	0.44
BMCI-1212	1/2	1/2	0.75	0.90	15/16	7/8	15/16	2.28	1.65	1.25	2.68	49/64	0.50

Metric Range

Dimensions in mm

PART NO.	T O/D	P	B	D	F BODY	G NUT	FX	H	AX	HX	A	PANEL HOLE	MAX PANEL
BMCM-0618	6	1/8	10	15.88	16	14	16	42.2	33.5	26.2	49.5	11.5	10.2
BMCM-0614	6	1/4	14	15.88	16	14	16	46.2	33.5	26.2	53.4	11.5	10.2
BMCM-1212	12	1/2	19	23.02	24	22	24	58.6	41.9	31.8	68.0	19.5	12.7

Bulkhead Female Connector



Imperial Range

Dimensions in Inches

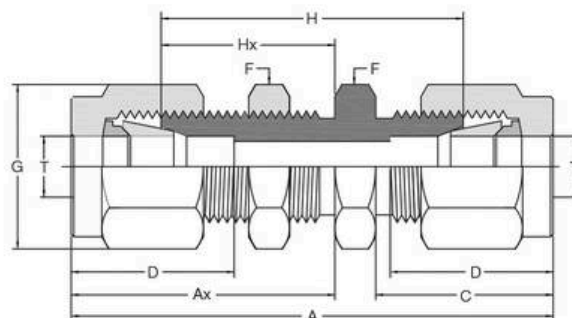
PART NO.	T O/D	P	B	D	F BODY	G NUT	FX	H	AX	HX	A	PANEL HOLE	MAX PANEL
BFCI-1418	1/4	1/8	0.39	0.60	5/8	9/16	5/8	1.56	1.32	1.32	1.85	0.453	0.40
BFCI-1414	1/4	1/4	0.59	0.60	3/4	9/16	5/8	1.75	1.32	1.32	2.04	0.453	0.40
BFCI-3814	3/8	1/4	0.59	0.66	3/4	11/16	3/4	1.88	1.45	1.45	2.17	0.579	0.44
BFCI-1238	1/2	3/8	0.59	0.90	15/16	7/8	15/16	2.03	1.65	1.65	2.43	0.768	0.50
BFCI-1212	1/2	1/2	0.78	0.90	1 1/16	7/8	1 1/16	2.22	1.65	1.65	2.62	0.768	0.50

Metric Range

Dimensions in mm

PART NO.	T O/D	P	B	D	F BODY	G NUT	FX	H	AX	HX	A	PANEL HOLE	MAX PANEL
BFCM-0614	6	1/4	14	15.88	19	14	16	44.5	33.5	26.2	51.9	11.5	10.2
BFCM-1212	12	1/2	19	23.02	27	22	24	56.4	41.9	31.8	66.5	19.5	12.7

Bulkhead Union



Imperial Range

Dimensions in Inches

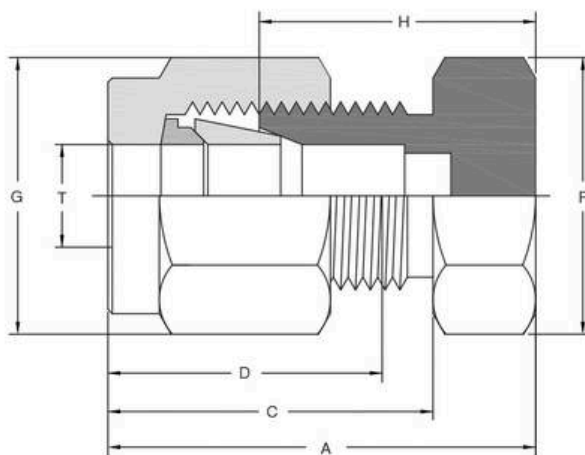
PART NO.	T O/D	A LENGTH	C	D	F BODY	G NUT	H
BUI-16	1/16	1.24	0.43	0.34	5/16	5/16	0.94
BUI-18	1/8	2.02	0.60	0.50	1/2	7/16	1.50
BUI-36	3/16	2.11	0.63	0.54	9/16	1/2	1.59
BUI-14	1/4	2.27	0.70	0.60	5/8	9/16	1.69
BUI-56	5/16	2.39	0.73	0.64	11/16	5/8	1.81
BUI-38	3/8	2.45	0.76	0.66	3/4	11/16	1.87
BUI-12	1/2	2.80	0.86	0.90	15/16	7/8	2.00
BUI-58	5/8	2.86	0.86	0.96	1-1/16	1	2.06
BUI-34	3/4	3.11	0.86	0.96	1-3/16	1-1/8	2.31
BUI-01	1	3.77	1.04	1.23	1-5/8	1-1/2	2.81

Imperial Range

Dimensions in mm

PART NO.	T O/D	A LENGTH	C	D	F BODY	G NUT	H BODY	AX	HX
BUM-03	3	51.3	15.3	12.9	14	12	38.1	31.0	24.6
BUM-04	4	53.6	16.1	13.7	14	12	40.4	32.0	25.4
BUM-06	6	57.7	17.7	15.3	16	14	42.9	33.6	26.2
BUM-08	8	61.0	18.6	16.2	18	16	46.0	36.1	28.6
BUM-10	10	63.7	19.5	17.2	22	19	48.5	37.0	29.4
BUM-12	12	71.0	22.0	22.8	24	22	50.8	41.9	31.8
BUM-16	16	72.5	22.0	24.0	27	25	52.3	42.6	32.5
BUM-20	20	84.5	22.0	26.0	35	32	64.3	53.0	42.9

Cap



Imperial Range

Dimensions in Inches

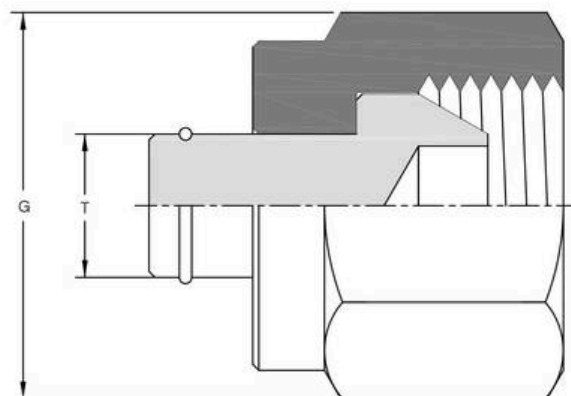
PART NO.	T O/D	A LENGTH	C	D	F BODY	G NUT	H BODY
CI-36	3/16	0.84	0.63	0.54	7/16	1/2	0.58
CI-14	1/4	0.92	0.70	0.60	1/2	9/16	0.63
CI-56	5/16	0.96	0.73	0.64	9/16	5/8	0.67
CI-38	3/8	1.01	0.76	0.66	5/8	1/16	0.72
CI-12	1/2	1.15	0.86	0.90	13/16	7/8	0.75
CI-58	5/8	1.18	0.86	0.96	15/16	1	0.78
CI-34	3/4	1.24	0.86	0.96	1-1/16	1-1/8	0.84
CI-01	1	1.51	1.04	1.23	1-3/8	1-1/2	1.03

Imperial Range

Dimensions in mm

PART NO.	T O/D	A LENGTH	C	D	F BODY	G NUT	H BODY
CM-04	4	21.3	16.1	13.70	12	12	14.70
CM-06	6	23.1	17.7	15.30	14	14	15.70
CM-08	8	24.5	18.6	16.20	15	16	17.00
CM-10	10	26.6	19.5	17.20	18	19	19.00
CM-12	12	29.1	22.0	22.00	22	22	19.00
CM-16	16	29.9	22.0	22.00	24	25	19.80
CM-18	18	31.4	22.0	22.00	27	30	21.30
CM-20	20	34.0	22.0	22.00	30	32	23.90
CM-25	25	38.5	26.5	26.50	35	38	26.20

Plug



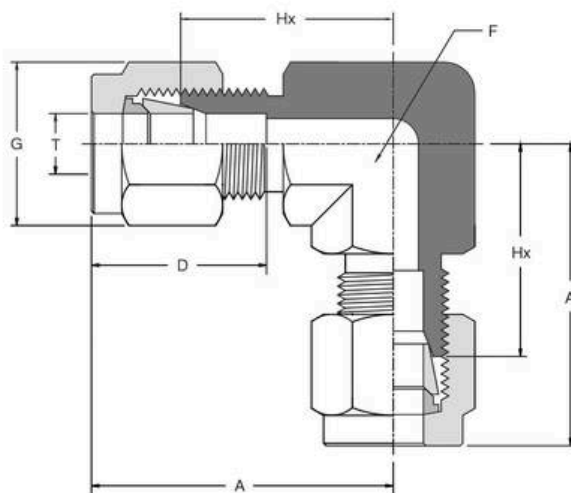
Imperial Range Dimensions in Inches

PART NO.	T O/D	G NUT
PI-16	1/16	5/16
PI-18	1/8	7/16
PI-36	3/16	1/2
PI-14	1/4	9/16
PI-56	5/16	5/8
PI-38	3/8	11/16
PI-12	1/2	7/8
PI-58	5/8	1
PI-34	3/4	1-1/8
PI-01	1	1-1/2

Matric Range Dimensions in mm

PART NO.	T O/D	G NUT
PM-04	4	12
PM-06	6	14
PM-08	8	16
PM-10	10	19
PM-12	12	22
PM-16	16	25
PM-18	18	30
PM-20	20	32
PM-25	25	38

Union Elbow



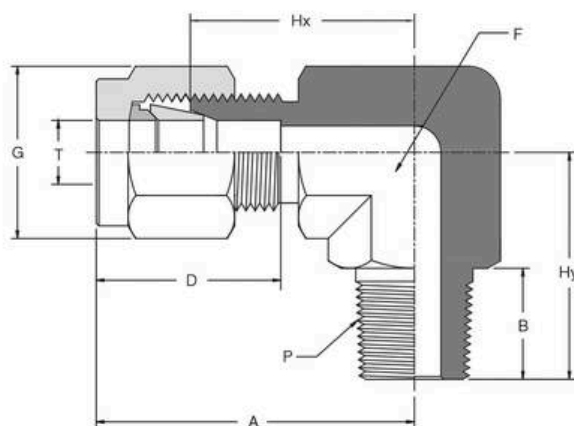
Imperial Range Dimensions in Inches

PART. NO	T O/D	A LENGTH	D	F BODY	G NUT	HX BODY
UEI-16	1/16	17.8	0.34	3/8	5/16	0.56
UEI-18	1/8	22.4	0.53	3/8	7/16	0.72
UEI-36	3/16	26	0.54	1/2	1/2	0.75
UEI-14	1/4	27	0.60	1/2	9/16	0.78
UEI-56	5/16	29	0.64	9/16	5/8	0.88
UEI-38	3/8	31	0.66	5/8	1/16	0.91
UEI-12	1/2	36	0.90	13/16	7/8	0.98
UEI-58	5/8	38	0.96	15/16	1	1.06
UEI-34	3/4	40	0.96	1-1/16	1-1/8	1.11
UEI-01	1	49	1.23	1-3/8	1-1/2	1.31

Matric Range Dimensions in mm

PART. NO	T O/D	A LENGTH	D	F BODY	G NUT	HX BODY
UEM-04	4	24.61	14.29	12.7	12	19.05
UEM-06	6	27	15.88	12.7	14	19.81
UEM-08	8	29	16.67	14.3	17	22.23
UEM-10	10	31	17.46	17.5	19	17.5
UEM-12	12	36	23.02	20.6	22	24.89
UEM-16	16	38	24.61	25.4	27	26.19
UEM-18	18	40	24.61	27	30	29.37
UEM-20	20	44	24.61	34.92	30	29.37
UEM-25	25	50	30.96	34.92	41	33.34

Male Elbow



Imperial Range

Dimensions in Inches

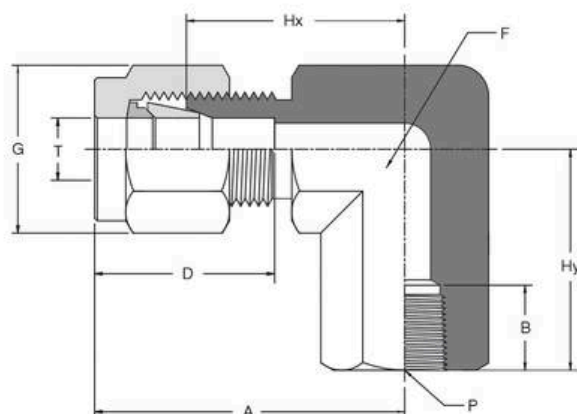
PART. NO	T O/D	P	A LENGTH	B	HY	D	F BODY	G NUT	HX BODY
MEI-1616	1/16	1/16	3/4	0.38	0.690	0.34	7/16	5/16	5/8
MEI-1618	1/16	1/8	3/4	0.38	0.708	0.34	7/16	5/16	5/8
MEI-1816	1/8	1/16	1	0.38	0.748	0.51	7/16	7/16	0.708
MEI-1818	1/8	1/8	0.945	0.38	0.748	0.53	1/2	7/16	0.708
MEI-1814	1/8	1/4	0.945	0.56	0.945	0.53	9/16	7/16	0.945
MEI-3618	3/16	1/8	1	0.38	0.768	0.54	1/2	1/2	3/4
MEI-1418	1/4	1/8	1.063	0.38	0.768	0.60	1/2	9/16	0.787
MEI-1414	1/4	1/4	1.063	0.56	0.945	0.60	9/16	9/16	0.945
MEI-1438	1/4	3/8	1.181	0.56	1.024	0.60	11/16	9/16	0.866
MEI-1412	1/4	1/2	1.260	0.75	1.299	0.60	7/8	9/16	0.945
MEI-5618	5/16	1/8	1.181	0.38	0.787	0.64	9/16	5/8	0.866
MEI-5614	5/16	1/4	1.181	0.56	1	0.64	9/16	5/8	0.866
MEI-3818	3/8	1/8	1.220	0.38	0.787	0.66	5/8	1/16	0.945
MEI-3814	3/8	1/4	1.220	0.56	1	0.66	5/8	11/16	0.945
MEI-3838	3/8	3/8	1.240	0.56	1.024	0.66	11/16	11/16	0.945
MEI-3812	3/8	1/2	1.338	0.75	1.299	0.66	7/8	11/16	1
MEI-1214	1/2	1/4	1.417	0.56	1.102	0.91	13/16	7/8	1
MEI-1238	1/2	3/8	1.417	0.56	1.102	0.91	13/16	7/8	1
MEI-1212	1/2	1/2	1.456	0.75	1.299	0.91	7/8	7/8	1
MEI-5838	5/8	3/8	1.496	0.56	1.181	0.96	15/16	1	1.063
MEI-5812	5/8	1/2	1.496	0.75	1.338	0.96	15/16	1	1.063
MEI-3412	3/4	1/2	1.535	0.75	1.417	0.96	1-1/16	1-1/8	1.141
MEI-3434	3/4	3/4	1.535	0.75	1.417	0.96	1-1/16	1-1/8	1.141
MEI-0134	1	3/4	1.929	0.94	1.654	1.23	1-3/8	1-1/2	1.338
MEI-0101	1	1	1.929	0.94	1.850	1.23	1-3/8	1-1/2	1.338

Matric Range

Dimensions in mm

PART. NO	T O/D	P	A LENGTH	B	HY	D	F BODY	G NUT	HX BODY
MEM-0418	4	1/8"	24.61	19.53	19.30	14.29	12.7	12	19.05
MEM-0618	6	1/8"	26.19	9.53	19.84	15.88	12.7	14	19.80
MEM-0614	6	1/4"	26.19	14.29	23.81	15.88	14.3	14	19.80
MEM-0638	6	3/8"	27.60	14.29	23.58	15.88	17.5	14	21.33
MEM-0612	6	1/2"	29.40	19.05	33.24	15.88	22	14	23.11
MEM-0818	8	1/8"	28.58	9.53	20.64	16.67	14.3	17	22.22
MEM-0814	8	1/4"	28.58	14.29	25.40	16.67	14.3	17	22.22
MEM-0838	8	3/8"	28.58	14.24	25.40	16.67	17.5	17	22.22
MEM-0812	8	1/2"	31.75	19.05	32.00	16.67	22	17	24.00
MEM-1018	10	1/8"	28.90	9.53	22.23	17.46	17.5	19	23.11
MEM-1014	10	1/4"	28.90	14.29	25.40	17.46	17.5	19	23.11
MEM-1038	10	3/8"	28.90	14.29	28.58	17.46	17.5	19	23.11
MEM-1012	10	1/2"	30.60	19.05	31.75	17.46	22	19	24.89
MEM-1214	12	1/4"	35.80	14.29	28.58	23.02	20.6	22	24.89
MEM-1238	12	3/8"	35.80	14.29	28.58	23.02	20.6	22	24.89
MEM-1212	12	1/2"	35.80	19.05	33.34	23.02	22	22	24.89
MEM-1638	16	3/8"	36.51	14.29	31.75	24.61	24	27	26.19
MEM-1612	16	1/2"	36.51	19.05	34.93	24.61	24	27	26.19
MEM-1812	18	1/2"	39.69	19.05	38.10	24.61	27	30	29.37
MEM-1834	18	3/4"	39.69	19.05	38.10	24.61	27	30	29.37
MEM-2012	20	1/2"	39.69	19.05	38.10	24.61	34.9	30	29.37
MEM-2034	20	3/4"	39.69	19.05	38.10	24.61	34.9	30	29.37
MEM-2534	25	3/4"	45.24	19.05	38.10	26.19	34.9	41	33.34
MEM-2501	25	1"	45.24	23.81	46.83	30.96	34.9	41	33.34

Female Elbow



Imperial Range

Dimensions in Inches

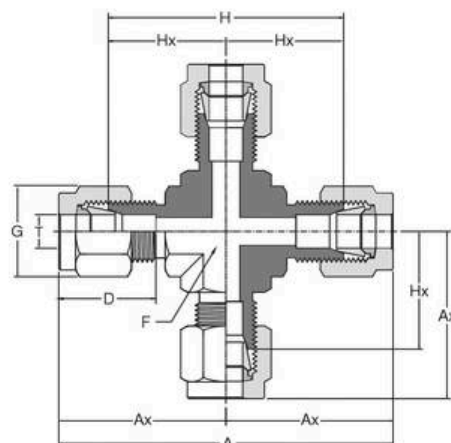
PART. NO	T O/D	P	A LENGTH	B	HY	D	F BODY	G NUT	HX BODY
FEI-1818	1/8	1/8	0.95	0.39	0.15	0.50	9/16	7/16	0.708
FEI-1814	1/8	1/4	1.07	0.59	0.88	0.50	3/4	7/16	0.825
FEI-3618	3/16	1/8	0.98	0.39	0.75	0.54	9/16	1/4	0.703
FEI-1418	1/4	1/8	1.04	0.39	0.75	0.60	9/16	9/16	0.749
FEI-1414	1/4	1/4	1.17	0.59	0.88	0.60	9/16	9/16	0.886
FEI-5618	5/16	1/8	1.17	0.39	0.75	0.64	3/4	5/8	0.866
FEI-5614	5/16	1/4	1.20	0.59	0.88	0.64	3/4	5/8	0.905
FEI-3818	3/8	1/8	1.20	0.39	0.75	0.66	5/8	11/16	0.905
FEI-3814	3/8	1/4	1.23	0.59	0.88	0.66	3/4	11/16	0.945
FEI-3838	3/8	3/8	1.32	0.59	0.88	0.66	7/8	11/16	1.027
FEI-3812	3/8	1/2	1.41	0.78	1.12	0.66	1-1/16	11/16	1.102
FEI-1214	1/2	1/4	1.38	0.59	0.88	0.90	13/16	7/8	1.000
FEI-1238	1/2	3/8	1.43	0.59	0.88	0.90	7/8	7/8	1.023
FEI-1212	1/2	1/2	1.52	0.78	1.12	0.90	1-1/16	7/8	1.142
FEI-5838	5/8	3/8	1.46	0.59	0.88	0.96	15/16	1	1.102
FEI-5812	5/8	1/2	1.52	0.78	1.12	0.96	1-1/16	1	1.142
FEI-3412	3/4	1/2	1.51	0.78	1.12	0.96	1-1/16	1-1/8	1.102
FEI-3434	3/4	3/4	1.65	0.81	1.25	0.96	1-3/8	1-1/8	1.260

Metric Range

Dimensions in mm

PART. NO	T O/D	P	A LENGTH	B	HY	D	F BODY	G NUT	HX BODY
FEM-0618	6	1/8	27.0	10	15.3	15.88	14.0	14	27.0
FEM-0614	6	1/4	29.8	14	15.3	15.88	19.0	14	29.8
FEM-0814	8	1/4	30.6	14	15.3	16.67	19.0	17	30.6
FEM-1018	10	1/8	31.5	10	15.3	17.46	17.0	19	31.5
FEM-1014	10	1/4	33.5	14	15.3	17.46	19.0	19	33.5
FEM-1214	12	1/4	36.0	14	15.3	23.02	20.5	22	36.0
FEM-1212	12	1/2	38.8	19	22.8	23.02	27.0	22	38.8
FEM-1612	16	1/2	39.5	19	22.8	23.02	27.0	27	39.5

Equal Union Cross



Imperial Range

Dimensions in Inches

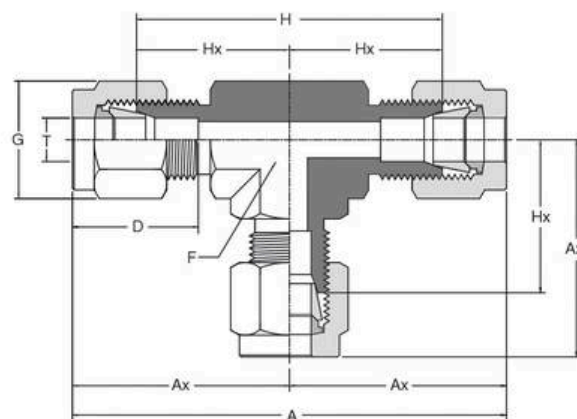
PART. NO	T O/D	AX LENGTH	D	F BODY	G NUT	HX BODY	H	A
UCI-18	1/8	0.99	0.53	3/8"	7/16	0.72	1.44	1.98
UCI-36	3/16	1.01	0.54	1/2"	1/2	0.75	1.50	2.02
UCI-14	1/4	1.07	0.60	1/2"	9/16	0.78	1.56	2.14
UCI-56	5/16	1.17	0.64	5/8	5/8	0.88	1.76	2.34
UCI-38	3/8	1.2	0.66	5/8	1/16	0.91	1.82	2.40
UCI-12	1/2	1.38	0.90	13/16"	7/8	0.98	1.96	2.76
UCI-58	5/8	1.46	0.96	15/16"	1	1.06	2.12	2.92

Metric Range

Dimensions in mm

PART. NO	T O/D	AX LENGTH	D	F Body	G NUT	Hx Body	H	A
UCM-04	4	26.15	14.29	12.7	12	19.00	38.00	52.30
UCM-06	6	27.20	15.88	12.7	14	19.80	39.60	54.40
UCM-08	8	29.70	16.67	14.3	17	22.20	44.40	59.40
UCM-10	10	30.80	17.46	17.5	19	23.10	46.20	61.60
UCM-12	12	35.00	23.02	20.6	22	24.90	49.80	70.00
UCM-16	16	36.39	24.61	24	27	26.19	52.39	72.78

Union Tee



Imperial Range

Dimensions in Inches

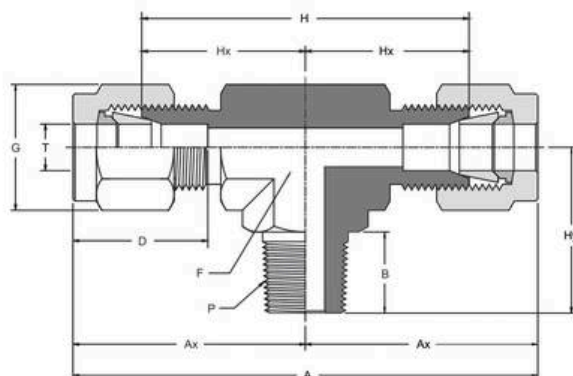
PART. NO	T O/D	AX LENGTH	D	F BODY	G NUT	HX BODY	H	A
UTI-16	1/16	0.72	0.34	3/8	5/16	0.56	1.12	1.44
UTI-18	1/8	0.99	0.50	3/8	7/16	0.72	1.44	1.98
UTI-36	3/16	1.01	0.54	1/2	1/2	0.75	1.50	2.02
UTI-14	1/4	1.07	0.60	1/2	9/16	0.78	1.56	2.25
UTI-56	5/16	1.17	0.64	9/16	5/8	0.88	1.76	2.34
UTI-38	3/8	1.20	0.66	5/8	11/16	0.91	1.82	2.40
UTI-12	1/2	1.30	0.90	13/16	7/8	0.98	1.96	2.76
UTI-58	5/8	1.46	0.96	15/16	1	1.06	2.12	2.92
UTI-34	3/4	1.51	0.96	1-1/16	1-1/8	1.11	2.22	3.02
UTI-01	1	1.79	1.23	1-3/8	1-1/2	1.31	2.62	3.58

Metric Range

Dimensions in mm

PART. NO	T O/D	AX LENGTH	D	F BODY	G NUT	Hx BODY	H	A
UTM-04	4	26.15	14.29	12.7	12	19.00	38.00	52.30
UTM-06	6	27.20	15.88	12.7	14	19.80	39.60	54.40
UTM-08	8	29.70	16.67	14.5	17	22.20	44.40	59.40
UTM-10	10	30.80	17.46	17.5	19	23.10	46.20	61.60
UTM-12	12	35.00	23.02	20.6	22	24.90	49.80	70.00
UTM-16	16	36.39	24.61	25.4	27	26.19	52.39	72.78
UTM-18	18	39.37	24.61	27	30	29.37	58.74	78.74
UTM-20	20	39.37	24.61	34.92	30	29.37	58.74	78.74
UTM-25	25	45.24	30.96	34.92	41	33.54	67.08	90.48

Male Branch Tee



Imperial Range

Dimensions in Inches

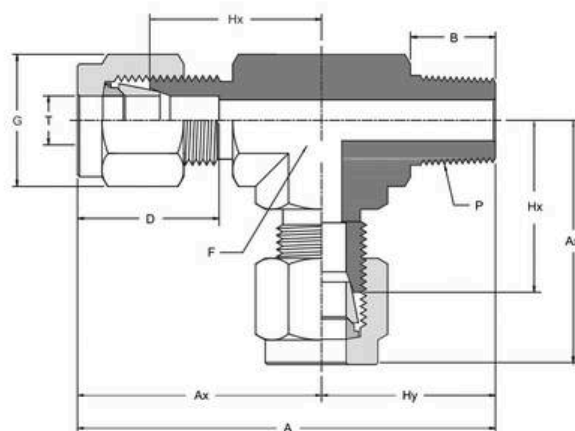
PART. NO.	T O/D	P	Ax Length	B	HY	F	G NUT	HX BODY	H	A
BTI-1818	1/8	1/8	0.99	0.38	0.76	1/2	7/16	0.72	1.44	1.98
BTI-1814	1/8	1/4	0.99	0.56	0.94	9/16	7/16	0.72	1.44	1.98
BTI-3618	3/16	1/8	1.01	0.38	0.76	1/2	1/2	0.75	1.50	2.02
BTI-1418	1/4	1/8	1.07	0.38	0.76	1/2	9/16	0.78	1.56	2.14
BTI-1414	1/4	1/4	1.07	0.56	0.94	9/16	9/16	0.78	1.56	2.14
BTI-1438	1/4	3/8	1.13	0.56	1.00	11/16	9/16	0.84	1.68	2.26
BTI-5618	5/16	1/8	1.16	0.38	0.82	9/16	5/8	0.87	1.74	2.32
BTI-3818	3/8	1/8	1.21	0.38	0.82	5/8	11/16	0.91	1.82	2.42
BTI-3814	3/8	1/4	1.21	0.56	1.00	5/8	11/16	0.91	1.82	2.42
BTI-3838	3/8	3/8	1.21	0.56	1.00	11/16	11/16	0.91	1.82	2.42
BTI-3812	3/8	1/2	1.21	0.75	1.26	7/8	11/16	0.91	1.82	2.42
BTI-1214	1/2	1/4	1.38	0.56	1.08	13/16	7/8	0.98	1.96	2.76
BTI-1238	1/2	3/8	1.38	0.56	1.08	13/16	7/8	0.98	1.96	2.76
BTI-1212	1/2	1/2	1.38	0.75	1.26	7/8	7/8	0.98	1.96	2.76
BTI-5812	5/8	1/2	1.46	0.75	1.38	15/16	1	1.06	2.12	2.92

Metric Range

Dimensions in mm

PART. NO.	T O/D	P	AX LENGTH	B	HY	F	G NUT	HX BODY	H	A
BTM-0416	4	1/16	26.15	9.53	19.30	12.7	12	19.05	38.10	52.30
BTM-0418	4	1/8	26.15	9.53	19.53	12.7	12	19.05	38.10	52.30
BTM-0414	4	1/4	26.15	14.29	23.90	14	12	19.05	38.10	52.30
BTM-0618	6	1/8	27.21	9.53	19.30	12.7	14	19.80	39.60	54.42
BTM-0614	6	1/4	27.21	14.29	23.90	14	14	19.80	39.60	54.42
BTM-0638	6	3/8	29.00	14.29	25.40	17.5	14	21.33	42.66	57.46
BTM-0612	6	1/2	30.51	19.05	32.00	22	14	23.10	46.20	61.00
BTM-0818	8	1/8	29.70	9.53	20.80	14	17	22.22	44.44	59.40
BTM-0814	8	1/4	29.70	14.29	25.40	14	17	22.22	44.44	59.40
BTM-0838	8	3/8	29.70	14.29	25.40	17.5	17	22.22	44.44	59.40
BTM-0812	8	1/2	31.50	19.05	32.00	22	17	24.00	48.00	63.00
BTM-1018	10	1/8	30.80	9.53	20.80	17.5	19	23.10	46.20	61.60
BTM-1014	10	1/4	30.80	14.29	25.40	17.5	19	23.10	46.20	61.60
BTM-1038	10	3/8	30.80	14.29	25.40	17.5	19	23.10	46.20	61.60
BTM-1012	10	1/2	32.59	19.05	32.00	22	19	24.89	49.78	65.18
BTM-1214	12	1/4	34.99	14.29	27.40	20.6	22	24.89	49.78	69.98
BTM-1238	12	3/8	32.59	14.29	27.40	20.6	22	24.89	49.78	65.18
BTM-1212	12	1/2	32.59	19.05	32.00	22	22	24.89	49.78	65.18
BTM-1612	16	1/2	36.39	19.05	34.90	24	27	26.19	52.39	72.78

Male Run Tee



Imperial Range

Dimensions in Inches

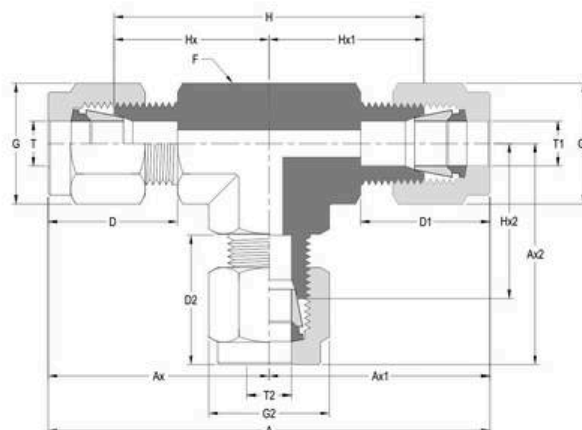
PART. NO	T O/D	P	AX LENGTH	B	HY	F	G NUT	HX BODY	H	A
RTI-1818	1/8	1/8	1.00	0.38	0.76	1/2	7/16	0.72	1.48	1.76
RTI-1814	1/8	1/4	1.00	0.56	0.94	9/16	7/16	0.72	1.66	1.94
RTI-3618	3/16	1/8	1.00	0.38	0.76	1/2	1/2	0.75	1.51	1.76
RTI-1418	1/4	1/8	1.07	0.38	0.76	1/2	9/16	0.78	1.54	1.83
RTI-1414	1/4	1/4	1.07	0.56	0.94	9/16	9/16	0.78	1.72	2.01
RTI-5618	5/16	1/8	1.16	0.38	0.82	9/16	5/8	0.87	1.70	1.98
RTI-3814	3/8	1/4	1.21	0.56	1.00	5/8	1/16	0.91	1.91	2.21
RTI-1238	1/2	3/8	1.38	0.56	1.08	13/16	7/8	0.98	2.06	2.46
RTI-1212	1/2	1/2	1.38	0.75	1.26	7/8	7/8	0.98	2.24	2.64
RTI-5812	5/8	1/2	1.47	0.75	1.38	15/16	1	1.06	2.40	2.85

Metric Range

Dimensions in mm

PART. NO	T O/D	P	AX LENGTH	B	HY	F	G NUT	HX BODY	H	A
RTM-0416	4	1/16	26.15	9.53	19.30	12	12.7	19.05	38.10	45.45
RTM-0418	4	1/8	26.15	9.53	19.30	12	12.7	19.05	38.10	45.45
RTM-0414	4	1/4	26.15	14.29	23.88	12	12.7	19.05	38.10	50.03
RTM-0618	6	1/8	27.21	9.53	19.30	12	14.7	19.81	39.62	46.51
RTM-0614	6	1/4	27.21	14.29	23.88	14	14	19.81	39.62	51.09
RTM-0638	6	3/8	28.74	14.29	25.40	19	17.5	21.34	42.68	54.14
RTM-0612	6	1/2	30.51	19.05	32.00	22	22	23.11	43.22	62.51
RTM-0818	8	1/8	29.73	9.53	20.83	19	14.3	22.23	44.46	50.56
RTM-0814	8	1/4	29.73	14.29	25.40	19	14.3	22.23	44.46	55.13
RTM-0838	8	3/8	29.73	14.29	25.40	19	7.5	22.23	44.46	55.13
RTM-0812	8	1/2	31.50	19.05	32.00	22	22	24.00	48.00	63.50
RTM-1018	10	1/8	30.81	9.53	20.83	19	17.5	23.11	46.22	51.64
RTM-1014	10	1/4	30.81	14.29	25.40	19	17.5	23.11	46.22	56.21
RTM-1038	10	3/8	30.81	14.29	25.40	19	17.5	23.11	46.22	56.21
RTM-1012	10	1/2	32.59	19.05	32.00	22	22	24.89	49.78	64.59
RTM-1214	12	1/4	34.99	14.29	27.43	22	20.6	24.89	49.78	62.42
RTM-1238	12	3/8	34.99	14.29	27.43	22	20.6	24.89	49.78	62.42
RTM-1212	12	1/2	34.99	19.05	32.00	22	22	24.89	49.78	66.99
RTM-1612	16	1/2	36.39	19.05	34.30	25	25.4	26.19	52.38	71.32

Unequal Union Tee



Imperial Range

Dimensions in Inches

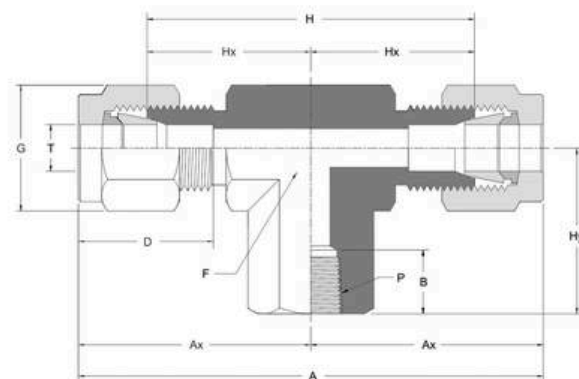
PART. NO	T O/D	T1 O/D	T2 O/D	AX LENGTH	Ax1 LENGTH	Ax2 LENGTH	D	D1	D2	F BODY	G NUT	G1 NUT	G2 NUT	HX BODY	Hx1 BODY	Hx2 BODY	H	A
UUTI-141418	1/4	1/4	1/8	1.06	1.06	0.96	0.70	0.70	0.60	1/2	9/16	9/16	7/16	0.77	0.77	0.70	1.54	2.12
UUTI-141438	1/4	1/4	3/8	1.15	1.15	1.20	0.70	0.70	0.76	5/8	9/16	9/16	11/16	0.86	0.86	0.91	1.73	2.31
UUTI-141212	1/4	1/2	1/2	1.25	1.42	1.42	0.70	0.86	0.86	13/16	9/16	7/8	7/8	0.96	1.02	1.02	1.98	2.67
UUTI-381438	3/8	1/4	3/8	1.20	1.15	1.20	0.76	0.70	0.76	5/8	11/16	9/16	11/16	0.91	0.86	0.91	1.77	2.35
UUTI-383814	3/8	3/8	1/4	1.20	1.20	1.15	0.76	0.76	0.70	5/8	11/16	11/16	9/16	0.91	0.91	0.86	1.82	2.40
UUTI-383812	3/8	3/8	1/2	1.31	1.31	1.42	0.76	0.76	0.86	13/16	11/16	11/16	7/8	1.02	1.02	1.02	2.04	2.62
UUTI-121438	1/2	1/4	3/8	1.42	1.31	1.31	0.86	0.70	0.76	13/16	7/8	9/16	11/16	1.02	1.02	1.02	2.04	2.73
UUTI-121412	1/2	1/4	1/2	1.42	1.25	1.42	0.86	0.70	0.86	13/16	7/8	9/16	7/8	1.02	0.96	1.02	1.98	2.67
UUTI-123838	1/2	3/8	3/8	1.42	1.31	1.31	0.86	0.76	0.76	13/16	7/8	11/16	11/16	1.02	1.02	1.02	2.04	2.73
UUTI-121214	1/2	1/2	1/4	1.42	1.42	1.25	0.86	0.86	0.70	13/16	7/8	7/8	9/16	1.02	1.02	0.96	2.04	2.84
UUTI-121238	1/2	1/2	3/8	1.42	1.42	1.31	0.86	0.86	0.76	13/16	7/8	7/8	11/16	1.02	1.02	1.02	2.04	2.84
UUTI-341234	3/4	1/2	3/4	1.57	1.57	1.57	0.86	0.86	0.86	1 - 1/16	1 - 1/8	7/8	1 - 1/8	1.17	1.17	1.17	2.34	3.14
UUTI-343414	3/4	3/4	1/4	1.57	1.57	1.40	0.86	0.86	0.70	1 - 1/16	1 - 1/8	1 - 1/8	9/16	1.17	1.17	1.11	2.34	3.14
UUTI-343438	3/4	3/4	3/8	1.57	1.57	1.40	0.86	0.86	0.76	1 - 1/16	1 - 1/8	1 - 1/8	11/16	1.17	1.17	1.17	2.34	3.14
UUTI-343412	3/4	3/4	1/2	1.57	1.57	1.51	0.86	0.86	0.86	1 - 1/16	1 - 1/8	1 - 1/8	7/8	1.17	1.17	1.17	2.34	3.14
UUTI-343401	3/4	3/4	1	1.93	1.93	1.80	0.86	0.86	1.04	1 - 3/8	1 - 1/8	1 - 1/8	1 - 1/2	1.36	1.36	1.45	2.71	3.86
UUTI-013434	1	3/4	3/4	1.93	1.80	1.80	1.04	0.86	0.86	1 - 3/8	1 - 1/2	1 - 1/8	1 - 1/8	1.45	1.40	1.40	2.85	3.73
UUTI-010114	1	1	1/4	1.93	1.93	1.59	1.04	1.04	0.70	1 - 3/8	1 - 1/2	1 - 1/2	9/16	1.45	1.45	1.30	2.90	3.86
UUTI-010138	1	1	3/8	1.93	1.93	1.65	1.04	1.04	0.76	1 - 3/8	1 - 1/2	1 - 1/2	11/16	1.45	1.45	1.36	2.90	3.86
UUTI-010112	1	1	1/2	1.93	1.93	1.76	1.04	1.04	0.86	1 - 3/8	1 - 1/2	1 - 1/2	7/8	1.45	1.45	1.36	2.90	3.86
UUTI-010134	1	1	3/4	1.93	1.93	1.80	1.04	1.04	0.86	1 - 3/8	1 - 1/2	1 - 1/2	1 - 1/8	1.45	1.45	1.36	2.90	3.86

Metric Range

Dimensions in Inches

PART. NO	T O/D	T1 O/D	T2 O/D	AX LENGTH	Ax1 LENGTH	Ax2 LENGTH	D	D1	D2	F BODY	G NUT	G1 NUT	G2 NUT	HX BODY	Hx1 BODY	Hx2 BODY	H	A
UUTM-080806	8	8	6	28.8	128.8	28.0	18.6	18.6	17.7	14.0	16.0	16.0	14.0	21.3	21.3	20.5	42.6	57.6
UUTM-101006	10	10	6	31.5	31.5	29.8	19.5	19.5	17.7	17.4	19.0	19.0	14.0	23.9	23.9	22.9	47.8	63.0
UUTM-101012	10	10	12	33.5	33.5	36.0	19.5	19.5	22.0	20.6	19.0	19.0	22.0	25.9	25.9	25.9	51.8	67.0
UUTM-120610	12	6	10	36.0	31.8	33.5	22.0	17.7	19.5	20.6	22.0	14.0	19.0	25.9	24.4	25.9	50.3	67.8
UUTM-120612	12	6	12	36.0	31.8	36.0	22.0	17.7	22.0	20.6	22.0	14.0	22.0	25.9	24.4	25.9	50.3	67.8
UUTM-121010	12	10	10	36.0	33.5	33.5	22.0	19.5	19.5	20.6	22.0	19.0	19.0	25.9	25.9	25.9	51.8	69.5
UUTM-121210	12	12	10	36.0	36.0	33.5	22.0	22.0	19.5	20.6	22.0	22.0	19.0	25.9	25.9	25.9	51.8	72.0
UUTM-121206	12	12	6	36.0	36.0	31.8	22.0	22.0	17.5	20.6	22.0	22.0	14.0	25.9	25.9	24.4	51.8	72.0
UUTM-161612	16	16	12	38.8	38.8	38.8	22.0	22.0	22.0	25.4	25.0	25.0	22.0	28.7	28.7	28.7	57.4	77.6
UUTM-181812	18	18	12	39.8	39.8	38.3	22.0	22.0	22.0	27.0	30.0	30.0	22.0	29.7	29.7	28.2	59.4	79.6
UUTM-201220	20	12	20	42.6	42.6	42.6	22.0	22.0	22.0	34.9	32.0	22.0	32.0	32.5	32.5	32.5	65.0	85.2
UUTM-202006	20	20	6	42.6	42.6	38.4	22.0	22.0	17.7	34.9	32.0	32.0	14.0	32.5	32.5	31.0	65.0	85.2
UUTM-202010	20	20	10	42.6	42.6	40.1	22.0	22.0	19.5	34.9	32.0	32.0	19.0	32.5	32.5	32.5	65.0	85.2
UUTM-202012	20	20	12	42.6	42.6	42.6	22.0	22.0	22.0	34.9	32.0	32.0	22.0	32.5	32.5	32.5	65.0	85.2
UUTM-202025	20	20	25	44.4	44.4	49.1	22.0	22.0	26.5	34.9	32.0	32.0	38.0	34.3	34.3	36.8	68.6	88.8
UUTM-222212	22	22	12	42.6	42.6	42.6	22.0	22.0	22.0	34.9	32.0	32.0	22.0	32.5	32.5	32.5	65.0	85.2
UUTM-252020	25	20	20	49.1	44.4	44.4	26.5	22.0	22.0	34.9	38.0	32.0	32.0	36.8	34.3	34.3	71.1	93.5
UUTM-252510	25	25	10	49.1	49.1	38.9	26.5	26.5	19.5	34.9	38.0	38.0	19.0	36.8	36.8	34.3	73.6	98.2
UUTM-252512	25	25	12	49.1	49.1	44.4	26.5	26.5	22.0	34.9	38.0	38.0	22.0	36.8	36.8	34.3	73.6	98.2
UUTM-252520	25	25	20	49.1	49.1	44.4	26.5	26.5	22.0	34.9	38.0	38.0	32.0	36.8	36.8	34.3	73.6	98.2

Female Branch Tee



Imperial Range

Dimensions in Inches

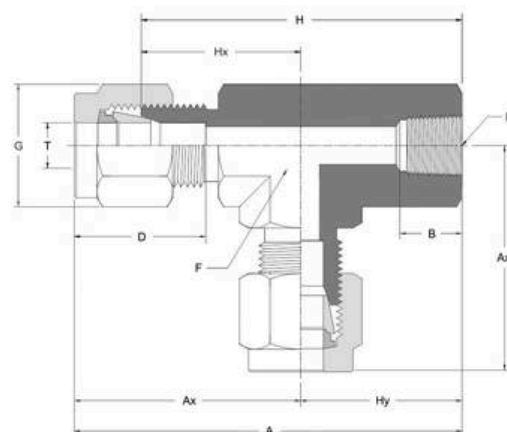
PART. NO	T O/D	P	AX LENGTH	B	HY	F	G NUT	HX BODY	H	A
FBTI-1818	1/8	1/8	0.97	0.38	0.75	1/2	7/16	0.72	0.70	11.94
FBTI-1418	1/4	1/8	1.06	0.38	0.75	1/2	9/16	0.78	0.77	2.12
FBTI-1414	1/4	1/4	1.17	0.56	0.88	3/4	9/16	0.78	0.88	2.34
FBTI-3814	3/8	1/4	1.23	0.56	0.88	3/4	11/16	0.91	0.94	2.46
FBTI-1214	1/2	1/4	1.42	0.56	0.88	13/16	7/8	0.98	1.02	2.84
FBTI-1238	1/2	3/8	1.42	0.56	0.88	7/8	7/8	0.98	1.02	2.84
FBTI-1212	1/2	1/2	1.53	0.75	1.12	1-1/16	7/8	0.98	1.13	3.06

Metric Range

Dimensions in mm

PART. NO	T O/D	P	AX LENGTH	B	HY	F	G NUT	HX BODY	H	A
FBTM-0618	6	1/8	26.95	10	19.0	14.0	14	19.6	39.2	53.9
FBTM-0614	6	1/4	29.75	14	22.4	19.0	14	22.4	44.8	59.5
FBTM-0818	8	1/8	29.85	10	19.0	15.0	17	22.4	44.8	59.7
FBTM-0814	8	1/4	30.60	14	22.4	19.0	17	22.4	44.8	61.2
FBTM-1014	10	1/4	33.50	14	22.4	20.6	19	25.9	51.8	67.0
FBTM-1214	12	1/4	36.00	14	22.4	20.6	22	25.9	51.8	72.0
FBTM-1238	12	3/8	36.00	14	22.4	22.0	22	25.9	51.8	72.0
FBTM-1212	12	1/2	38.85	19	28.5	27.0	22	28.7	57.4	77.7

Female Run Tee



Imperial Range

Dimensions in Inches

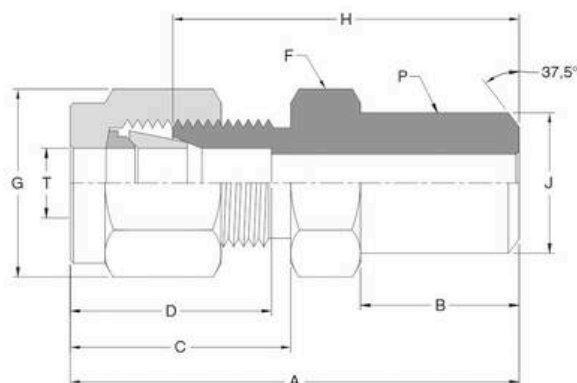
PART. NO	T O/D	P	AX LENGTH	B	HY	F	G NUT	HX BODY	H	A
FRTI-1818	1/8	1/8	1.00	0.38	0.75	1/2	7/16	0.72	1.45	1.75
FRTI-1418	1/4	1/8	1.07	0.38	0.81	1/2	9/16	0.78	1.52	1.88
FRTI-1414	1/4	1/4	1.07	0.56	0.88	3/4	9/16	0.78	1.76	1.95
FRTI-3814	3/8	1/4	1.21	0.56	0.88	3/4	11/16	0.91	1.82	2.09
FRTI-1214	1/2	1/4	1.38	0.56	0.88	13/16	7/8	0.98	1.90	2.26
FRTI-1238	1/2	3/8	1.38	0.56	0.88	7/8	7/8	0.98	1.90	2.26
FRTI-1212	1/2	1/2	1.38	0.75	1.12	1-1/16	7/8	0.98	2.26	2.50

Metric Range

Dimensions in mm

PART. NO	T O/D	P	AX LENGTH	B	HY	F	G NUT	HX BODY	H	A
FRTM-0618	6	1/8	27.2	10	19.0	14.0	14	19.81	38.6	46.2
FRTM-0614	6	1/4	27.2	14	22.4	19.0	14	19.81	44.8	49.6
FRTM-0818	8	1/8	29.7	10	19.0	15.0	17	22.23	41.4	48.2
FRTM-0814	8	1/4	29.7	14	22.4	19.0	17	22.23	44.8	52.1
FRTM-1014	10	1/4	30.8	14	22.4	20.6	19	23.11	48.3	53.2
FRTM-1214	12	1/4	35.0	14	22.4	20.6	22	24.89	48.3	57.4
FRTM-1238	12	3/8	35.0	14	22.4	22.0	22	24.89	48.3	57.4
FRTM-1212	12	1/2	35.0	19	28.5	27.0	22	24.89	57.2	63.5

Weld Connector



Imperial Range

Dimensions in Inches

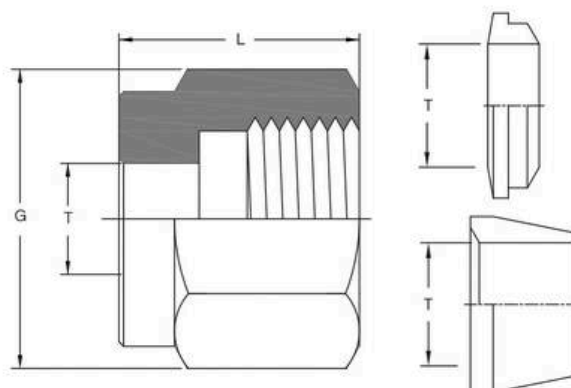
PART. NO	T O/D	P	AX LENGTH	B	C	D	F BODY	G NUT	H BODY	J
WCI-1818	1/8	1/8	1.20	0.38	0.63	0.53	7/16	7/16	0.91	0.405
WCI-1418	1/4	1/8	1.29	0.38	0.70	0.60	9/16	9/16	1.00	0.405
WCI-1414	1/4	1/4	1.46	0.56	0.72	0.60	5/8	9/16	1.19	0.540
WCI-3814	3/8	1/4	1.49	0.56	0.76	0.66	11/16	11/16	1.28	0.540
WCI-3838	3/8	3/8	1.60	0.56	0.76	0.66	7/8	11/16	1.28	0.675
WCI-3812	3/8	1/2	1.82	0.75	0.76	0.66	1-1/16	11/16	1.50	0.840
WCI-3834	3/8	3/4	1.88	0.75	0.76	0.66	13/16	11/16	1.53	1.050
WCI-1238	1/2	3/8	1.71	0.56	0.86	0.90	7/8	7/8	1.31	0.675
WCI-1212	1/2	1/2	1.93	0.75	0.86	0.90	1-1/16	7/8	1.50	0.840
WCI-1234	1/2	3/4	1.99	0.75	0.86	0.90	1-1/16	7/8	1.53	1.050
WCI-3434	3/4	3/4	1.99	0.75	0.86	0.96	1-3/8	1-1/8	1.59	1.050
WCI-0101	1	1	2.46	0.94	1.04	1.23		1-1/2	1.97	1.310

Metric Range

Dimensions in mm

PART. NO	T O/D	P	AX LENGTH	B	C	D	F BODY	G NUT	H BODY	J
WCM-0618	6	1/8	32.8	10	18.26	15.88	14	14	25.40	10.3
WCM-0614	6	1/4	37.9	14	18.26	15.88	14	14	30.16	13.7
WCM-0818	8	1/8	34.2	10	19.05	16.67	14	17	26.59	10.3
WCM-0814	8	1/4	38.7	14	19.05	16.67	14	17	31.35	13.7
WCM-0812	8	1/2	45.6	19	19.05	16.67	22	17	37.30	21.3
WCM-1014	10	1/4	40.9	14	19.84	17.46	17.5	19	32.54	13.7
WCM-1038	10	3/8	40.9	14	19.84	17.46	17.5	19	32.54	17.2
WCM-1012	10	1/2	46.5	19	19.84	17.46	22	19	38.10	21.3
WCM-1214	12	1/4	43.4	14	22.23	23.02	20.6	22	33.34	13.7
WCM-1238	12	3/8	43.4	14	22.23	23.02	20.6	22	33.34	17.2
WCM-1212	12	1/2	49.0	19	22.23	23.02	22	22	38.10	21.3
WCM-1612	16	1/2	49.0	19	22.23	24.61	24	27	38.89	21.3

Nut & Ferrules



Nut

Imperial Range Dimensions in Inches

PART. NO	T O/D	G NUT A/F	L
NI-16	1/16	5/16	0.31
NI-18	1/8	7/16	0.47
NI-36	3/16	1/2	0.50
NI-14	1/4	9/16	0.50
NI-56	5/16	5/8	0.53
NI-38	3/8	11/16	0.56
NI-12	1/2	7/8	0.69
NI-58	5/8	1	0.69
NI-34	3/4	1-1/8	0.69
NI-01	1	1-1/2	0.81

Nut

Metric Range Dimensions in mm

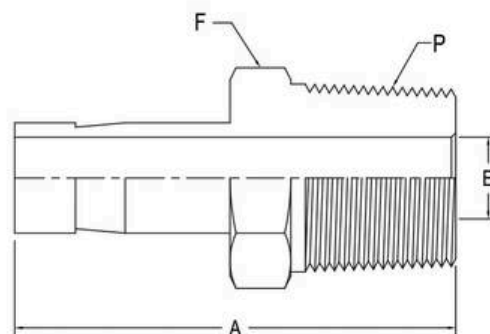
PART. NO	T O/D	G NUT A/F	L
NM-04	4	12	12.00
NM-06	6	14	12.70
NM-08	8	17	14.00
NM-10	10	19	14.00
NM-12	12	22	17.00
NM-16	16	27	17.46
NM-18	18	30	17.46
NM-20	20	30	17.46
NM-25	25	41	20.64

Ferrules

Imperial Range

PART. NO	T O/D
FI-16	1/16
FI-18	1/8
FI-36	3/16
FI-14	1/4
FI-56	5/16
FI-38	3/8
FI-12	1/2
FI-58	5/8
FI-34	3/4
FI-01	1

Tube Adapter (Male & Female)



Tube Male Adapter

Imperial Range

Dimensions in Inches

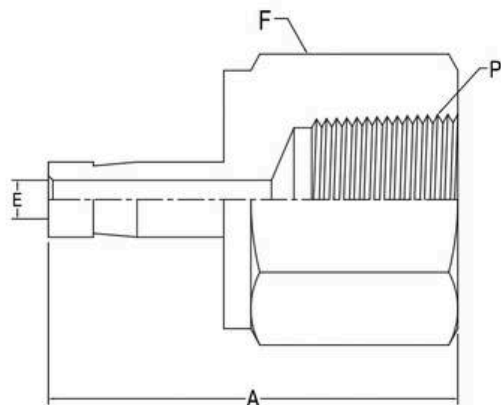
PART. NO	T O/D	P	A	E	F
MAI-1818	1/8"	1/8"	1.16	0.08	7/16"
MAI-1814	1/8"	1/4"	1.37	0.08	9/16"
MAI-1418	1/4"	1/8"	1.25	0.17	7/16"
MAI-1414	1/4"	1/4"	1.46	0.17	9/16"
MAI-1438	1/4"	3/8"	1.49	0.17	11/16"
MAI-1412	1/4"	1/2"	1.71	0.17	7/8"
MAI-3818	3/8"	1/8"	1.32	0.19	7/16"
MAI-3814	3/8"	1/4"	1.53	0.27	9/16"
MAI-3838	3/8"	3/8"	1.56	0.27	11/16"
MAI-3812	3/8"	1/2"	1.78	0.27	7/8"
MAI-1214	1/2"	1/4"	1.75	0.28	9/16"
MAI-1238	1/2"	3/8"	1.78	0.37	11/16"
MAI-1212	1/2"	1/2"	2.00	0.37	7/8"
MAI-3412	3/4"	1/2"	2.06	0.47	7/8"
MAI-3434	3/4"	3/4"	2.06	0.58	1 1/16"
MAI-0134	1"	3/4"	2.31	0.62	1 1/16"
MAI-0101	1"	1"	2.60	0.80	1 3/8"

Metric Range

Dimensions in Inches

PART. NO	T O/D	P	A	E	F
MAM-0618	6	1/8"	32.8	4.1	12
MAM-0614	6	1/4"	38.1	4.1	14
MAM-0814	8	1/4"	39.1	5.6	14
MAM-0838	8	3/8"	39.9	5.6	17.5
MAM-1014	10	1/4"	39.9	7.1	14
MAM-1038	10	3/8"	40.6	7.1	17.5
MAM-1012	10	1/2"	46.2	7.1	22
MAM-1214	12	1/4"	46.5	7.1	14
MAM-1212	12	1/2"	52.1	8.8	22

Tube Female Adapter



Imperial Range

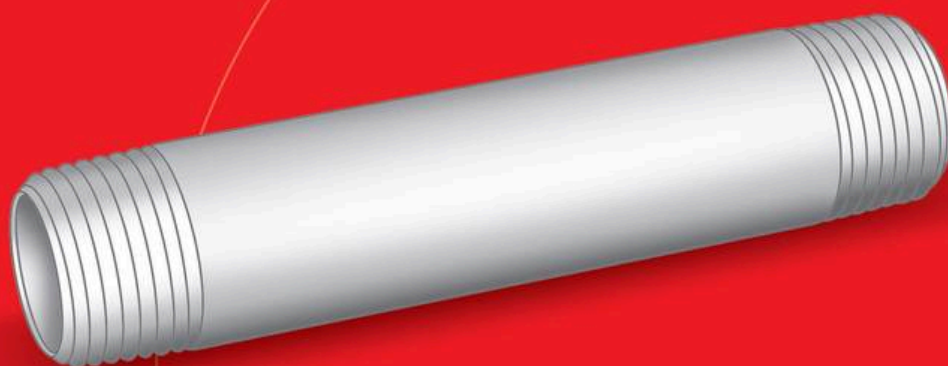
Dimensions in Inches

PART. NO	T O/D	P	A	E	F
FAI-1818	1/8"	1/8"	1.24	0.08	9/16"
FAI-1814	1/8"	1/4"	1.39	0.08	3/4"
FAI-1418	1/4"	1/8"	1.30	0.17	9/16"
FAI-1414	1/4"	1/4"	1.46	0.17	3/4"
FAI-1438	1/4"	3/8"	1.55	0.17	7/8"
FAI-1412	1/4"	1/2"	1.79	0.17	1 1/16"
FAI-3818	3/8"	1/8"	1.35	0.27	9/16"
FAI-3814	3/8"	1/4"	1.50	0.27	3/4"
FAI-3838	3/8"	3/8"	1.59	0.27	7/8"
FAI-3812	3/8"	1/2"	1.84	0.27	1 1/16"
FAI-1214	1/2"	1/4"	1.71	0.37	3/4"
FAI-1238	1/2"	3/8"	1.79	0.37	7/8"
FAI-1212	1/2"	1/2"	2.05	0.37	1 1/16"
FAI-3412	3/4"	1/2"	2.08	0.58	1 1/16"
FAI-3434	3/4"	3/4"	2.16	0.58	1 5/16"
FAI-3401	3/4"	1"	2.30	0.58	1 5/8"
FAI-0134	1"	3/4"	2.39	0.80	1 5/16"
FAI-0101	1"	1"	2.53	0.80	1 5/8"

Metric Range

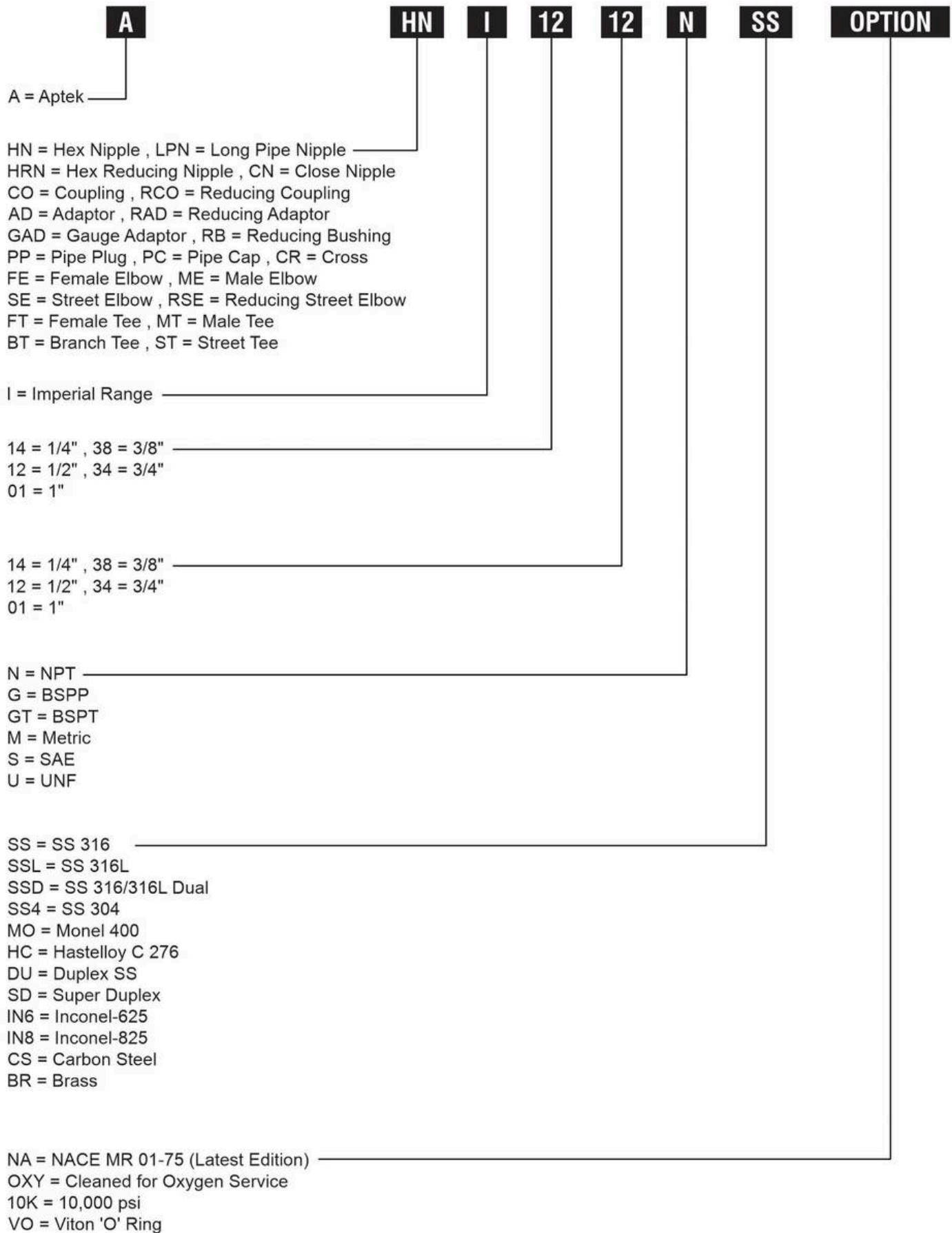
Dimensions in Inches

PART. NO	T O/D	P	A	E	F
FAM-0618	6	1/8"	32.5	4.1	14
FAM-0614	6	1/4"	37.1	4.1	19
FAM-0814	8	1/4"	37.6	5.6	19
FAM-1014	10	1/4"	38.1	7.1	19
FAM-1038	10	3/8"	40.1	7.1	22
FAM-1012	10	1/2"	46.7	7.1	27
FAM-1214	12	1/4"	43.7	8.8	19
FAM-1212	12	1/2"	52.3	8.8	27

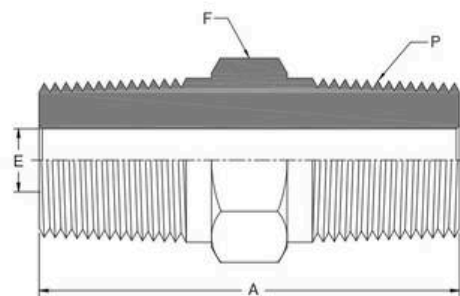


Instrumentation Pipe Fittings

Pipe Fittings Selection Chart

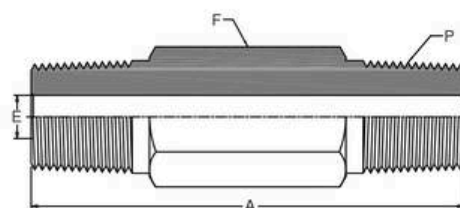


Hex Nipple



PART NO.	PIPE SIZE	DIMENSIONS			
	P	A	E	F	
HN-16	1/16	25.7	3	10	
HN-18	1/8	25.7	4.8	11	
HN-14	1/4	35.6	7.1	14	
HN-38	3/8	36.3	9.6	17	
HN-12	1/2	46.7	11.9	22	
HN-34	3/4	46.7	15.8	28	
HN-01	1	58.9	22.4	36	

Hex Long Nipple



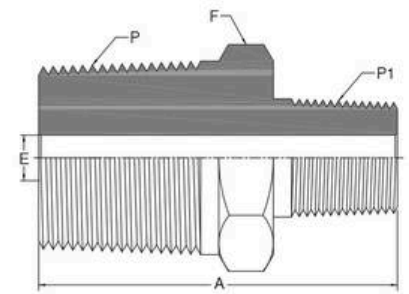
PART NO.	PIPE SIZE	DIMENSIONS							
	P	A (INCH)				E	F		
HLN-18	1/8	1.5	2	2.5	3	4	4.8	11	
HLN-14	1/4	1.5	2	2.5	3	4	7.1	14	
HLN-38	3/8	1.5	2	2.5	3	4	9.6	17	
HLN-12	1/2	1.5	2	2.5	3	4	11.9	22	
HLN-34	3/4	1.5	2	2.5	3	4	15.8	28	
HLN-01	1	-	2	2.5	3	4	22.4	36	

Long Pipe Nipple



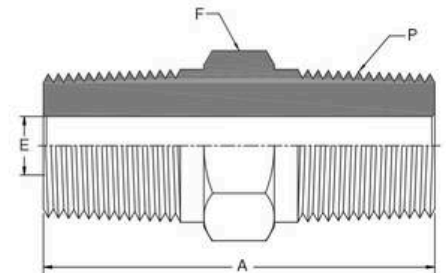
PART NO.	PIPE SIZE	DIMENSIONS							
	P	A (Inches)				E			
LPN-18	1/8	1.5	2	2.5	3	4	6.80		
LPN-14	1/4	1.5	2	2.5	3	4	9.24		
LPN-38	3/8	1.5	2	2.5	3	4	12.52		
LPN-12	1/2	1.5	2	2.5	3	4	15.80		
LPN-34	3/4	1.5	2	2.5	3	4	20.43		
LPN-01	1	-	2	2.5	3	4	26.65		

Hex Reducing Nipple



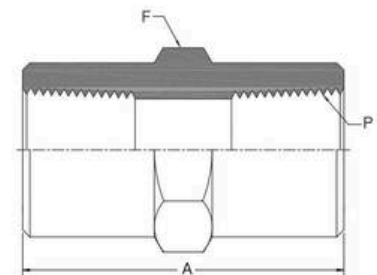
PART NO.	PIPE SIZE		DIMENSIONS		
	P	P1	A	E	F
HRN-1816	1/8	1/16	25.7	3.0	11
HRN-1418	1/4	1/8	31.0	4.8	14
HRN-3818	3/8	1/8	31.8	4.8	17
HRN-3814	3/8	1/4	36.3	7.1	17
HRN-1218	1/2	1/8	37.3	4.8	22
HRN-1214	1/2	1/4	41.9	7.1	22
HRN-1238	1/2	3/8	41.9	9.6	22
HRN-3414	3/4	1/4	41.9	7.1	28
HRN-3412	3/4	1/2	46.7	11.9	28
HRN-0112	1	1/2	54.1	11.9	36
HRN-0134	1	3/4	54.1	15.8	36

Close Nipple



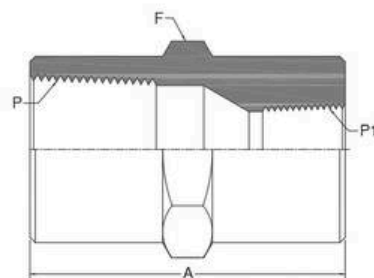
PART NO.	PIPE SIZE	DIMENSIONS		
	P	A	E	F
CN-18	1/8	19.00	4.8	11
CN-14	1/4	28.40	7.1	14
CN-38	3/8	28.40	9.6	17
CN-12	1/2	38.10	11.9	22
CN-34	3/4	38.10	15.8	28
CN-01	1	47.70	22.4	36

Coupling



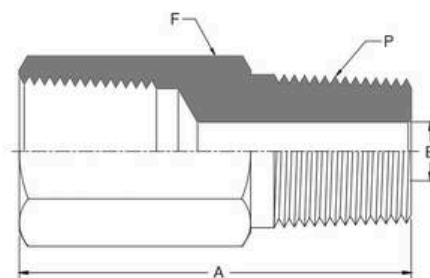
PART NO.	PIPE SIZE	DIMENSIONS		
	P	A	E	F
CO-18	1/8	20.6	8.6	14
CO-14	1/4	30.2	11.41	19
CO-38	3/8	33.3	15.0	22
CO-12	1/2	39.6	18.5	28
CO-34	3/4	41.1	23.9	36
CO-01	1	50.8	29.7	42

Reducing Coupling



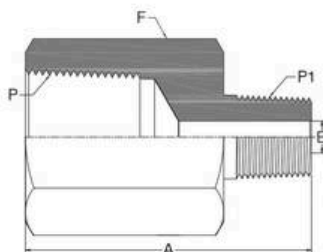
PART NO.	PIPE SIZE		DIMENSIONS		
	P	P1	A	E	F
RCO-1418	1/4	1/8	31.0	8.6	19
RCO-3814	3/8	1/4	35.1	11.4	22
RCO-1218	1/2	1/8	39.6	8.6	28
RCO-1214	1/2	1/4	44.5	11.4	28
RCO-1238	1/2	3/8	45.2	15.0	28
RCO-3414	3/4	1/4	42.9	11.4	36
RCO-3412	3/4	1/2	52.3	18.5	36
RCO-0112	1	1/2	55.6	18.5	42
RCO-0134	1	3/4	57.2	23.9	42

Adapter



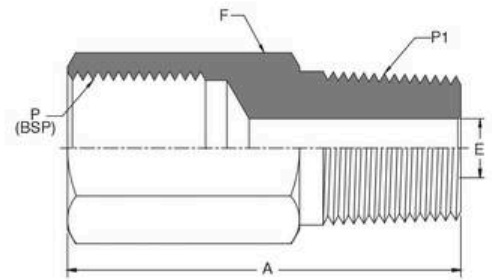
PART NO.	PIPE SIZE	DIMENSIONS		
	P	A	E	F
AD-18	1/8	27.9	4.8	14
AD-14	1/4	35.6	7.1	19
AD-38	3/8	38.4	9.6	22
AD-12	1/2	49.3	11.9	28
AD-34	3/4	51.3	15.7	36
AD-01	1	57.9	22.1	42

Reducing Adapter



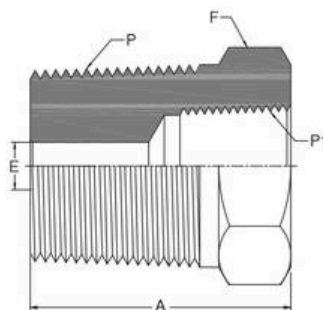
PART NO.	PIPE SIZE		DIMENSIONS		
	P	P1	A	E	F
RAD-1816	1/8	1/16	26.1	3	14
RAD-1418	1/4	1/8	31.0	4.8	19
RAD-3818	3/8	1/8	33.6	4.8	22
RAD-3814	3/8	1/4	38.0	7.1	22
RAD-1218	1/2	1/8	40.1	4.8	28
RAD-1214	1/2	1/4	44.7	7.1	28
RAD-1238	1/2	3/8	44.5	9.5	28
RAD-3414	3/4	1/4	46.2	7.1	36
RAD-3438	3/4	3/8	46.2	9.5	36
RAD-3412	3/4	1/2	51.3	11.9	36
RAD-0114	1	1/4	49.8	7.1	42
RAD-0112	1	1/2	54.9	11.9	42
RAD-0134	1	3/4	55.1	15.7	42

Gauge Adapter



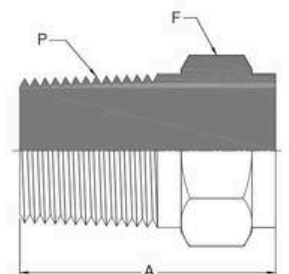
PART NO.	PIPE SIZE		DIMENSIONS		
	P	P1	A	E	F
GAD-14	1/4-19	1/4	27.7	3.0	19
GAD-38	3/8-19	3/8	32.0	4.8	22
GAD-12	1/2-14	1/2	41.9	7.1	28

Reducing Bushing



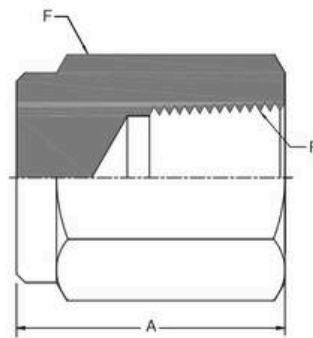
PART NO.	PIPE SIZE		DIMENSIONS		
	P	P1	A	E	F
RB-1816	1/8	1/16	26.2	4.8	12
RB-1418	1/4	1/8	26.9	7.1	14
RB-3818	3/8	1/8	21.8	8.6	19
RB-3814	3/8	1/4	30.2	9.5	19
RB-1218	1/2	1/8	27.4	8.6	22
RB-1214	1/2	1/4	27.4	11.4	22
RB-1238	1/2	3/8	35.8	11.9	22
RB-3414	3/4	1/4	27.4	11.4	28
RB-3438	3/4	3/8	27.4	15.0	28
RB-3412	3/4	1/2	41.4	15.7	28
RB-0114	1	1/4	34.8	11.4	36
RB-0138	1	3/8	34.8	15.0	36
RB-0112	1	1/2	34.8	17.45	36
RB-0134	1	3/4	47.0	22.4	36

Pipe Plug



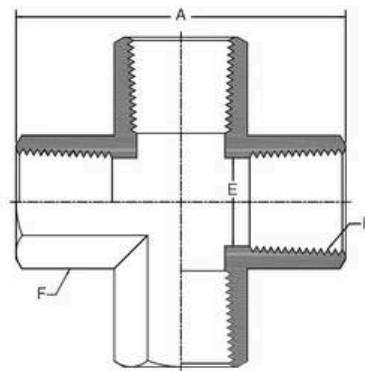
PART NO.	PIPE SIZE	DIMENSIONS	
	P	A	F
PP-16	1/16	19.0	8
PP-18	1/8	19.0	12
PP-14	1/4	24.5	14
PP-38	3/8	25.0	19
PP-12	1/2	31.0	22
PP-34	3/4	31.0	28
PP-01	1	38.0	36

Pipe Cap



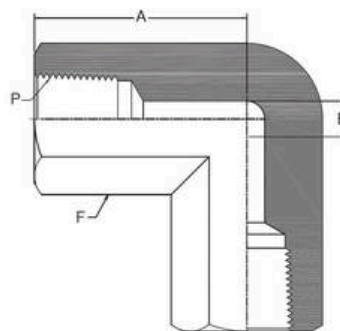
PART NO.	PIPE SIZE	DIMENSIONS		
	P	A	F	
PC-18	1/8	19.1	14	
PC-14	1/4	23.1	19	
PC-38	3/8	26.2	22	
PC-12	1/2	34.0	28	
PC-34	3/4	36.6	36	
PC-01	1	41.1	42	

Cross



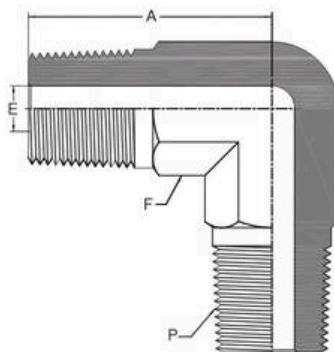
PART NO.	PIPE SIZE	DIMENSIONS			
	P	A	E	F	
CR-18	1/8	52.8	8.6	14	
CR-14	1/4	59.4	11.4	19	
CR-38	3/8	72.2	15.0	22	
CR-12	1/2	79.2	17.45	28	
CR-34	3/4	97.6	22.8	36	
CR-01	1	97.0	29.7	42	

Female Elbow



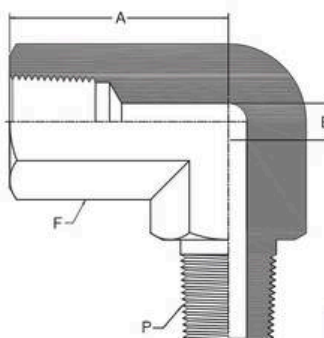
PART NO.	PIPE SIZE	DIMENSIONS			
	P	A	E	F	
FE-18	1/8	26.4	8.6	14	
FE-14	1/4	30.0	11.4	19	
FE-38	3/8	36.0	15.0	22	
FE-12	1/2	39.6	17.45	28	
FE-34	3/4	48.8	22.8	36	
FE-01	1	48.5	29.7	42	

Male Elbow



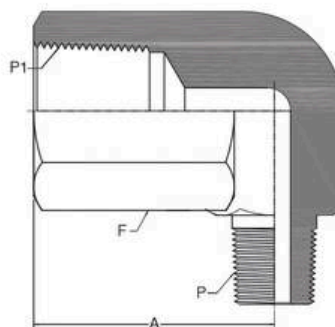
PART NO.	PIPE SIZE	DIMENSIONS			
	P	A	E	F	
ME-18	1/8	22.4	4.8	12	
ME-14	1/4	26.6	7.1	14	
ME-38	3/8	29.7	9.6	19	
ME-12	1/2	36.8	11.9	22	

Street Elbow



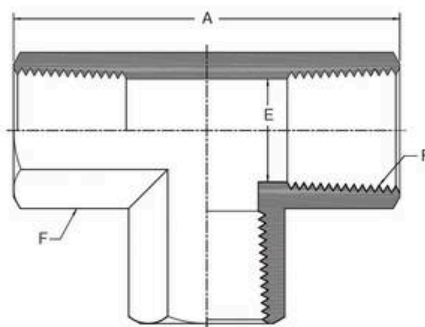
PART NO.	PIPE SIZE	DIMENSIONS				
	P	A	E	F	S	
SE-16	1/16	21.8	3.1	12	22.3	
SE-18	1/8	26.4	4.8	14	22	
SE-14	1/4	29.7	7.1	19	29.7	
SE-38	3/8	36.0	9.6	22	32	
SE-12	1/2	39.6	11.9	28	39.6	
SE-34	3/4	48.8	15.8	36	42.4	
SE-01	1	48.5	22.4	42	49.2	

Reducing Street Elbow



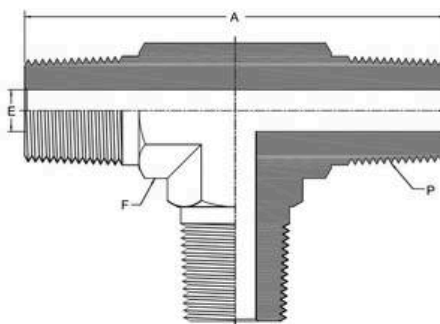
PART NO.	PIPE SIZE		DIMENSIONS				
	P	P1	A	E	F	S	
RSE-1418	1/4	1/8	29.7	4.8	14	25.4	
RSE-3814	3/8	1/4	36.0	7.1	19	32.0	
RSE-1214	1/2	1/4	39.6	7.1	22	35.0	
RSE-1238	1/2	3/8	39.6	9.6	22	35.0	
RSE-3412	3/4	1/2	40.3	11.9	27	42.4	
RSE-0112	1	1/2	48.5	11.9	36	49.2	

Female Tee



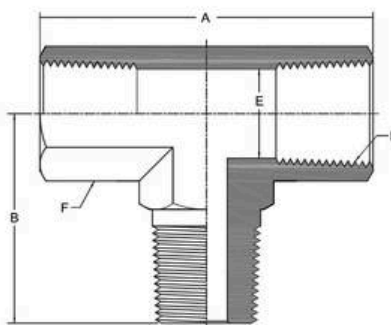
PART NO.	PIPE SIZE	DIMENSIONS			
	P	A	E	F	
FT-18	1/8	40.5	8.6	14	
FT-14	1/4	48.5	11.4	19	
FT-38	3/8	53.0	13.8	22	
FT-12	1/2	59.5	17.4	28	

Male Tee



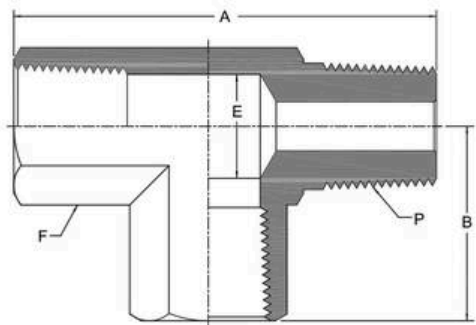
PART NO.	PIPE SIZE	DIMENSIONS			
	P	A	E	F	
MT-18	1/8	40.5	4.8	12	
MT-14	1/4	48.5	7.1	14	
MT-38	3/8	59.5	9.6	19	
MT-12	1/2	66.0	11.6	22	

Branch Tee

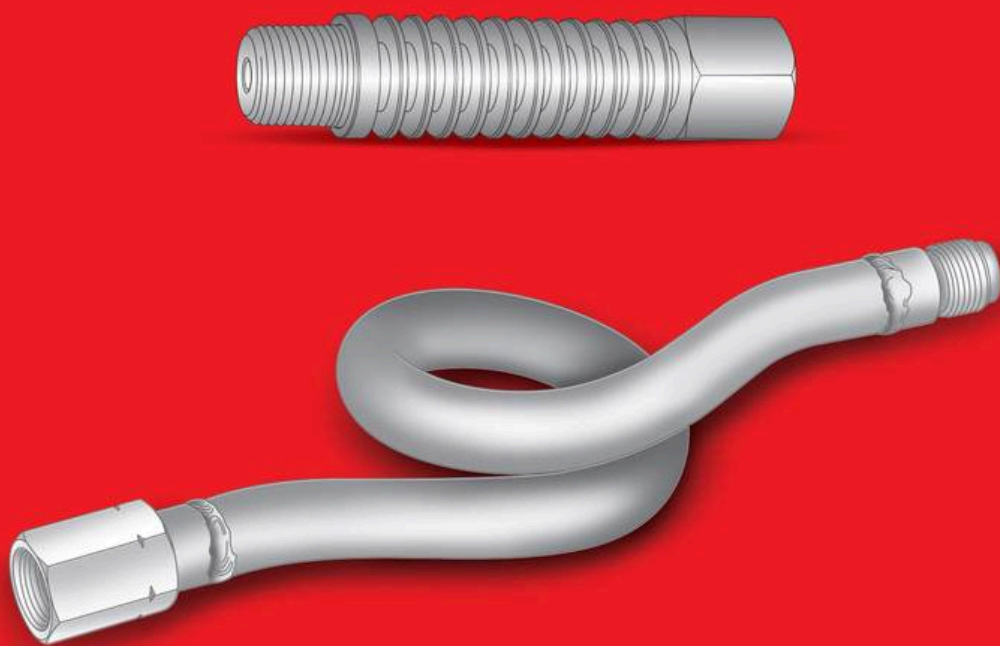


PART NO.	PIPE SIZE	DIMENSIONS				
	P	A	B	E	F	
BT-18	1/8	40.5	19.8	4.8	14	
BT-14	1/4	44.5	27.5	7.1	19	
BT-38	3/8	52.0	31.0	9.6	22	
BT-12	1/2	62.5	37.4	11.6	28	

Street Tee

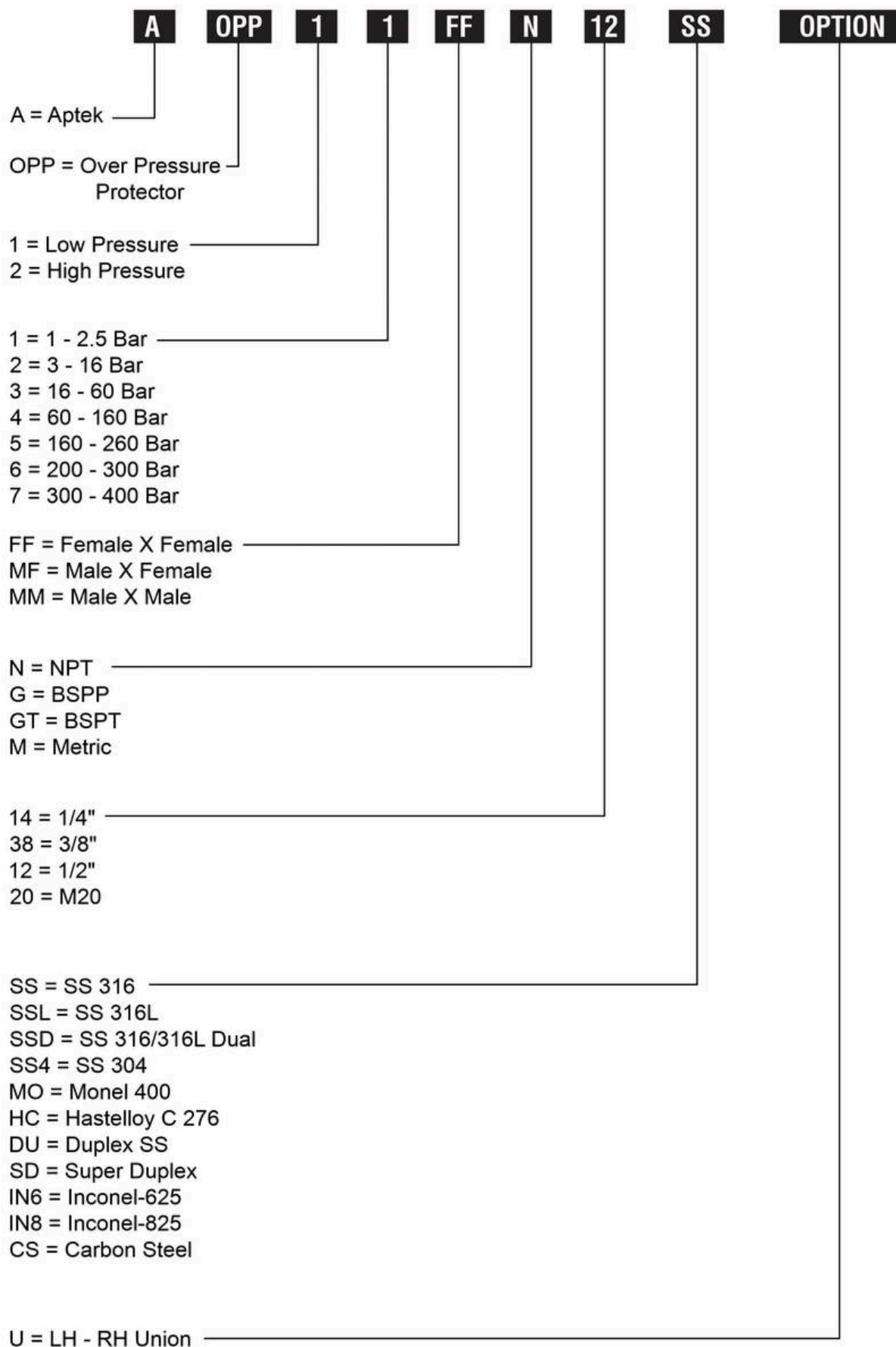


PART NO.	PIPE SIZE	DIMENSIONS				
	P	A	B	E	F	S
ST-18	1/8	39.6	16.8	4.8	14	22.0
ST-14	1/4	55.4	22.4	7.1	19	29.7
ST-38	3/8	62.0	25.9	9.6	22	32.0
ST-12	1/2	74.7	31.2	11.6	28	39.6



Pressure Gauge Accessories

Over Pressure Protector Selection Chart



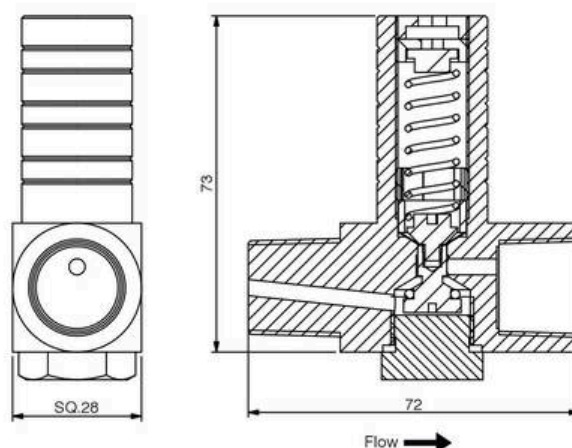
As the name indicates, this device saves instruments from the over pressure. Gauge savers close at a set point protects instruments from over load.

A OPP 1



Standard Specifications:

Size: 1/4" NPT M X F, G 1/2 LH-RH Union
Material: 316 SS
Seal: Viton
Set Pressure: 0.4 bar to 2.5 bar
Maximum working Pressure: 400 Bar
Over Pressure Safety up to: 1000 Bar
Max. Temp. : +80°C

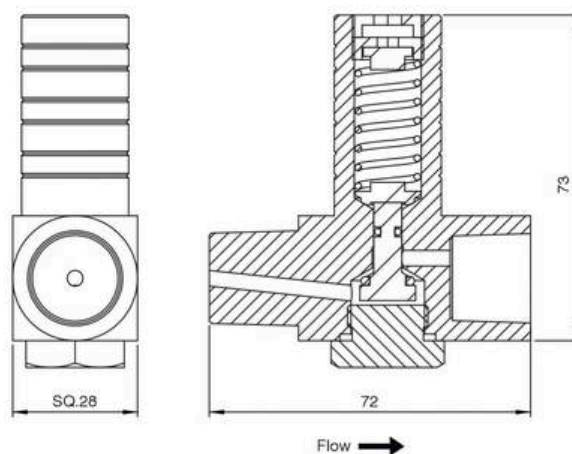


A OPP 2

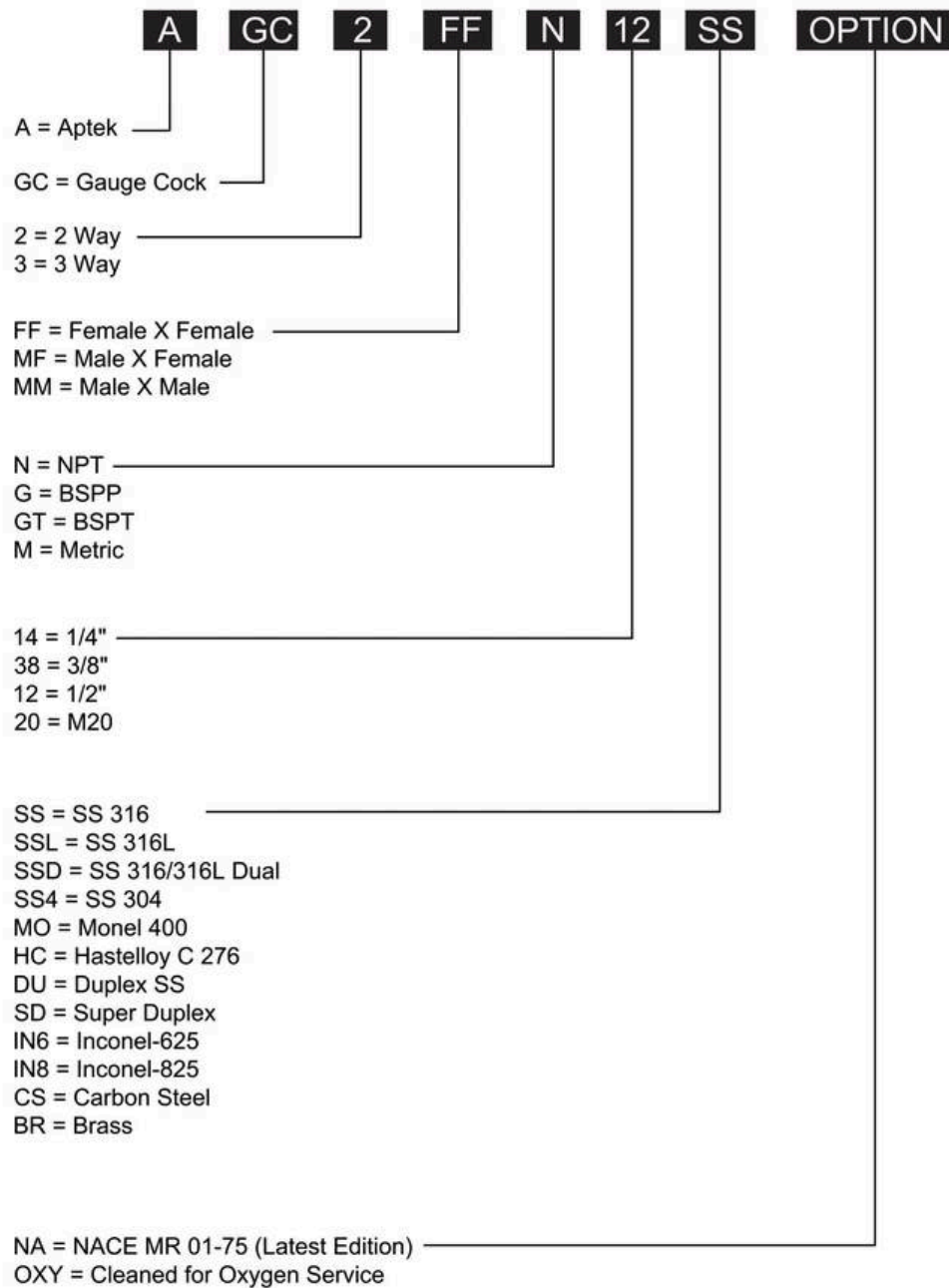


Standard Specifications:

Size: 1/4" NPT M X F, G 1/2 LH-RH Union
Material: 316 SS
Seal: Viton
Set Pressure: 3 to 16 bar, 16 to 60 bar, 60 to 160 bar, 160 to 260 bar, 200 to 300 bar, 300 to 400 bar
Maximum working Pressure: 400 Bar
Over Pressure Safety up to: 1000 Bar
Max. Temp. : +80°C



Gauge Cock Selection Chart



This cock is use with the pressure gauge for isolation and drain purpose.

Standard Specifications:

Size: 1/2" to 1/4"

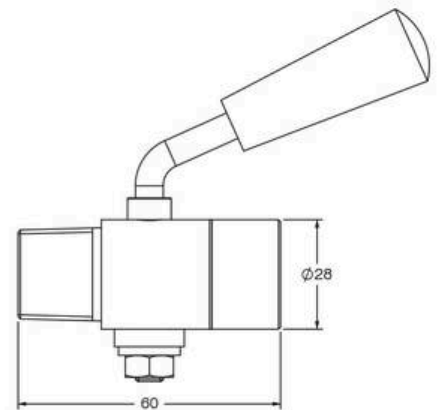
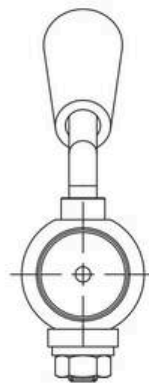
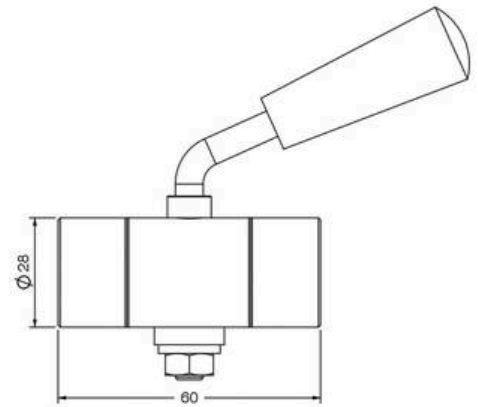
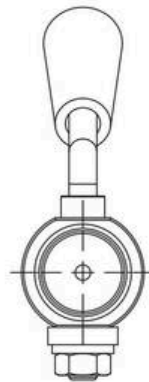
End Connections: NPT / BSPP / BSPT

Material: 316 SS / 316L SS / 304 SS / Brass

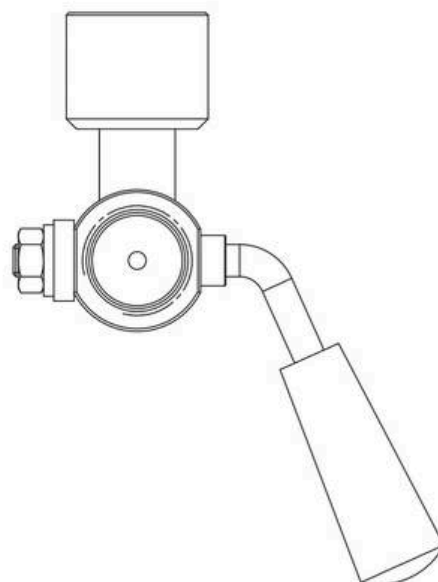
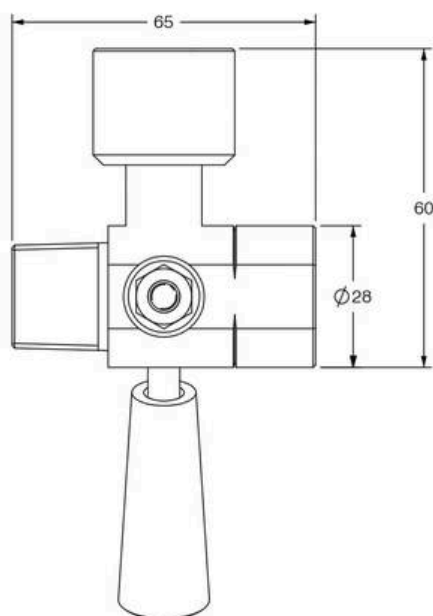
Seal: PTFE

Maximum working Pressure: 1500 PSI

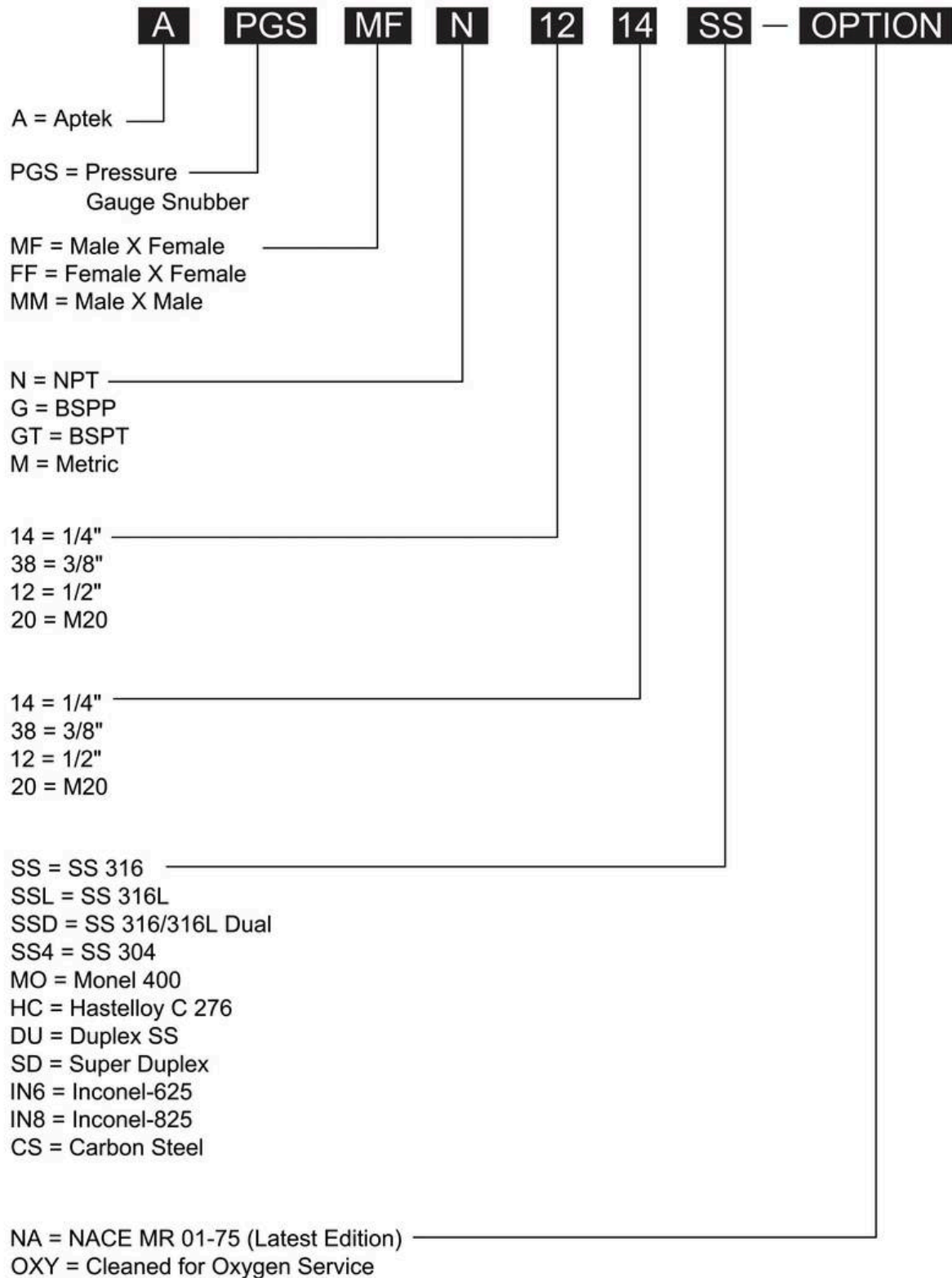
Gauge Cock 2-Way



Gauge Cock 3-Way



Pressure Gauge Snubber Selection Chart



With the help of variable orifice, these gauge snubbers protect pressure gauges by reducing fluctuations.

Standard Specifications :

Size: 1/2" to 1/4"

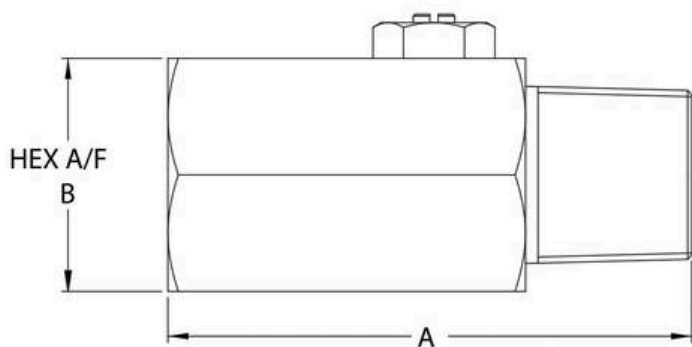
End Connections: NPT / BSPP / BSPT

Material: 316 SS / 316L SS / 304 SS / Brass

Seal: Viton

Maximum working Pressure: 6000 PSI

Maximum working Temperature: 120° C



Size	A	B
1/4" NPT	55	25
3/8" NPT	55	25
1/2" NPT	63	28

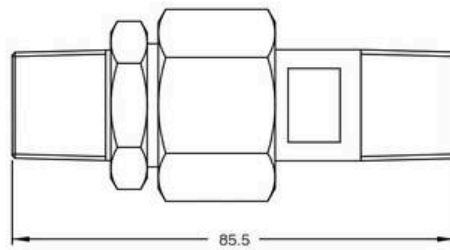
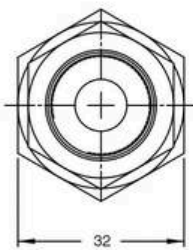
Standard Specifications:

Size: 1/4" NPT / BSPP / BSPT

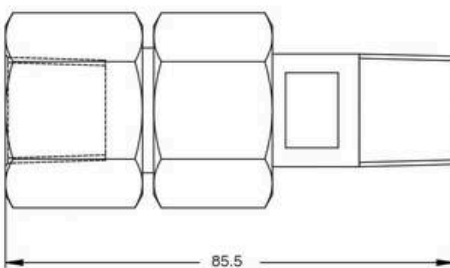
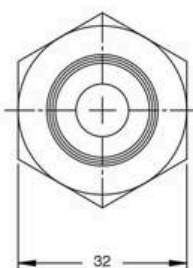
Material: 316 SS / 316L SS / 304 SS

Maximum working Pressure: 6000 PSI

Gauge Swivel Adapter M X M



Gauge Swivel Adapter M X F



The siphon is used to protect the pressure gauges from the steam or high temperature fluid. Siphon reduces the temperature of the process fluid by dissipating the heat. It also helps to reduce the line surges fluctuations.

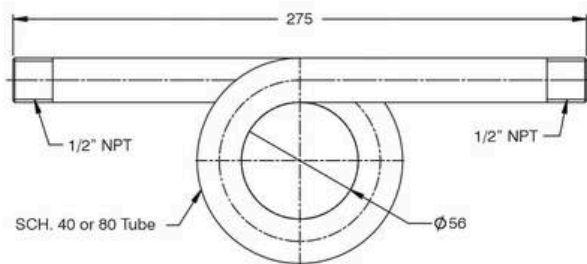
Standard Specifications:

Size: 1/2" to 1/4"

End Connections: NPT / BSPP / BSPT

Material: 316 SS / 316L SS / 304 SS / CS

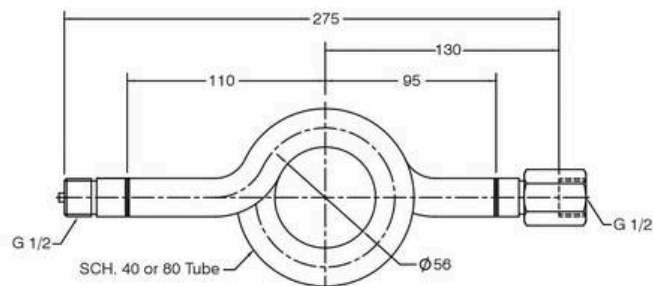
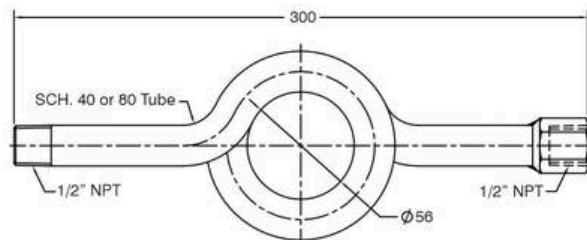
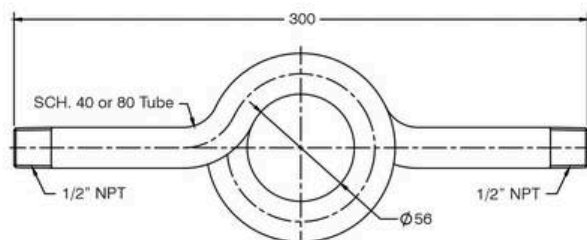
Pigtail Type Siphon



Pigtail Type Siphon



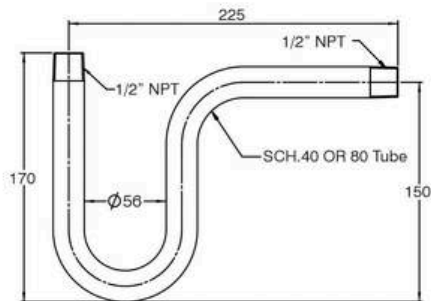
Coil Type Siphon



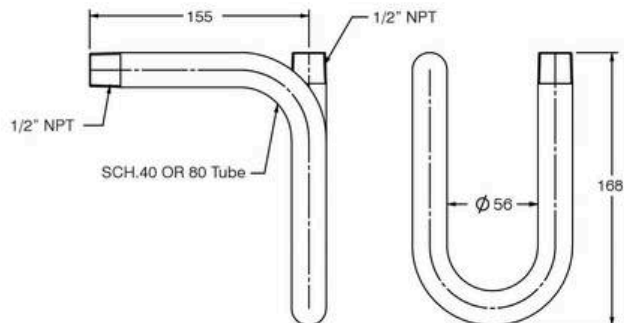
Coil Type Siphon



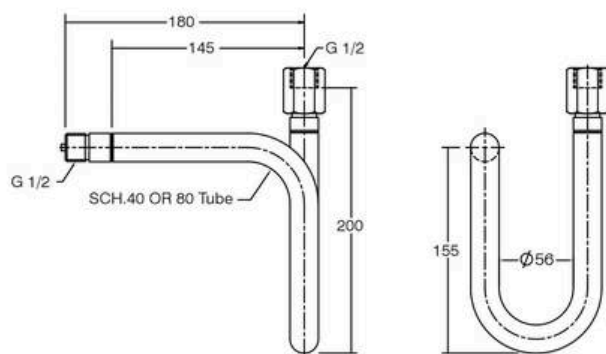
"U" Type Siphon



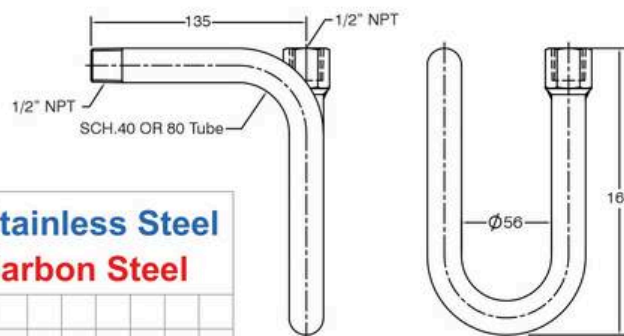
Flat "U" Type Siphon



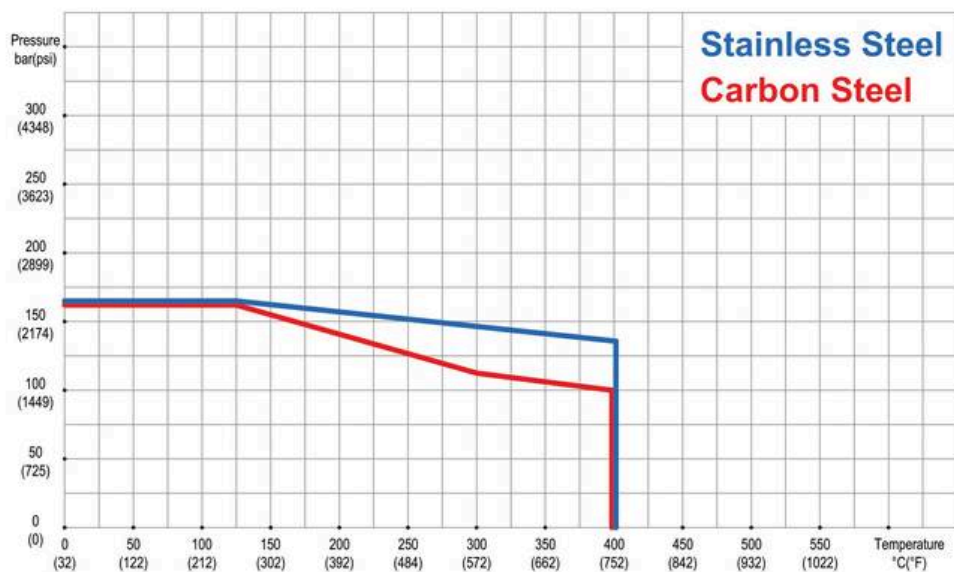
"U" Type Siphon



"U" Type Siphon DIN



"U" Type Siphon MXF



P-T CHART FOR SYPHON (COIL TYPE & U TYPE)

(Above P&T chart is for 1/2" SCH. 40 pipe, for other sizes and thickness contact factory)

Cooling Towers are designed to protect the pressure instrument from high process temperatures. This compact design uses cooling fins to disperse heat and cool the process temperature before it reaches the instrument. These can be fitted in conjunction with a chemical seal which cools the system fill fluid before it reaches the instrument.

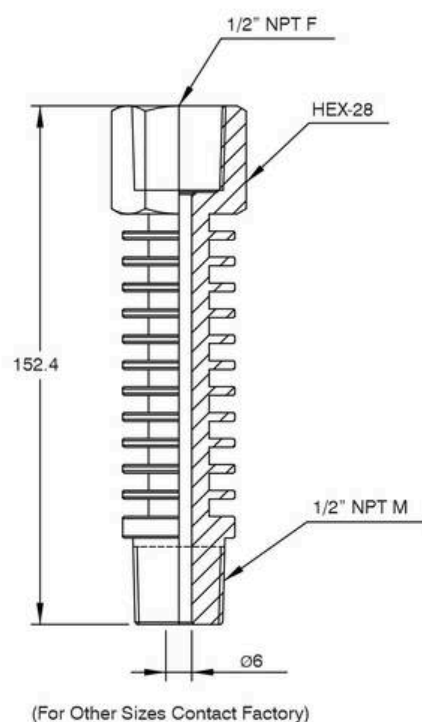
Standard Specifications :

1/4" or 1/2" NPT / BSP connections

Standard Lengths are 6", 8" & 10".

Standard MOC is SS 316/SS 316L, special materials available on request.

Cooling Towers



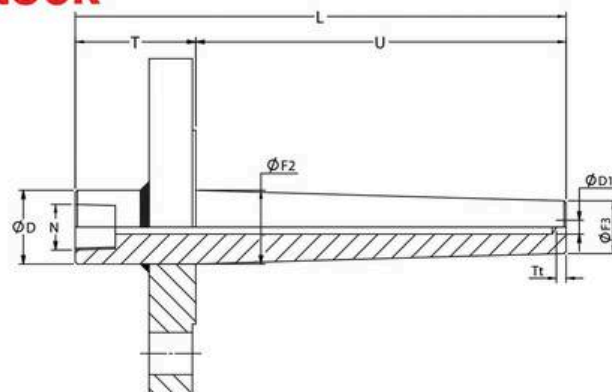


Thermowells

A thermowell acts as a barrier between a process medium and the sensing element of a temperature measuring device.

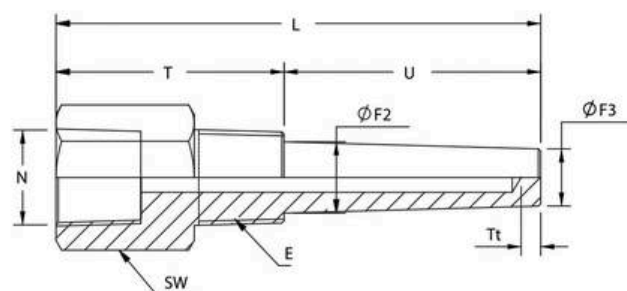
Material: 316 SS / 316L SS / 304 SS OR any special material

Thermowell - Flanged Bar Stock



SIZE	PRESSURE CLASS	T	ØF2	ØF3	ØD1	ØD		
1"	150	2 ½" (57 mm)	22	16	6.6 or 8.5	30		
	300							
	600	3 ½" (83 mm)						
	1500							
1/2"	150	2 ½" (57 mm)	25	19				
	300							
	600	3 ½" (83 mm)						
	1500							
2"	150	2 ½" (57 mm)	25	19				
	300							
	600	3 ½" (83 mm)						
	1500							

Thermowell - Screwed Bar Stock



SIZE	N	ØD1	ØF2	ØF3	HI	T	SW
1/2" NPT	1/2" NPT	6.6	16	13	19	1 ¾" OR 50 MM	27
		8.5					
3/4" NPT		6.6	22	16			36
		8.5					
1" NPT		6.6	27	19			
		8.5					

The flushing flange or ring is used for flushing / cleaning of diaphragm seals. This type of flushing flange or ring is clamped between process flange and diaphragm seal flange. The flush flange or ring is made of bar stock or forged material. The ring contains one or two flush holes. The flush holes can be selected in most common sizes. Flushing assemblies are having Needle valve or ball valve or flanged connections depending on the requirement.

Standard Specifications:

(Type ASME B16.5)

Size: 1" to 4"

Pressure Class: 150 to 2500

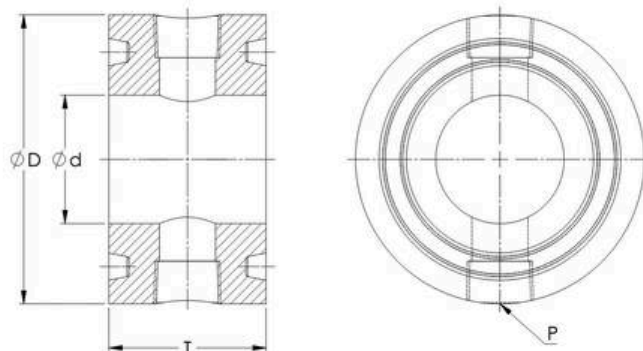
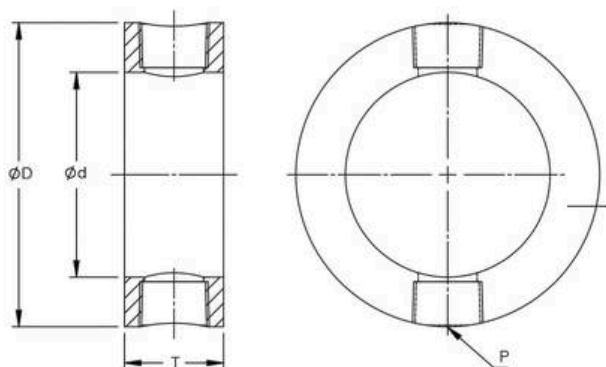
Flange Facing: RF, RTF, LMF, LFF, SMF, LTF, STF, LGF, SGF, SFF

MOC: SS 316, SS 316L, SS 316/316L Dual, SS 304, Monel 400, Hastelloy C 276, Duplex SS, Super Duplex,

Inconel-625, Inconel-825, Carbon Steel

Note: Flushing Rings & Flushing Flanges are also available for EN-1092 and API flanges.

Flushing Ring



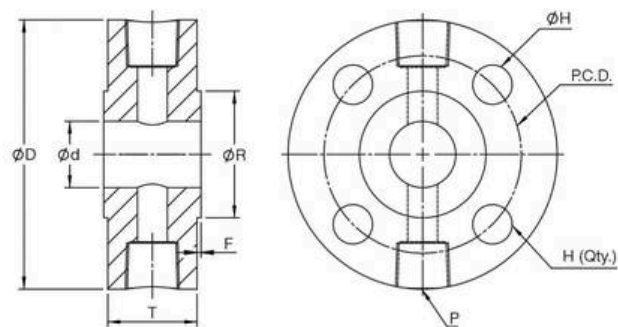
RTJ Flushing Ring

Size	Class	ØD	Ød	T	P	Ring		
1"	150	63.5	26.6	50.0	1/2" NPT	15		
	300	70.0				16		
	400-600	70.0				16		
	900-1500	71.5				16		
	2500	82.6				18		
1.5"	150	82.6	40.9					19
	300	90.5				20		
	400-600	90.5				20		
	900-1500	92.0				20		
	2500	114.0				23		
2"	150	102.0	52.5					22
	300	108.0				23		
	400-600	108.0				23		
	900-1500	124.0				24		
	2500	133.0				26		
3"	150	133.0	77.9					29
	300	146.0				31		
	400-600	146.0				31		
	900	156.0				31		
	1500	168.0				35		
4"	2500	168.0	102.3					32
	150	171.0				36		
	300	175.0				37		
	400	175.0				37		
	600	175.0				37		
	900	181.0				37		
	1500	194.0				39		
	2500	203.0					38	

RF Flushing Ring

SIZE	CLASS	ØD	Ød	T	P
3/4"	150 to 2500	42.9	21.0	35	1/2" NPT
1"		50.8	26.6		
1.25"		63.5	35.1		
1.5"		73.0	40.9		
2"		91.9	52.5		
2.5"		104.8	62.7		
3"		127.0	77.9		
4"		157.2	102.3		

Flushing Flange



SIZE	CLASS	øD	ød	T	P	P.C.D	øH	H(Qty.)	øR	F	
1"	150	108.0	26.6	39.0	1/2" NPT	79.2	15.9	4	50.8	1.5	
	300	124.0		39.0		88.9	19.1	4		1.5	
	400-600	124.0	24.3	49.0		88.9	19.1	4		6.4	
	900-1500	149.0				101.6	25.4	4			
	2500	159.0				108.0	25.4	4			
1.5"	150	124.0	40.9	39.0		98.6	15.9	4	73.0	1.5	
	300	155.0		39.0		114.3	22.3	4		1.5	
	400-600	155.0	38.1	49.0		114.3	22.3	4		6.4	
	900-1500	178.0				124.0	28.6	4			
	2500	203.0				146.0	31.8	4			
2"	150	152.0	52.5	39.0		120.6	19.1	4	91.9	1.5	
	300	165.0		39.0		127.0	19.1	8		1.5	
	400-600	165.0	49.2	49.0		127.0	19.1	8		6.4	
	900-1500	216.0				165.1	25.4	8			
	2500	235.0				171.4	28.6	8			
3"	150	190.0	77.9	39.0		152.4	19.1	8	127.0	1.5	
	300	210.0		39.0		168.1	22.3	8		1.5	
	400-600	210.0	73.7	49.0		168.1	22.3	8		6.4	
	900	241.0				190.5	25.4	8			
	1500	267.0				203.2	31.9	8			
	2500	305.0				228.6	35.0	8			
4"	150	229.0	102.3	39.0		190.5	19.1	8	157.2	1.5	
	300	254.0		39.0		200.2	22.3	8		1.5	
	400	254.0		49.0		200.2	25.5	8		6.4	
	600	273.0	215.9			25.5	8				
	900	292.0	235.0			31.8	8				
	1500	311.0	241.3			34.9	8				
	2500	356.0	87.3			273.0	41.3	8			



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Instrumentation Valves,
Fittings and Accessories

Aptek Instrumentation Pvt. Ltd.

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